

GATES LEARJET CORPORATION
maintenance manual

CHAPTER

28

FUEL

Gates Learjet Corporation

maintenance manual

Chapter Section Subject	Page	Date	Chapter Section Subject	Page	Date
28-Title			28-12-00	201	Dec 2/77
28-Record of Temp.			28-12-01	1	Dec 2/77
Revisions			28-12-01	201	Dec 2/77
*28-List of			28-12-01	202	Dec 2/77
Effective Pages	1	May 25/84	28-12-02	201	Dec 2/77
*28-List of			28-12-02	202	Dec 2/77
Effective Pages	2	May 25/84	28-12-03	1	Dec 2/77
*28-Contents	1	May 25/84	28-12-03	201	Dec 18/81
*28-Contents	2	May 25/84	28-12-03	202	Dec 2/77
*28-Contents	3	May 25/84	28-12-03	203	Dec 18/81
*28-Contents	4	May 25/84	28-12-04	201	Dec 18/81
28-Contents	5	Dec 18/81	28-12-05	201	Dec 18/81
28-00-00	1	Dec 2/77	28-12-05	202	Dec 18/81
28-00-00	2	Dec 2/77	28-13-00	1	Dec 18/81
28-00-00	201	Dec 2/77	28-13-00	2	Dec 18/81
28-00-00	202	Dec 18/81	28-13-00	3	Dec 18/81
*28-00-00	203	May 25/84	28-13-00	4	Dec 18/81
*28-00-00	204	May 25/84	28-13-00	201	Dec 18/81
28-10-00	1	Dec 18/81	28-13-01	201	Dec 18/81
28-10-00	2	Dec 2/77	28-13-01	202	Dec 2/77
28-10-00	201	Dec 18/81	28-13-01	203	Dec 2/77
28-10-00	202	Dec 18/81	28-13-01	204	Dec 2/77
28-10-00	203	Dec 18/81	28-13-01	205	Dec 2/77
28-10-00	204	Dec 18/81	28-13-01	206	Dec 2/77
28-11-00	1	Dec 18/81	28-13-01	207	Dec 18/81
28-11-00	2	Dec 18/81	28-13-01	208	Dec 18/81
28-11-00	3	Dec 18/81	28-13-01	209	Dec 18/81
28-11-00	4	Dec 18/81	28-13-01	210	Dec 2/77
28-11-00	201	Dec 18/81	28-13-01	211	Dec 18/81
28-11-00	202	Dec 18/81	28-13-01	212	Dec 18/81
28-11-00	203	Dec 18/81	28-13-01	213	Dec 2/77
28-11-00	204	Dec 18/81	28-13-01	214	Dec 2/77
28-11-01	201	Dec 18/81	28-13-02	201	Dec 2/77
28-11-02	201	Dec 18/81	28-13-02	202	Dec 2/77
28-11-02	202	Dec 18/81	28-13-02	203	Dec 2/77
28-11-03	201	Dec 18/81	28-13-03	1	Dec 2/77
28-11-04	201	Dec 18/81	28-13-03	201	Dec 2/77
28-12-00	1	Dec 2/77	28-13-04	1	Dec 18/81
28-12-00	2	Dec 2/77	28-13-04	201	Dec 18/81
28-12-00	3	Dec 18/81			

Insert Latest Revised Pages; Destroy Superseded or Deleted Pages.

* Asterisk indicates pages revised, added or deleted by current revision.

The portion of the text affected by the current revision is indicated by a vertical line in the outer margin of the page.

28-LIST OF EFFECTIVE PAGES

Page 1

May 25/84

Gates Learjet Corporation

maintenance manual

Chapter Section Subject	Page	Date	Chapter Section Subject	Page	Date
28-13-04	202	Dec 2/77	28-20-05	204	Dec 18/81
28-13-04	203	Dec 18/77	28-20-05	205	Dec 18/81
28-13-05	1	Dec 2/77	28-20-05	206	Dec 18/81
28-13-05	201	Dec 2/77	28-20-06	201	Dec 18/81
28-13-05	202	Dec 2/77	28-20-07	201	Dec 2/77
28-14-00	1	Dec 2/77	28-20-07	202	Dec 2/77
28-14-00	2	Dec 2/77	28-20-08	201	Dec 18/81
28-14-00	3	Dec 2/77	28-20-08	202	Dec 18/81
28-14-00	4	Dec 18/81	28-30-00	1	Dec 2/77
28-14-00	5	Dec 18/81	28-30-00	201	Dec 2/77
28-14-00	6	Dec 18/81	28-40-00	1	Dec 18/81
28-14-00	201	Dec 2/77	28-40-00	2	Dec 2/77
28-14-01	201	Dec 2/77	28-40-00	101	Dec 18/81
28-14-01	202	Dec 2/77	28-40-00	102	Dec 18/81
28-14-01	203	May 25/84	28-40-00	103	Dec 18/81
28-14-01	204	May 25/84	28-40-00	201	Dec 18/81
28-14-02	201	Dec 2/77	28-40-00	202	Dec 18/81
28-14-03	201	Dec 2/77	28-40-00	203	Dec 18/81
28-14-03	202	Dec 2/77	28-40-00	204	Dec 18/81
28-20-00	1	Dec 2/77	28-40-00	205	Dec 18/81
28-20-00	2	Dec 18/81	28-40-00	206	Dec 18/81
28-20-00	3	Dec 18/81	28-40-00	207	Dec 18/81
28-20-00	4	Dec 18/81	28-40-00	208	Dec 18/81
28-20-00	5/6	Dec 18/81	28-40-00	209	Dec 18/81
28-20-00	7	Dec 18/81	28-40-00	210	Dec 18/81
28-20-00	8	Dec 18/81	28-40-00	211	Dec 18/81
28-20-00	9	Dec 18/81	28-40-00	212	Dec 18/81
28-20-00	10	Dec 18/81	28-40-00	213	Dec 18/81
28-20-00	11/12	Dec 18/81	28-40-00	214	Dec 18/81
28-20-00	13	Dec 18/81	28-40-00	215	Dec 18/81
28-20-01	1	Dec 2/77	28-40-01	201	Dec 18/81
28-20-01	201	Dec 2/77	28-40-02	201	Dec 18/81
28-20-02	201	Dec 2/77	28-40-02	202	Dec 18/81
28-20-02	202	Dec 2/77	28-40-02	203	Dec 18/81
28-20-03	1	Dec 18/81	28-40-02	204	Dec 18/81
28-20-03	201	Dec 18/81			
28-20-03	202	Dec 18/81			
28-20-03	203	Dec 18/81			
28-20-03	204	Dec 18/81			
28-20-04	1	Dec 18/81			
28-20-04	201	Dec 18/81			
28-20-04	202	Dec 18/81			
28-20-05	1	Dec 18/81			
28-20-05	201	Dec 18/81			
28-20-05	202	Dec 18/81			
28-20-05	203	Dec 18/81			

28-LIST OF EFFECTIVE PAGES

Page 2

May 25/84

Gates Learjet Corporation

RECORD OF TEMPORARY REVISIONS

[illegible]

Gates Learjet Corporation

maintenance manual

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
GENERAL	28-00-00	
General	28-00-00	1
Description	28-00-00	1
General - Maintenance Practices	28-00-00	
General	28-00-00	201
Tools and Equipment	28-00-00	201
Safety Practices	28-00-00	201
Preventing Contamination of Tanks	28-00-00	202
Fuel Leak Classification	28-00-00	202
Fuel Hose Installation	28-00-00	203
STORAGE	28-10-00	
General	28-10-00	1
Description	28-10-00	1
Storage - Maintenance Practices	28-10-00	
General	28-10-00	201
Leak Path Analysis	28-10-00	201
Inspection of Tank Interior	28-10-00	202
Leak Test	28-10-00	203
Fuel Tank Sealing Repairs	28-10-00	203
Tip Tank - Description and Operation	28-11-00	
Description	28-11-00	1
Operation	28-11-00	3
Tip Tank Maintenance Practices	28-11-00	
Removal/Installation	28-11-00	201
Inspection/Check	28-11-00	205
Cleaning/Painting	28-11-00	205
Jet Pump - Maintenance Practices	28-11-01	
Removal/Installation	28-11-01	201
Drain Valves - Maintenance Practices	28-11-02	
Approved Repairs	28-11-02	201
Pressure Relief Valves - Maintenance Practices	28-11-03	
Removal/Installation	28-11-03	201
Vacuum Relief Valve - Maintenance Practices	28-11-04	
Removal/Installation	28-11-04	201
Wing Tank - Description and Operation	28-12-00	
Description	28-12-00	1
Wing Tank - Maintenance Practices	28-12-00	
General	28-12-00	201
Lightning Strike Protection Seal Application	28-12-00	201
Wing Fuel Pressure Switch - Description and Operation ..	28-12-01	
General	28-12-01	1
Operation	28-12-01	1

Gates Learjet Corporation

maintenance manual

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
Wing Fuel Pressure Switch - Maintenance Practices . .	28-12-01	
General	28-12-01	201
Removal/Installation	28-12-01	201
Inspection/Check	28-12-01	201
Drain Valve - Maintenance Practices	28-12-02	
General	28-12-02	201
Approved Repairs	28-12-02	201
Wing Float Switch - Description and Operation	28-12-03	
Operation	28-12-03	1
Wing Float Switch - Maintenance Practices	28-12-03	
Removal/Installation	28-12-03	201
Inspection/Check	28-12-03	201
Check Valve - Maintenance Practices	28-12-04	
General	28-12-04	201
Removal/Installation	28-12-04	201
Crossflow Valve - Maintenance Practices	28-12-05	
General	28-12-05	201
Removal/Installation	28-12-05	201
Approved Repairs	28-12-05	201
Fuselage Tank - Description and Operation	28-13-00	
Description	28-13-00	1
Operation	28-13-00	1
Fuselage Tank - Maintenance Practices	28-13-00	
General	28-13-00	201
Fuel Bladder - Maintenance Practices	28-13-01	
Removal/Installation	28-13-01	201
Fuel Transfer Valve - Maintenance Practices	28-13-02	
Description	28-13-02	201
Operation	28-13-02	201
Removal/Installation	28-13-02	201
Fuselage Pressure Switch - Description and Operation . .	28-13-03	
General	28-13-03	1
Operation	28-13-03	1
Fuselage Pressure Switch - Maintenance Practices . . .	28-13-03	
Removal/Installation	28-13-03	201
Fuselage Float Switch - Description and Operation	28-13-04	
Description	28-13-04	1
Operation	28-13-04	1
Fuel Float Switch - Maintenance Practices	28-13-04	
Removal/Installation	28-13-04	201
Fuel Boost Pump - Description and Operation	28-13-05	
Description	28-13-05	1
Operation	28-13-05	1

28 - CONTENTS

Page 2

May 25/84

Gates Learjet Corporation

maintenance manual

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
Fuel Boost Pump - Maintenance Practices	28-13-05	
General	28-13-05	201
Removal/Installation	28-13-05	201
Fuel Vent System - Description and Operation	28-14-00	
General	28-14-00	1
Operation	28-14-00	1
Fuel Vent System - Maintenance Practices	28-14-00	
General	28-14-00	201
Adjustment/Test	28-14-00	201
Float Valve - Maintenance Practices	28-14-01	
General	28-14-01	201
Description	28-14-01	201
Removal/Installation	28-14-01	201
Approved Repairs	28-14-01	202
Sump Assembly - Maintenance Practices	28-14-02	
General	28-14-02	201
Removal/Installation	28-14-02	201
Flame Arrestor - Maintenance Practices	28-14-03	
General	28-14-03	201
Removal/Installation	28-14-03	201
Airscoop - Maintenance Practices	28-14-04	
General	28-14-04	201
Removal/Installation	28-14-04	201
DISTRIBUTION - DESCRIPTION AND OPERATION	28-20-00	
General	28-20-00	1
Operation	28-20-00	1
Adjustment/Test	28-20-00	17
Jet Pump - Description and Operation	28-20-01	
General	28-20-01	1
Description	28-20-01	1
Jet Pump - Maintenance Practices	28-20-01	
Removal/Installation	28-20-01	201
Fuel Boost Pump - Maintenance Practices	28-20-02	
General	28-20-02	201
Removal/Installation	28-20-02	201
Fuel Filter - Description and Operation	28-20-03	
Description	28-20-03	1
Operation	28-20-03	1
Fuel Filter - Maintenance Practices	28-20-03	
General	28-20-03	201
Removal/Installation	28-20-03	201
Adjustment/Test	28-20-03	203
Cleaning/Painting	28-20-03	204

Gates Learjet Corporation

maintenance manual

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
Fuel Supply Shutoff Valve - Description and Operation	28-20-04	
General	28-20-04	1
Operation	28-20-04	1
Fuel Supply Shutoff Valve - Maintenance Practices	28-20-04	
Removal/Installation	28-20-04	201
Inspection/Check	28-20-04	203
Motive Flow Valve - Description and Operation	28-20-05	
Description	28-20-05	1
Operation	28-20-05	1
Motive Flow Valve - Maintenance Practices	28-20-05	
Removal/Installation	28-20-05	201
Relief Valve - Maintenance Practices	28-20-06	
General	28-20-06	201
Removal/Installation	28-20-06	201
Fuel Control Relay Panel - Maintenance Practices	28-20-07	
General	28-20-07	201
Removal/Installation	28-20-07	201
Fuel Supply Line Check Valves - Maintenance Practices	28-20-08	
Description	28-20-08	201
Inspection/Check	28-20-08	201
FUEL JETTISON - DESCRIPTION AND OPERATION	28-30-00	
General	28-30-00	1
Description	28-30-00	1
Operation	28-30-00	1
Fuel Jettison - Maintenance Practices	28-30-00	
General	28-30-00	201
Removal/Installation	28-30-00	201
INDICATING - DESCRIPTION AND OPERATION	28-40-00	
Description	28-40-00	1
Indicating Trouble Shooting	28-40-00	
Description	28-40-00	101
Trouble Shooting	28-40-00	101
Indicating - Maintenance Practices	28-40-00	
Adjustment/Test	28-40-00	201
Inspection/Check	28-40-00	215
Fuel Quantity Indicator - Maintenance Practices	28-40-01	
Removal/Installation	28-40-01	201
Fuel Probe - Maintenance Practices	28-40-02	
Removal/Installation	28-40-02	201
Inspection/Check	28-40-02	203
Cleaning/Painting	28-40-02	204

Gates Learjet Corporation
maintenance manual

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
Pressure Switch - Description and Operation	28-41-00	
General	28-41-00	1
Operation	28-41-00	1
Pressure Switch - Maintenance Practices	28-41-00	
General	28-41-00	201
Removal/Installation	28-41-00	201
Fuel Counter System - Maintenance Practices	28-42-00	
General	28-42-00	201
Operation	28-42-00	201



GATES LEARJET CORPORATION

maintenance manual

GENERAL - DESCRIPTION AND OPERATION

1. General

- A. The aircraft fuel system consists of fuel storage, fuel tank venting, fuel distribution and fuel quantity indicating. Fuel is stored in integral wing tanks, the tip tanks and the fuselage tank.

2. Description

- A. The tip tanks are attached to each wing at two attach points, one forward of spar 7 and one aft of spar 1. Baffles are incorporated to minimize slosh and prevent adverse fuel effects on the aircraft center of gravity. Five access covers, installed in the tip tank, provide entry for inspection and maintenance. Fuel is transferred to the wing tank by gravity flow and a jet pump.
- B. The full span, integral wing tanks comprise a large portion of each wing. The inboard end of the tank is at the aircraft centerline and the outboard end is at wing station 181.10. The forward side of the tank ends at wing spar No. 1 and the aft side terminates at wing spar No. 7. Fuel is not stored in the main landing gear wheel wells and wing leading edge areas. Small vent holes are incorporated in the upper and lower portion of each spar and rib. This allows fuel to move from one compartment to the other due to fuel expansion. It also allows the individual compartments to completely fill and drain. The wing dihedral of 2.5° causes the inboard portion of the wing to be the lowest point of the fuel system. This assures that the primary fuel pumps will be submerged in fuel until the tanks are practically empty. Wing rib 0.0 incorporates two pressure relief valves (allowing flow in opposite directions). The relief valves are set at approximately 1 psi.
- C. On 25B/D aircraft, the fuselage tank consists of two bladder-type cells located in the cavity between frames 22 and 25. The cavity is a sealed compartment with hangers installed in the upper portion. The hangers mate with fasteners on the bladder to provide hanging support. Additional bladder support is gained by the interconnect fittings, access covers and plumbing fittings incorporated in the bladder. Two drain ports are located in the aft portion of the cavity. Interconnecting tubes and a crossover tube connect the bladders to form a single tank. Access to the bladders is gained through two access covers in the tailcone; an access cover on the top of the fuselage and through the keel beam access covers.
- D. On 25C/F aircraft the fuselage tank, consisting of four bladder-type cells, is located in cavities between frames 18 and 25. The cavity is a sealed compartment with hangers installed in the upper portion. The hangers mate with fasteners on the bladder to provide hanging support. Additional bladder support is gained by the interconnect fittings, access covers and plumbing fittings incorporated in the bladder. The cavity also incorporates four cavity drain ports.



GATES LEARJET CORPORATION

maintenance manual

Two are located in the forward cavity area and two in the aft area. Interconnecting tubes and a crossover tube connect the bladders to form a single tank. Access to the bladders is gained through two large access covers in the tailcone; two access covers behind frame 18 pressure bulkhead; an access cover on the top of the fuselage and through the keel beam access covers.

- E. The fuel vent system provides continuous ram air pressure to the tip tanks, wing tanks and the fuselage tank while the aircraft is in flight. Ram air is admitted through the ram air scoops to the tip tanks and the fuselage tank. The ram air pressure is vented to the wing tanks through the wing-to-tip tank interconnects. Ram air pressure is approximately 1 psi. A sump drain valve is located on the aircraft centerline at approximately F. S. 421.42 (frame 22).
- F. Engine fuel flow is provided by two jet pumps, one located in each wing near wing rib 0.0. Two electrically operated boost pumps are also installed, adjacent to the jet pumps, for engine starting and as a standby in case of jet pump failure.
- G. A capacitance-type fuel quantity indicating system is installed in the aircraft. The system components consist of an indicator and a selector switch on the fuel control panel and seven tank unit probes. The tank unit probes are located one in each tip tank, three in each wing tank, and one in the fuselage tank.
- H. When aircraft is stored for an extended period of time the fuel system must be preserved in accordance with procedures outlined in 10-10-00.

GATES LEARJET CORPORATION
maintenance manual

GENERAL - MAINTENANCE PRACTICES

1. General

- A. Prior to performing maintenance on the aircraft fuel system, maintenance personnel should read, thoroughly understand and carefully adhere to the following instructions. These instructions will aid in maintaining a functional and trouble-free system and provide maximum safety for the aircraft and personnel.
- B. Check condition and replace, if necessary, seals and gaskets removed during maintenance. Fuel system gaskets are neoprene and do not require lubrication prior to installation.

2. Tools and Equipment

NAME	NUMBER	MANUFACTURER	USE
Explosion-Proof Lights		Commercially Available	For use in tank.
Air Mover			To ventilate tank.

3. Safety Practices

CAUTION: DO NOT USE SAFETY WIRE TO SECURE ANY UNIT OR FITTING WITHIN THE FUEL TANKS. SAFETY WIRE ENDS CAN BECOME THE POINT OF ELECTROSTATIC DISCHARGE.

- A. Perform fuel system maintenance in areas which permit free movement of fire fighting and other emergency equipment.
- B. Maintenance area selected shall not be upwind of any building in which fuel vapor may accumulate, any open flame or spark producing equipment is located, or smoking is permitted.

WARNING: DO NOT WEAR NYLON OR OTHER SYNTHETIC CLOTHES; THEY WILL GENERATE STATIC ELECTRICITY.

- C. Personnel should wear clean, lint-free cotton clothes with nonspark producing zippers or buttons when working in fuel tanks.
- D. Remove all spark producing items such as electrically powered tools or electronic test equipment from the aircraft and vapor hazard area.
- E. Use explosion-proof lights and air-driven tools in fuel tanks.
- F. Connect grounding cable to an approved ground and aircraft, refer to 12-10-10.

EFFECTIVITY: ALL

Gates Learjet Corporation

maintenance manual

- G. Ensure that no high frequency radio transmitters shall be operated within 200 feet and that no radar equipment shall be operated within 400 feet of the aircraft during fuel maintenance or while fuel tank access covers are removed.
- H. Air ventilate fuel tanks until fuel vapor concentration is determined to be below explosive limit before removing access covers.

CAUTION: ALTHOUGH FUEL VAPOR WITHIN A TANK MAY BE TOO RICH TO BURN WHEN TANK IS OPENED, THE VAPOR CONCENTRATION WILL PASS THROUGH THE EXPLOSIVE MIXTURE LEVEL DURING TANK VENTILATION. MIXTURES TOO LEAN TO BURN, IF ALLOWED TO ACCUMULATE IN AN UNVENTILATED SPACE, MAY FORM AN EXPLOSIVE MIXTURE WITHIN THE SPACE DUE TO STRATIFICATION.

- I. Ground air mover to the aircraft and to the same ground as the aircraft.
- J. Support air mover to prevent exhaust from contacting any object; a static electrical charge buildup can result.

4. Preventing Contamination of Tanks

WARNING: USE ONLY AN AIR-DRIVEN VACUUM.

- A. All equipment used in the fuel tanks shall be free of dirt and dust. Thoroughly clean air ducts used for ventilation internally and externally. Exercise every precaution to prevent contamination of fuel tanks at all times while tanks are open. Close all tanks when work is not in progress to prevent entry of dirt and dust, or other foreign matter. Vacuum clean fuel tank in which extensive work has been accomplished.
- B. When fuel lines are disconnected and/or components are removed, provide suitable protection to prevent foreign matter from entering lines or components.

5. Fuel Leak Classification

- A. Fuel leaks are classified as stains, seeps, heavy seeps, or running leaks. (See figure 201.)
- B. Turbine fuel dye, Type II yellow, MIL-D-812986 (Automate Yellow 662), manufactured by Morton Chemical Company of Chicago, Illinois, may be used as a fuel additive for detecting leaks. The dye should be added in the proportion of 1.6 or fewer ounces per 100 U.S. gallons. It is recommended that a portion of fuel be in the tank before addition of the dye. Agitation incurred during refueling is enough to give a uniform shade throughout the tank.
- C. Measure the size of the wetted area around the source of the leak on the aircraft skin to classify the leak.

Gates Learjet Corporation
maintenance manual

NOTE: For a more accurate measurement, wipe the aircraft skin dry and apply talcum powder to the leak area. At the end of 30 minutes inspect the leak area and classify the leak. (See figure 201.)

- D. Fuel leaks must be further classified as to whether they occur in an open area or an enclosed area to determine the need for immediate repair or leaks not considered potential flight hazards.
- E. A leak classified as a stain, seep, or heavy seep occurring in open areas must be repaired at such time the aircraft is grounded for other maintenance.
- F. A leak classified as a stain, seep, or heavy seep occurring in an enclosed area requires immediate repair.
- G. A running leak requires immediate repair regardless of the aircraft environment.

WARNING: LEAKS WHICH ARE PROGRESSING INTO CRITICAL AREAS OF THE AIRCRAFT (SUCH AS THE WHEEL WELL) OR LEAKS THAT WILL CAUSE FUEL TO BE BLOWN INTO CRITICAL FUSELAGE AREAS, MUST BE REPAIRED IMMEDIATELY REGARDLESS OF THE LEAK CLASSIFICATION.

6. **Fuel Hose Installation**

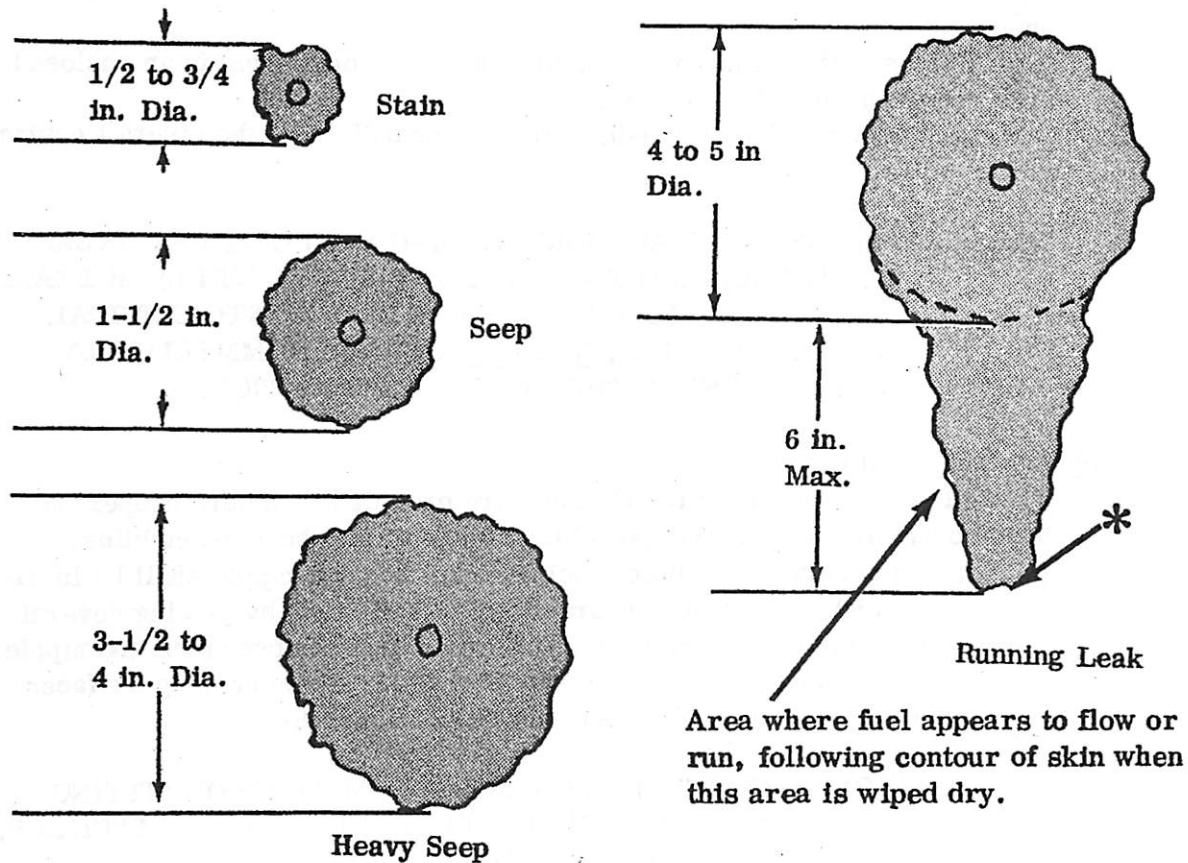
- A. The following installation instructions are provided to assure proper fuel hose connection and prevent possible damage to fuel hose assemblies.
 - (1) Bearing surfaces between fuel hose nut hex and nipple shall be lubricated with MIL-H-5606 hydraulic oil. Lubricate by placing several drops on nut retaining wire and several drops between nut and nipple.
 - (2) Hold fuel hose assembly so that oil will drain into bearing surfaces between nut and nipple. Spin nut to distribute oil.

CAUTION: DO NOT ALLOW THREAD COMPOUND ON FITTING ENDS, CONE SURFACES, IN THE BORE OF FITTINGS, OR INSIDE ANY SYSTEM COMPONENT.

- (3) Align hose assembly with fitting and engage threads 1/2 to 1 turn. Apply thread compound (Rector Seal, 100-00) sparingly to exposed male threads.
- (4) Torque fuel hose nuts to minimum torque value. (Refer to Chapter 20 for hose torque values.)
- (5) Fuel hoses shall be held firmly to prevent twisting while torquing fuel nuts.
- (6) Fuel hoses that are twisted shall be removed and the internal surfaces inspected for permanent deformation. Any hose that is permanently damaged or questionable shall be replaced.

Gates Learjet Corporation

maintenance manual



* Fuel will usually begin to drip after reaching this point.

Fuel Leak Classification
Figure 201

Gates Learjet Corporation

maintenance manual

STORAGE - DESCRIPTION AND OPERATION

1. General

- A. The fuel storage system consists of the individual fuel tanks and fuel vent system.

2. Description

- A. The tip tanks are attached to each wing at two attach points, one forward of spar 7 and one aft of spar 1. The metal-to-metal joints are faying surface and fillet sealed. The internal surfaces of the tanks are coated with epoxy for corrosion resistance. Baffles are incorporated to minimize slosh and prevent adverse fuel effects on the aircraft center of gravity. Five access covers are installed to provide entry for inspection and maintenance. Fuel is transferred to the wing tank by gravity flow and a jet pump. Two drain valves are installed in each tank to drain any accumulation of moisture and sediment.
- B. The full span, integral wing tanks, comprise a large portion of each wing. The tank inboard end is at aircraft centerline and the outboard end at wing station 181.10. The forward side of the tank ends at wing spar No. 1 and the aft side terminates at wing spar No. 7. Fuel is not stored in the main landing gear wheel wells and wing leading edge areas. The metal-to-metal joints are faying surface and fillet sealed and the entire inner surfaces are coated with epoxy for corrosion resistance. Small vent holes are incorporated in the upper and lower portion of each spar and rib. This allows fuel to move from one compartment to the other due to fuel expansion. It also allows the individual compartments to completely fill and drain. The wing dihedral of 2.5° causes the inboard portion of the wing to be the lowest point of the fuel system. This assures that the primary fuel pumps will be submerged in fuel until the tanks are practically empty. Wing rib 0.0 incorporates two pressure relief valves, allowing flow in opposite directions to equalize tank pressure.
- C. On 25B/D aircraft the fuselage tank consists of two bladder-type cells located in the cavity between frames 22 and 25. The cavity is a sealed compartment with hangers installed in the upper portion. The hangers mate with fasteners on the bladder to provide hanging support. Additional bladder support is gained by the interconnect fittings, access covers, and plumbing fittings incorporated in the bladder. Two drain ports are located in the aft portion of the cavity. Interconnecting tubes and a crossover tube connect the bladders to form a single tank. Access to the bladders is gained through two access covers in the tailcone; an access cover on the top of the fuselage and through the keel beam access covers.



GATES LEARJET CORPORATION

maintenance manual

- D. On 25C/F aircraft the fuselage tank consists of four bladder-type cells located in the cavity between frame 18 and 25. The cavity is a sealed compartment with hangers installed in the upper portion. The hangers mate with fasteners on the bladder to provide hanging support. Additional bladder support is gained by the interconnect fittings, access covers and plumbing fittings incorporated in the bladder. The cavity also incorporates four cavity drain ports. Two are located in the forward cavity area and two in the aft area. Interconnecting tubes and a crossover tube connect the bladders to form a single tank. Access to the bladders is gained through two large access covers in the tailcone; two access covers behind frame 18 pressure bulkhead; an access cover on the top of the fuselage and through the keel beam access covers.
- E. The fuel vent system provides continuous ram air pressure to the tip tanks, wing tanks and the fuselage tank while the aircraft is in flight. Ram air is admitted through the ram air scoops to the tip tanks and the fuselage tank. The ram air pressure is vented to the wing tanks through the wing-to-tip tank interconnects. Ram air is maintained at 1.0 psi by two pressure relief valves, one in each wing tank. A sump drain valve is located at approximately the aircraft centerline (lowest point in the vent system).

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maintenance manual

STORAGE - MAINTENANCE PRACTICES

1. General

- A. Maintenance practices consist of replacement of defective component, inspection and repairing of sealed areas if deterioration or leakage occurs.
- B. For maintenance practices on components refer to applicable section in this chapter.
- C. Fuel tank sealing must be applied so that absolutely no leakage exists. Since the fuel has low surface tension and viscosity, it can flow through extremely small openings. Sealing of the structure that has sealant applied to it must be kept to a minimum to avoid adding excessive weight to the airplane. These factors make sealing an important process. In order to maintain the sealing, it is necessary to understand thoroughly how sealant materials are used and the methods of applying sealant to repair leaking fuel tanks. Damage to integral fuel tank sealing, which has been evaluated as being in need of repair, must be repaired using approved repair procedures. All repairs must be accomplished by the application of approved sealing compounds with special tools.

2. Leak Path Analysis

- A. Leak path analysis is one of the most important steps performed in fuel system repair. Repairs without finding the entrance and exit point of a leak is useless, as this type of repair is only temporary.
- B. The seal plane is the barrier that prevents escape of the fuel from the tank. When this seal plane is penetrated the faying surfaces of all structure on the opposite side of the seal plane becomes wet. The wetted area extends in all directions from the point of penetration until it is stopped by an injection, faying surface or pre-pack type hidden seal. Fillet seals act as sides to a channel formed in the structure through which the fuel will flow. Since there are no seals on the opposite side of the seal plane, any fuel that has penetrated the seal plane will leak out where the least resistance exists. For instance a row of fasteners in the channel may be wet, but the leak will appear at only one fastener due to the looseness at that point. If the leaking fastener were sealed, then the fastener having the next least resistance will leak and so on.
- C. After the leak is localized, the next step is to study the structure and sealing of the suspected area. Construct cross sectional views to locate the seal plane. Study these views and trace all potential leak paths from the external leak point to the tank interior, assuming failure of sealant at various locations. This will define the area of the tank in which the leak source or sources will be found.

EFFECTIVITY: ALL

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- D. The area where the bubbles appear when locating the interior leak point as described in "Leak Test" should indicate the true point of seal plane penetration, except in the case of failure of a hidden seal. In many cases the failure of an injection, pre-pack or hidden type of seal will allow fuel to enter and travel within the structure to appear at a point some distance from the leak point. Repairing the apparent leak area, where the bubbles appear, and not the failed hidden seal may mean a temporary fix which will have to be redone frequently until the true point of penetration of the seal plane is located. To prevent this situation, a fuel leak should be thoroughly analyzed for all possible leak paths between external and internal appearance points.

3. Inspection of Tank Interior

- A. Thoroughly inspect the interior of the tank in the area determined ("Leak Path Analysis") to contain the leak source. The following instructions will help to reveal the leak source.
- (1) Inspect the area carefully for defective sealant, such as deep cracks, air bubbles, loose fasteners, or loose fillets. Use mirrors to inspect areas which are not otherwise completely visible.
 - (2) Inspect location where injection seals have been installed to provide continuity of fillet seals.
 - (3) Inspect for topcoat breaks, tears, scuffs or nicks in sealant. Fuel will penetrate any break in the film of topcoat and cause the sealant to leach. It will not cause a leak, however, until the base sealant has deteriorated to the point where an edge of the structure is bare. A chalky material under the cracked topcoat will indicate leaching of the sealant. All leached sealant shall be removed and replaced with new sealant.
 - (4) Inspect fillet seals for adhesion. Fillets suspected of poor adhesion shall be checked by applying air pressure at a maximum of 100 psi with an air gun, placed approximately 1/2 inch from the fillet. This check will loosen a fillet which has poor adhesion. Cut a section through the loose fillet and strip it off by pulling away from the structure. Continue to strip the fillet until it breaks apart within itself. A fillet with poor adhesion will pull away from tank structure rather than pull apart.
 - (5) Inspect tank structure for cracks, distortion, corrosion and loose fasteners.
 - (6) Visible defects in sealant or the structure are not necessarily the source of a true leak. Continue inspection until the entire suspected leak area has been carefully inspected. Mark all defective sealant.

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maintenance manual

- (7) Watch closely for the following:
- (a) Excessive shrinkage.
 - (b) Loss of luster or discoloration.
 - (c) Loss of elasticity.
 - (d) Crumbling sealant.
 - (e) Loss of cohesive strength.

4. Leak Test

- A. If the leak source has not been definitely isolated by a visual inspection it may be revealed by the application of a bubble solution to the inside of the tank in the suspected leakage area.
- (1) Apply air pressure to the external leak point. Air pressure should be 100 psi with an air hose held approximately 1/2 inch from the fillet. If leak is present small bubbles will appear.
 - (2) If bubbles are observed, mark the spot internally and externally, and continue the bubble test until entire area has been covered.
 - (3) Recheck marked area.
 - (4) After completion of test, wipe bubble solution from interior and exterior surfaces. Assure that defective areas are marked.

5. Fuel Tank Sealing Repairs

- A. After the leak path has been carefully analyzed to determine which seal, or seals, have failed and after the actual leak point is known, an effective method of repair can be decided upon. The following paragraphs describe the repair methods used in a few typical situations.
- (1) If leak path tracing procedures determine the actual leak point is a failed fillet or fastener seal, the correct procedure is to remove and replace the faulty seal. All faulty sealant material must be cut away, surfaces thoroughly cleaned, and new sealant applied to produce a seal of the same size and shape as the original seal.
 - (2) Raising the seal plane can be employed whenever leak path tracing procedures determine that the actual leak point is in a failed pre-pack or faying surface seal. The structure surrounding the failed seal must be carefully reviewed to determine the point to which the seal plane must be raised. The new seal plane must completely isolate the failed seal from fuel but the seal plane should not be raised any more than necessary to accomplish this. As an option, the failed preassembly seal may be replaced as directed in step 4. Raising the seal plane can involve an extensive amount of sealant addition so in many cases structure removal and replacement of the failed seal will be the best action.

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maintenance manual

- (3) When removing a failed injection seal, all the sealant material must be removed through the injection channel. A satisfactory seal cannot be formed if the channel is not clear since trapped air will prevent complete filling of the channel with sealant material. As an option, the failed injection seal may be repaired by raising the seal plane.
- (4) Replacing a failed pre-pack or faying surface seal involves disassembly of structure to gain access to the faulty seal.

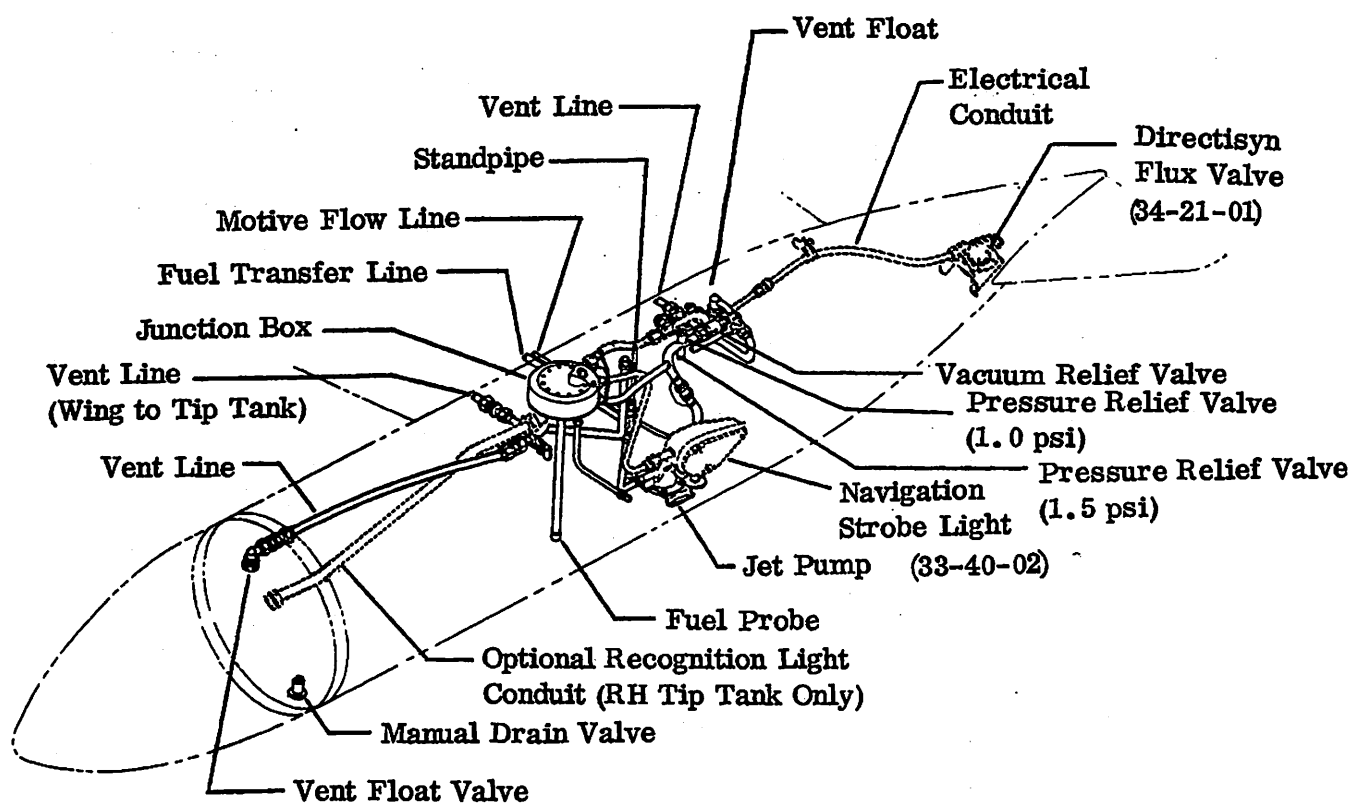
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TIP TANK - DESCRIPTION AND OPERATION

1. Description (See figure 1)

- A. The tip tanks are part of the fuel storage system. They are attached to each wing at two attach points, one forward of spar 7 and one aft of spar 1. Access covers on the top of the tank provide entry for inspection and maintenance. Baffles are installed to minimize slosh and to prevent adverse fuel effects on the aircraft center of gravity during extreme flight attitudes. A small vent hole, located forward and below the fuel filler cap, is connected by a tube to the tip tank junction box. This vent hole is used to vent and drain any moisture accumulation from the junction box. Assure that the hole is open at all times.
- B. Each tip tank is equipped with an optional fuel jettison system. For further information on the fuel jettison system, refer to 28-30-00.
- C. Each tip tank consists of a jet pump, a drain valve, two pressure relief valves, a vacuum relief valve, two vent float valves, a standpipe assembly, a fuel probe and a filler cap.
 - (1) The jet pump, installed in the bottom middle of the tip tank, transfers fuel into the inboard portion of the wing tank (approximately W.S. 23.00) during engine operation.
 - (2) The drain valve, located at the bottom forward end of the tank, is used to drain residual fuel or sediment from the tank.
 - (3) The pressure relief valves, located in the vacuum relief tube just forward of tank station 113, protect the tip tank against excessive pressure. One valve is set to operate at 1.0 psi. The other, installed as a backup to the 1.0 psi valve, is set to operate at 1.5 psi.
 - (4) The vacuum relief valve, secured to the bulkhead web at tank station 113, opens and admits air to the tip tanks and vent system when a vacuum condition exists.
 - (5) The vent float valves, located in the forward and aft ends of the tank, close when the fuel level reaches the vent ports. They thus prevent fuel from entering the vent system.
 - (6) The standpipe assembly, installed in the jet pump transfer line, prevents fuel from being siphoned down from the wing tank to the tip tank when the applicable engine is shut down.
 - (7) The fuel probe, mounted in the junction box, provides a capacitive signal to the fuel quantity indicating system. For more information on the fuel probe, refer to 28-40-00.
 - (8) The filler cap, located on the outer surface of the tip tank, is used to service the entire aircraft fuel system.

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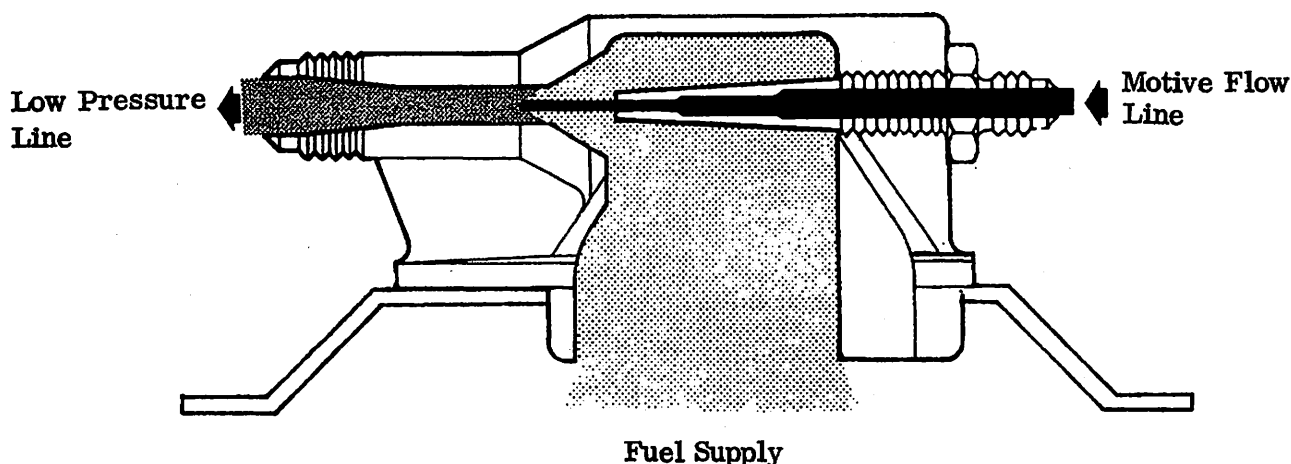
Tip Tank Components Locator
Figure 1

EFFECTIVITY: ALL

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2. Operation

- A.** All fueling operations for the aircraft are accomplished through the tip tanks. Fuel is transferred by the jet pump into the wing tank. Fuel can also gravity flow through flapper check valves into the wing tank. However, any fuel at a lower level than the check valves must be transferred using the jet pump. Further information on tip tank operation is furnished under component operation.
- B. Component Operation (See figure 2)**
- (1) The jet pumps have no moving parts and operate on the venturi principle. When the Starter-Generator switch is set to GEN position and the Jet Pump Switch is set to ON, the motive flow valve opens and high pressure fuel from the engine-driven fuel pump is directed through a nozzle in the jet pump drawing fuel from the tank. The motive flow pressure ranges from approximately 300 psi during high fuel flow rates at 45,000 feet altitude to approximately 250 psi at low fuel flow rates at sea level when the engine rpm is at idle. Jet pump discharge pressure increases as engine fuel flow increases; for example, at idle rpm, the fuel flow is 150 pounds per hour, the jet pump discharge pressure is approximately 10 psi with a pump inlet pressure (motive flow) of approximately 250 psi. At full power settings (Standard Day Conditions) the fuel flow is approximately 1700 pounds per hour, the jet pump discharge pressure is approximately 12 psi with a pump inlet pressure (motive flow) of approximately 300 psi. On Aircraft certified for 51,000 foot operation, there are no appreciable fuel flow pressure changes.



Fuel Jet Pump Schematic
Figure 2

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- (2) The drain valve is spring loaded to the closed position and sealed with an internal O-ring. To open the valve requires the use of a Phillips screwdriver. On Aircraft 25-061, 25-070 thru 25-318, the poppet assembly is removable. On Aircraft 25-319 and Subsequent, the poppet assembly is not removable. If a valve should develop a leak, the valve should be fully depressed to clear any sediment which could be preventing the valve from completely closing. If a valve has developed a leak not caused from sediment holding the valve open, the O-ring seal shall be replaced.

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TIP TANK - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: The following tip tank removal and installation procedure applies to removing the tip tank for repairs and reinstallation. This procedure does not include the installation of a new tip tank. New tip tank installation requires the locating and fitting of the tip tank to the wing which can be accomplished only at the Gates Learjet Factory or an authorized Service Center. Tip tanks that are removed for repairs should be pressure tested prior to installation. After the tip tank is installed, the tanks should be fully fueled and checked for leaks.

A. Remove Tip Tank (See figure 201)

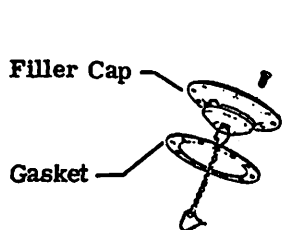
- (1) Defuel the aircraft. (Refer to Chapter 12.)
- (2) Support tip tank on a movable dolly.
- (3) Remove three lower outboard wing access panels and the access covers.
- (4) Disconnect motive flow and fuel discharge lines through the access openings and remove clamps from boot seals. Disconnect fuel vent lines from tip tank.
- (5) Remove flapper valves and clamps from tip tank interconnect boot seals. Loosen boot seals.
- (6) Remove screws from wing leading edge gap cover. Remove gap cover.
- (7) Remove screws from upper and lower access straps.
- (8) Remove access covers from tip tank and disconnect wiring from all electrical components. Pull wiring from tip tank through aft wing access opening.
- (9) Remove forward and aft attachment bolts and washers.
- (10) Remove tip tank from wing.
- (11) Remove retainer ring and boot seals from wing rib. Discard old boot seals.
- (12) Perform repairs to tip tank structure as required.
- (13) Thoroughly clean interior of tip tank.

B. Install Tip Tank (See figure 201)

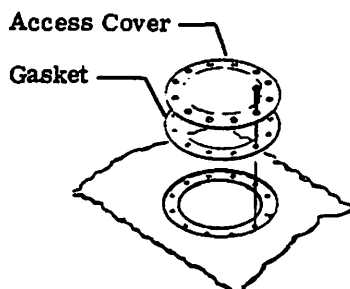
- (1) Install new boot seals with retainer rings. Torque bolts to 10 inch-pounds.
- (2) Apply bead of sealant around interconnect tubes.
- (3) Position tip tank on wing and install washer and forward attach bolt.
- (4) Check for gap between aft tip tank fitting and wing fitting. If gap exists, add AN960-716 or AN960-716L washer as required.
- (5) Install upper and lower access straps with screws.
- (6) Connect wiring to electrical components in accordance with applicable wiring diagram in the Wiring Manual.
- (7) Check that all components and plumbing are secured in the tip tank.

EFFECTIVITY: ALL

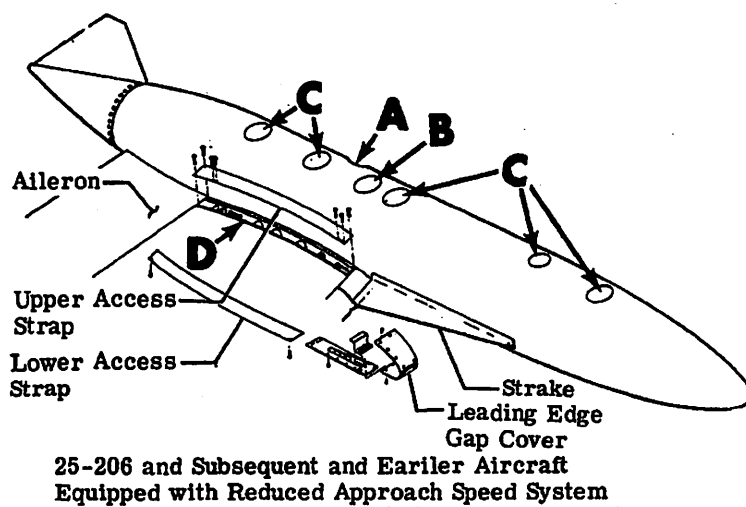
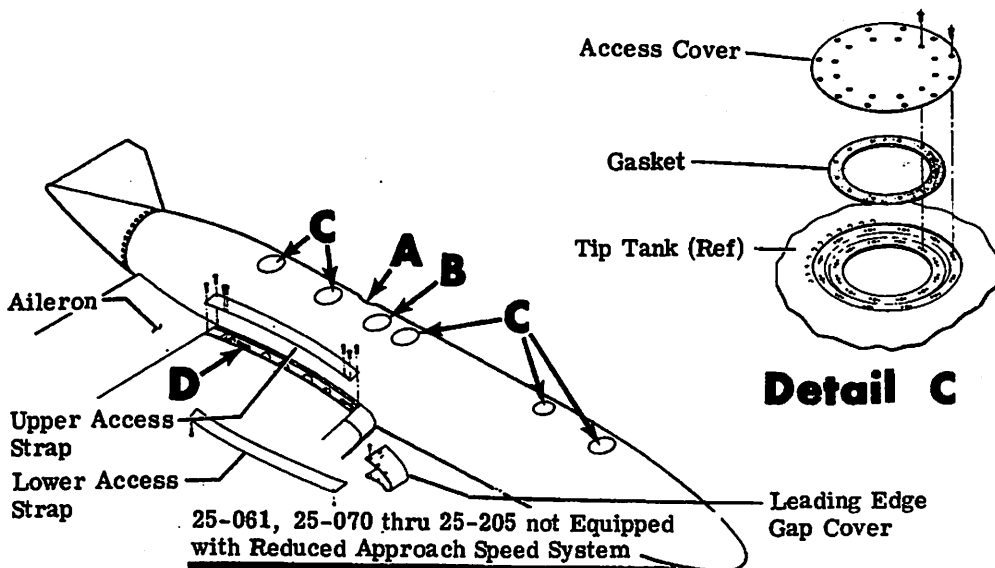
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Detail A



Detail B

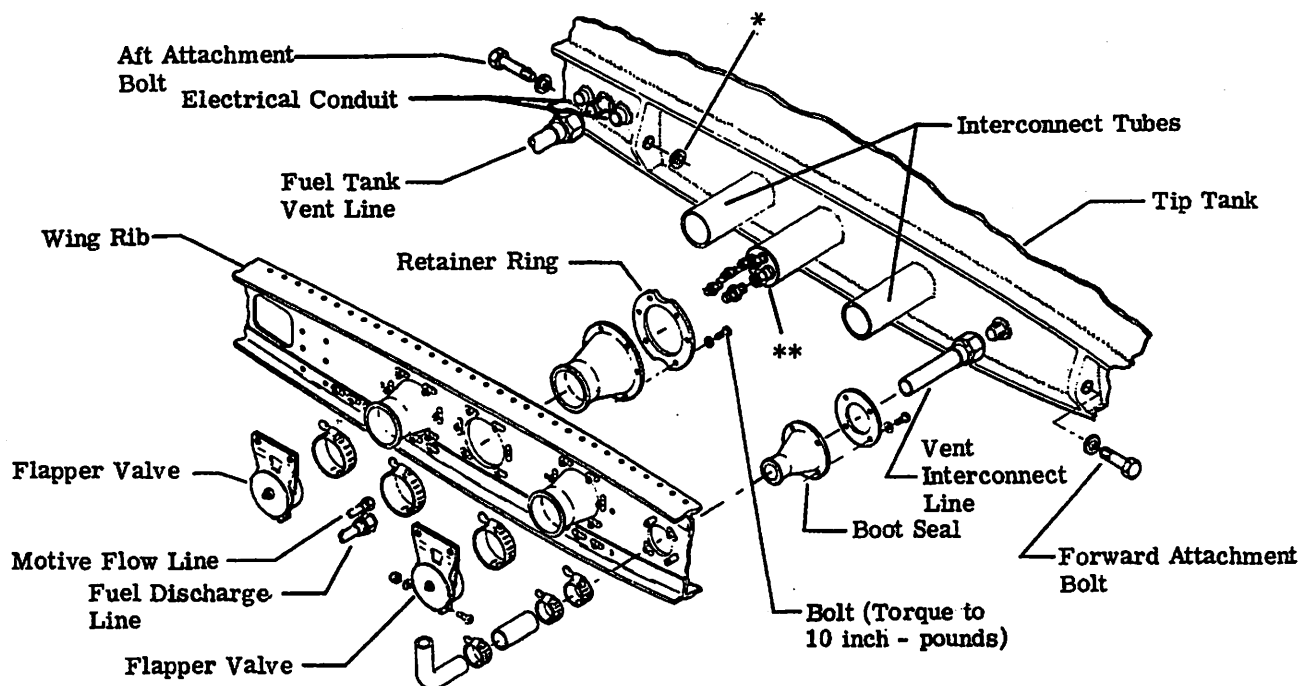


Tip Tank Installation
Figure 201 (Sheet 1 of 2)

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*Before installing aft attachment bolt, check for gap between aft tip tank fitting and wing fitting. If gap exists, add AN960-716 or AN960-716L washer between tip tank fitting and wing fitting.

**On Aircraft 25-346 and Subsequent, apply a fillet of Coast ProSeal 890 around plate, tubes, and bolt heads after ejector jet pump fuel line connections are complete.



INSTALLATION IS TYPICAL FOR BOTH TIP TANKS.

Detail D

Tip Tank Installation
Figure 201 (Sheet 2 of 2)

EFFECTIVITY: ALL

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28-11-00
 Page 203
 Dec 18/81

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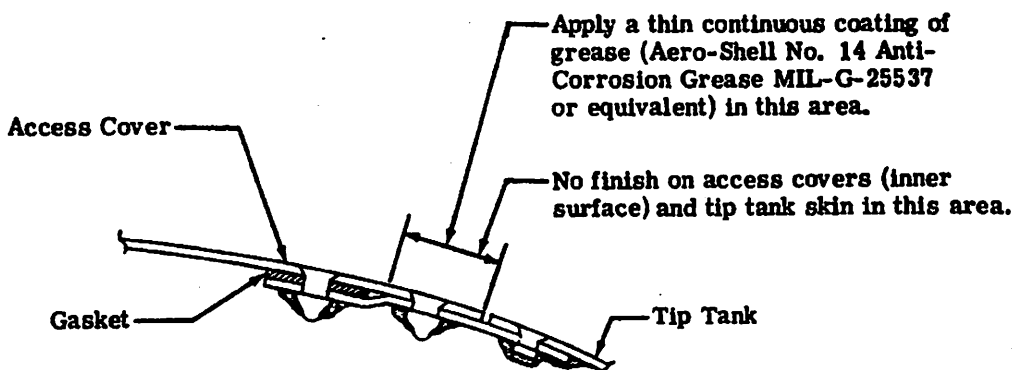
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- (8) Apply a thin continuous coat of anti-corrosion grease to access covers. (See figure 202.)
- (9) Install gaskets and access covers and secure with attaching parts.
- (10) Perform operational check of electrical components.
- (11) Install clamps over boot seals and torque to 25 inch-pounds.
- (12) Connect fuel tank vent, motive flow and fuel discharge line and tip tank vent interconnect line.
- (13) Connect flapper valves to interconnect tubes.
- (14) Install leading edge and trailing edge gap covers.
- (15) Install wing access panels.
- (16) Service aircraft fuel tanks and check for leaks. (Refer to Chapter 12.)
- (17) Fill gaps between leading edge, tip tank attaching straps, and wing with body putty.
- (18) Paint all affected areas. (Refer to Chapter 20.)

3. Inspection/Check

A. Tip Tank Pressure Check

- (1) Perform an air pressure leakage check by connecting a hose to an air source and static hose connected to a water manometer. Plug remaining openings with plugs having red streamers attached.
- (2) Apply air pressure slowly to tip tank not to exceed 4 psi and check for leaks with a soap solution at all fittings.
- (3) At the completion of the air pressure check; depressurize tip tank, disconnect hose connections and remove all plugs previously installed.



Tip Tank Access Cover Anti-Corrosion Protection
Figure 202

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JET PUMP - MAINTENANCE PRACTICES

1. Removal/Installation

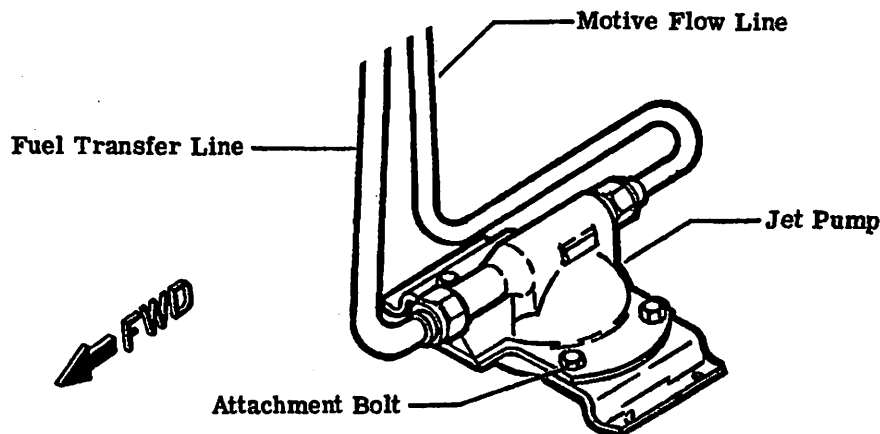
A. Remove Jet Pump (See figure 201)

CAUTION: MAINTENANCE PERSONNEL SHOULD ADHERE TO THE SAFETY PRACTICES AS OUTLINED IN 28-00-00, "GENERAL - MAINTENANCE PRACTICES".

- (1) Drain the tip tank.
- (2) Remove tip tank access covers.
- (3) Disconnect motive flow and fuel transfer lines from jet pump.
- (4) Remove attaching parts and jet pump from tip tank.

B. Install Jet Pump (See figure 201)

- (1) Install jet pump and secure with attaching part.
- (2) Connect motive flow and fuel transfer lines to jet pump.
- (3) Apply a thin continuous coat of anti-corrosion grease to access covers. (Refer to 28-11-00.)
- (4) Install gaskets and access covers and secure with attaching parts.



Jet Pump Installation (Typical)
Figure 201

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DRAIN VALVE - MAINTENANCE PRACTICES

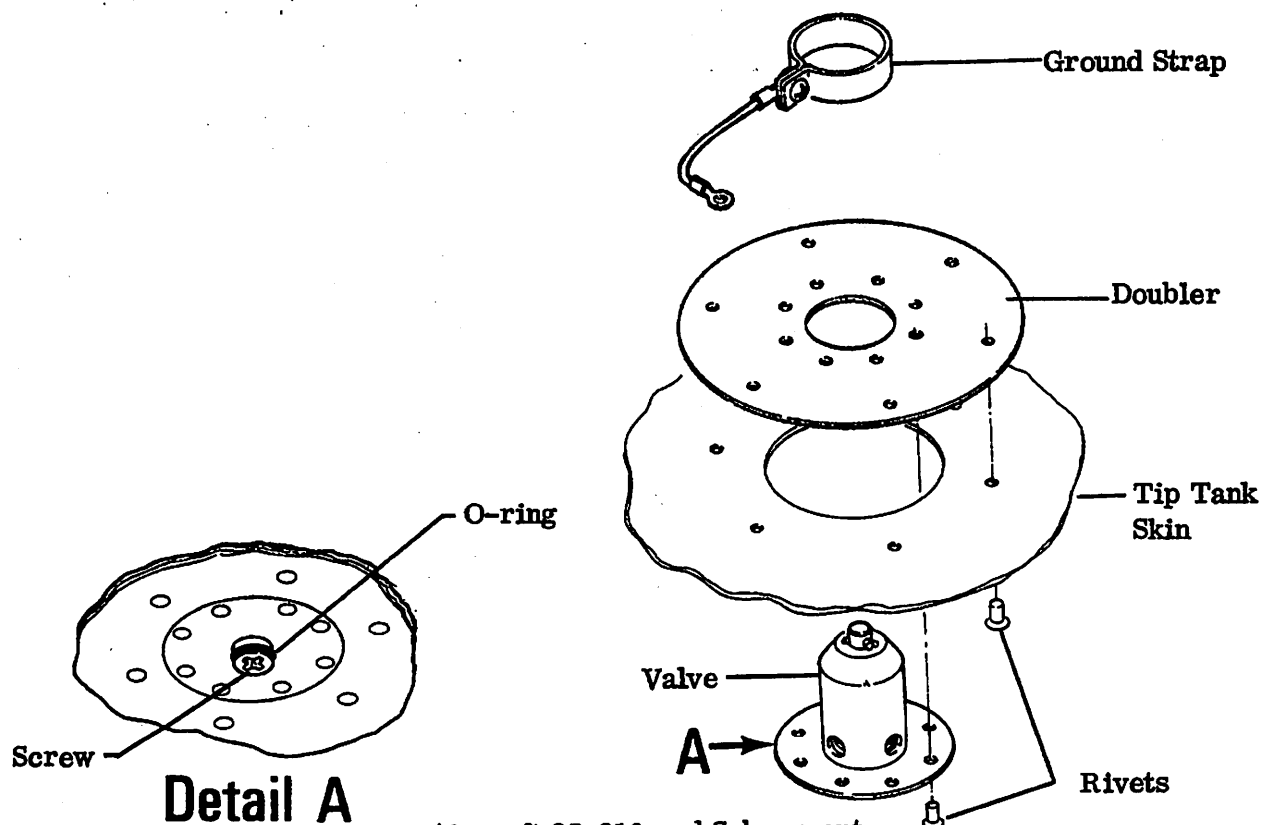
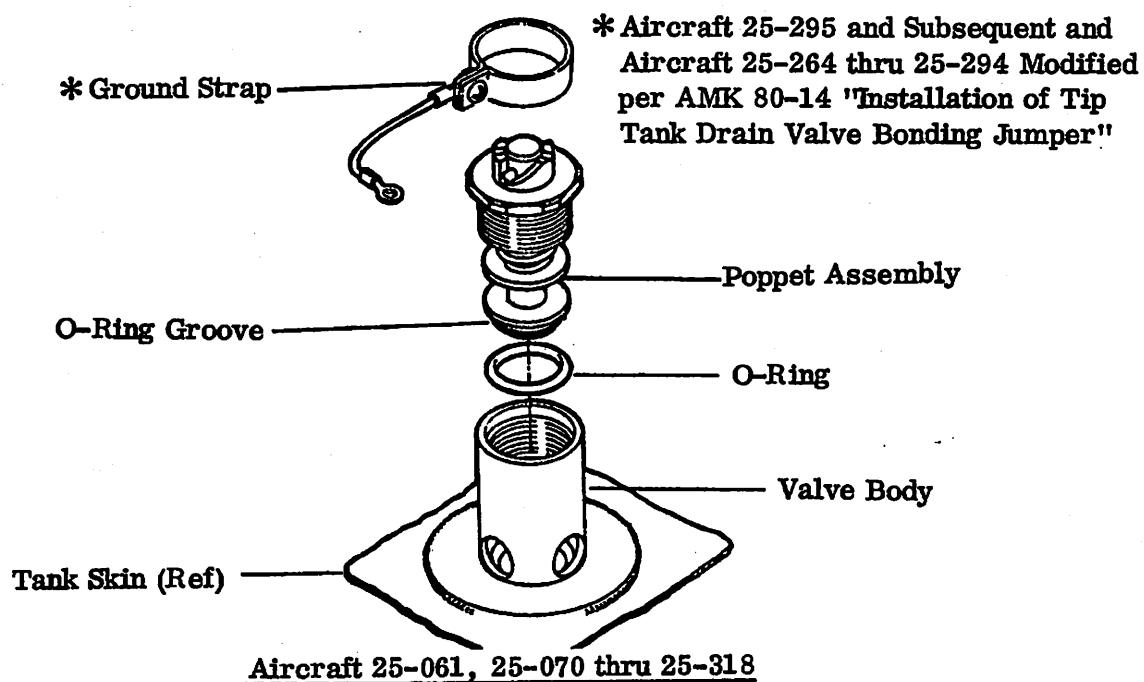
1. Approved Repairs

WARNING: OBSERVE ALL SAFETY PRACTICES DESCRIBED IN 28-00-00. SPECIAL CARE SHOULD BE TAKEN FOR PROPERLY VENTILATING THE FUEL CELL PRIOR TO PERFORMING MAINTENANCE PRACTICES.

- A. Drain Valve O-Ring Replacement (On Aircraft 25-061, 25-070 thru 25-318)
(See figure 201)
- (1) Defuel aircraft as necessary. (Refer to 12-00-00.)
 - (2) Remove tip tank access cover nearest the drain valve to be repaired.
 - (3) Reach into tip tank through access opening and unscrew poppet assembly from valve body.
 - (4) Using a suitable tool, remove old O-ring.
 - (5) Inspect O-ring groove for scratches, burrs or dirt.
 - (6) Install a new O-ring.
 - (7) Inspect seat in valve body for scratches, burrs, or dirt.
 - (8) Install poppet assembly in valve body.
 - (9) Apply a thin continuous coat of anti-corrosion grease to access cover. (Refer to 28-11-00, figure 202.)
 - (10) Install gasket, access cover and secure with attaching parts.
 - (11) Refuel aircraft and check for leaks.
- B. Drain Valve O-ring Replacement (Aircraft 25-319 and Subsequent) (See figure 201)
- (1) Place a suitable container for catching leaks directly under the drain valve.
 - (2) Using a Phillips screwdriver, turn valve screw located on the outer surface of drain valve slightly to the right. This will allow the screw and O-ring to pop down into view.
 - (3) Using a suitable tool, remove old O-ring.
 - (4) Inspect O-ring groove for scratches, burrs or dirt.
 - (5) Install new O-ring.
 - (6) Return valve to normal by pushing upward on screw with a Phillips screwdriver and turning slightly to the left.
 - (7) Check for leaks.

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Aircraft 25-319 and Subsequent
Drain Valve O-Ring Replacement
Figure 201

EFFECTIVITY: ALL

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PRESSURE RELIEF VALVES - MAINTENANCE PRACTICES

1. Removal/Installation

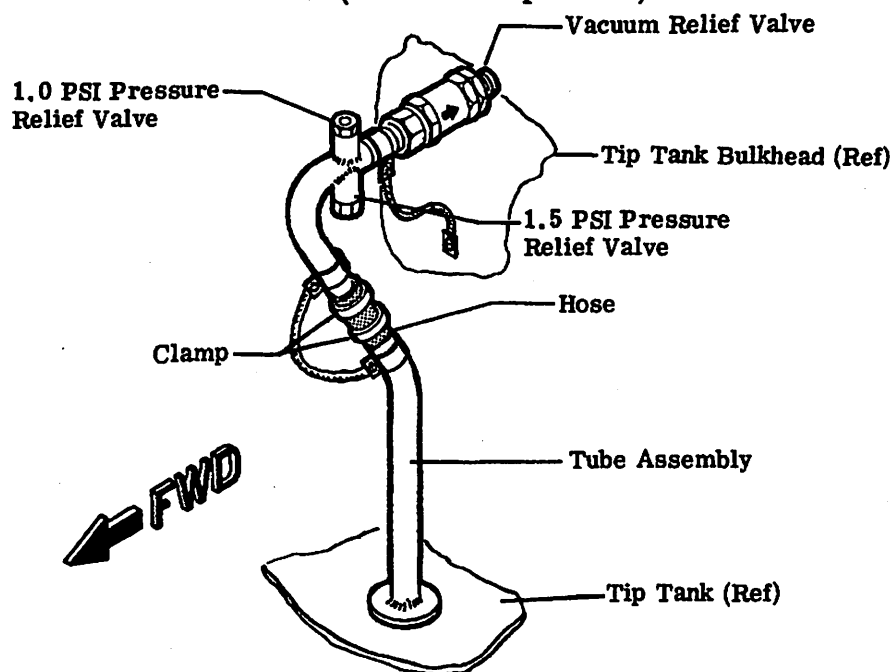
NOTE: The following removal and installation procedures are applicable to both tip tanks and either relief valve.

A. Remove Pressure Relief Valve (See figure 201)

- (1) Drain the tip tank.
- (2) Remove attaching parts and access covers from tip tank.
- (3) Remove attaching parts and bonding jumper from vent tubes.
- (4) Loosen clamps securing hose to vent tubes.
- (5) Disconnect vent tube from vacuum relief valve and remove vent tube with pressure relief valves installed from aircraft.
- (6) Remove pressure relief valve and O-ring from vent tube.
- (7) Inspect O-ring for serviceability, replace if required.

B. Install Pressure Relief Valve (See figure 201)

- (1) Install O-ring and pressure relief valve in vent tube.
- (2) Install vent tube and connect to vacuum relief valve.
- (3) Install hose on vent tubes and secure with clamps.
- (4) Install bonding jumper and secure with attaching parts.
- (5) Apply a thin coat of anti-corrosion grease to access cover. (Refer to 28-11-00.)
- (6) Install gaskets, access covers, and secure with attaching parts.
- (7) Refuel the aircraft. (Refer to Chapter 12.)



**Tip Tank Pressure Relief Valve Installation
Figure 201**

EFFECTIVITY: ALL

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VACUUM RELIEF VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

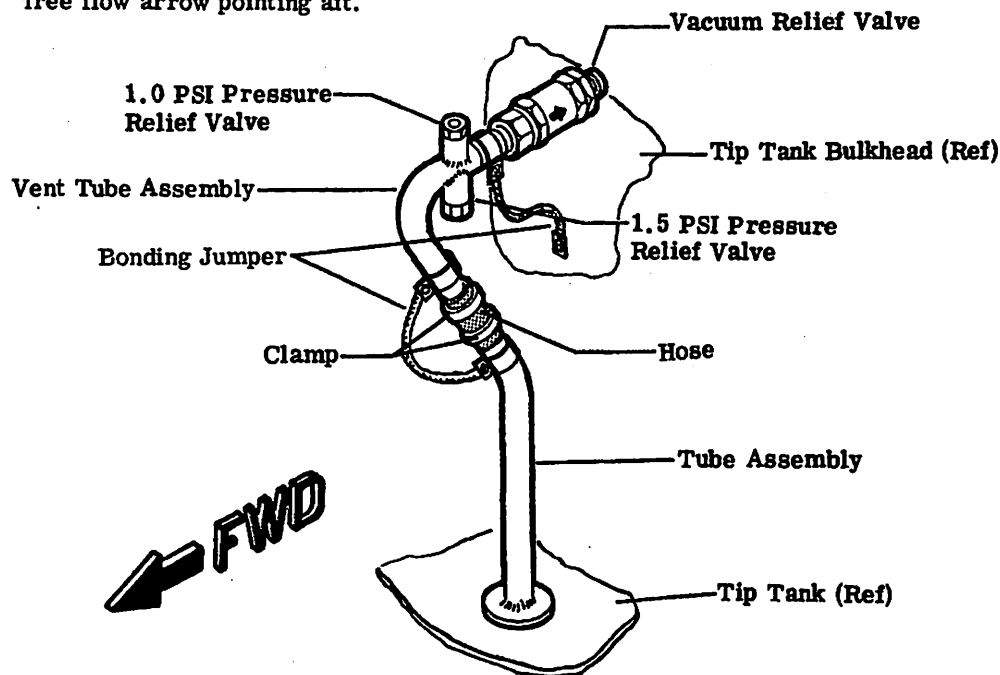
A. Remove Vacuum Relief Valve (See figure 201)

- (1) Drain the tip tank.
- (2) Remove attaching parts and access covers from tip tank.
- (3) Remove attaching parts and bonding jumpers from vent tubes.
- (4) Loosen clamps securing hose to vent tubes.
- (5) Disconnect vent tube from relief valve.
- (6) Remove attaching parts and relief valve from tip tank.

B. Install Vacuum Relief Valve (See figure 201)

- (1) Install relief valve (arrow towards bulkhead web) and secure with attaching parts.
- (2) Connect vent tube to relief valve.
- (3) Install bonding jumpers and secure with attaching parts.
- (4) Tighten clamps securing hose to vent tubes.
- (5) Apply a thin coat of anti-corrosion grease to access covers. (Refer to 28-11-00.)
- (6) Install gaskets, access covers, and secure with attaching parts.
- (7) Refuel the aircraft. (Refer to Chapter 12.)

NOTE: Install vacuum relief valve with free flow arrow pointing aft.



Tip Tank Vacuum Relief Valve Installation
Figure 201

EFFECTIVITY: ALL



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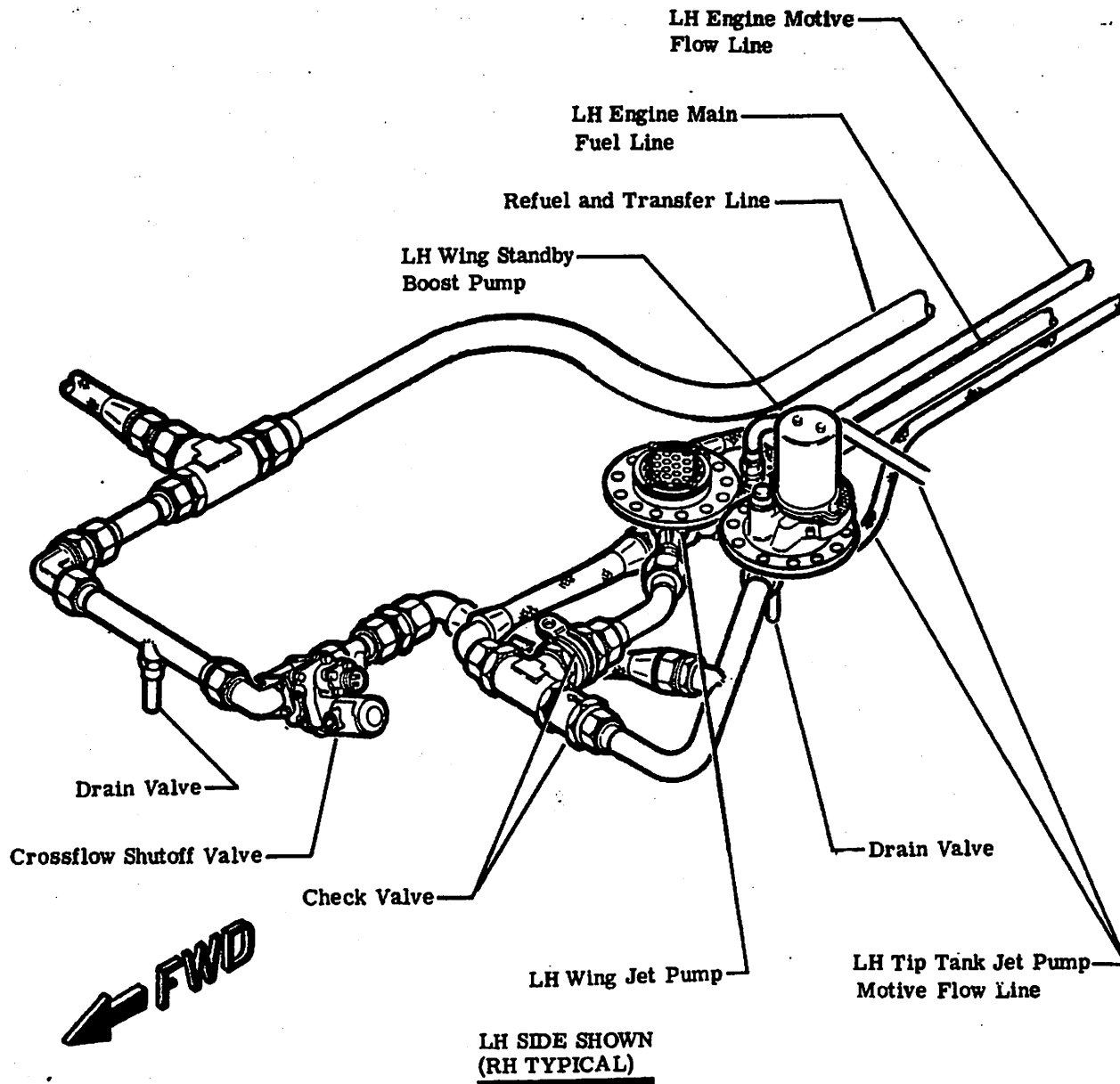
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WING TANK - DESCRIPTION AND OPERATION

1. Description

- A. The full span, integral wing tanks comprise a large portion of each wing. The inboard end of the tank is at the aircraft centerline and the outboard end is at wing station 181.10. The forward side of the tank ends at wing spar No. 1 and the aft side terminates at wing spar No. 7. Fuel is not stored in the main landing gear wheel well and wing leading edge areas. The tanks are sealed by applying a faying surface seal between the wing skins and all adjacent structure. A bead of sealant is then applied around the entire perimeter of the skins as added insurance against fuel tank leakage.
- B. Each wing tank incorporates one fuel boost pump (standby), one jet pump (main), one float switch, a shutoff valve (crossflow) and two fuel probes. For further information on the jet pumps and boost pumps, refer to 28-20-01 and 28-20-02. The applicable float switch energizes the LOW FUEL indicator light when wing tank fuel reaches approximately 400 to 500 pounds. The fuel probes are mounted on removable access panels. In addition, a sump drain valve and a de-fuel valve are installed on each inboard fuel probe access door. Flapper-type check valves, located in the wing ribs, allow free fuel flow inboard but restrict outboard fuel flow. Wing rib 0.0 incorporates two pressure relief valves set at approximately 1 psi (allowing flow in opposite direction) to prevent wing tank overpressurization during fuel crossflow operation. A wing pressure switch installed in the RH wheel well, senses wing tank pressure during fuselage fuel transfer and deenergizes the fuselage pump if an overpressurization condition should occur. The wings are vented to the tip tanks which are vented overboard through the ram air fuel vent system. A vacuum relief valve is installed in each tip tank and prevents negative pressure from damaging the tank. Compartment-to-compartment vents are also installed in the wing spars. Fuel flow between wing tanks is controlled by a shutoff valve installed in the crossflow tube.
- C. Fuel crossflow from one wing to the other is accomplished by opening the crossflow shutoff valve and energizing the applicable standby pump and deenergizing the opposite pump. For further information on the transfer operation, refer to 28-20-00.

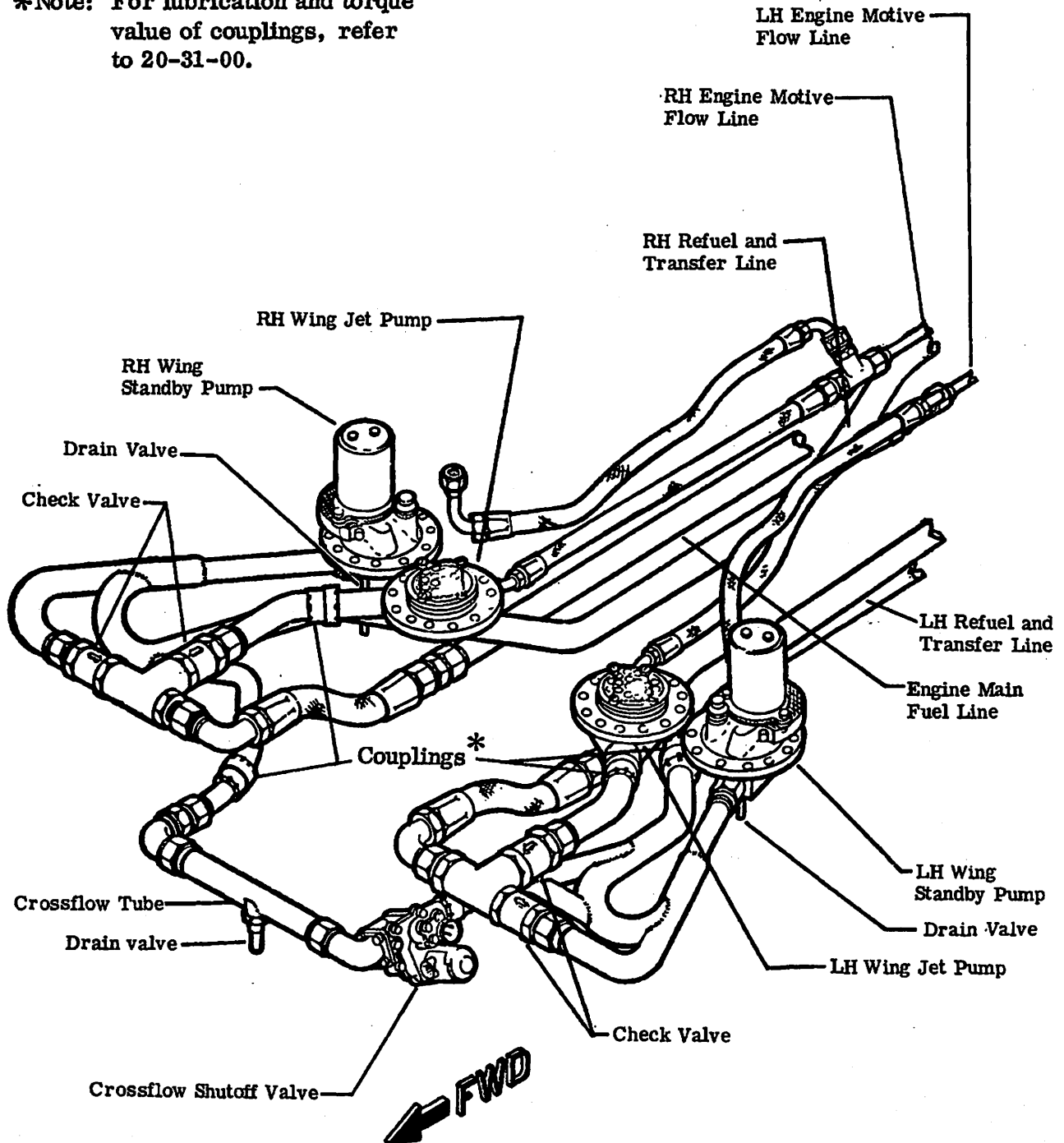
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maintenance manual



Wing Fuel Plumbing Installation
Figure 1

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***Note:** For lubrication and torque value of couplings, refer to 20-31-00.



**Wing Fuel Plumbing Installation
Figure 2**

EFFECTIVITY: 25C/F AIRCRAFT

MM-98

28-12-00

Page 3

Dec 18/81

GATES LEARJET CORPORATION
maintenance manual

WING TANK - MAINTENANCE PRACTICES

1. **General**

A. Maintenance practices on the wing tanks consist of inspection of integral tank sealing and replacement of defective component.

2. **Lightning Strike Protection Seal Application** (See figure 201)

A. When removal of the wing fuel access covers is required, special installation procedures are required to provide lightning protection.

- (1) Inspect and thoroughly clean the wing skin and access cover. Remove all grease and old sealant.

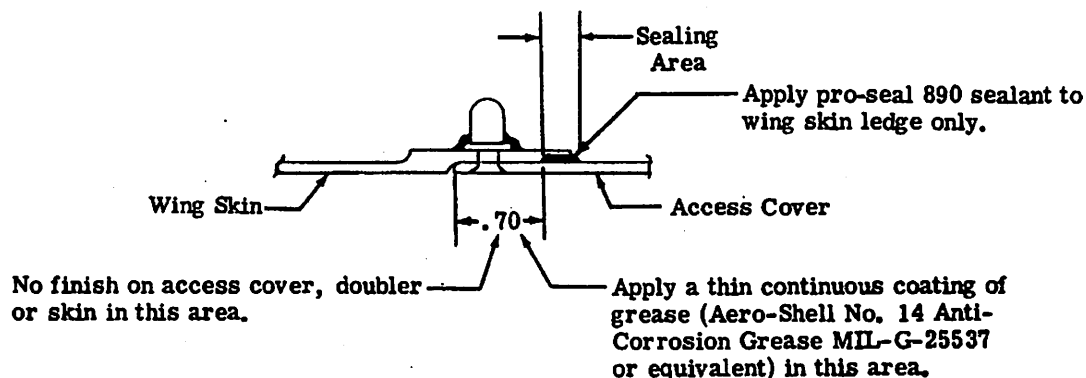
CAUTION: USE SPECIAL CARE WHEN APPLYING SEALANT TO PREVENT SEALANT FROM EXTRUDING INTO THE SURFACE BETWEEN THE ACCESS COVER AND WING SKIN.

- (2) Apply an even coat of pro-seal 890 sealant to the wing skin ledge.

CAUTION: EXCESSIVE GREASE WILL CAUSE THE SEALANT TO BE INEFFECTIVE AND REQUIRE COMPLETE RECLEANING AND RESEALING OF THE WING SKIN AND COVER.

- (3) Apply a thin continuous coating of grease (Aeroshell No. 14 Anti-Corrosion Grease or equivalent) around the outer edge of the wing access cover.

- (4) Install access cover.



**Lightning Strike Protection Seal Application
Figure 201**

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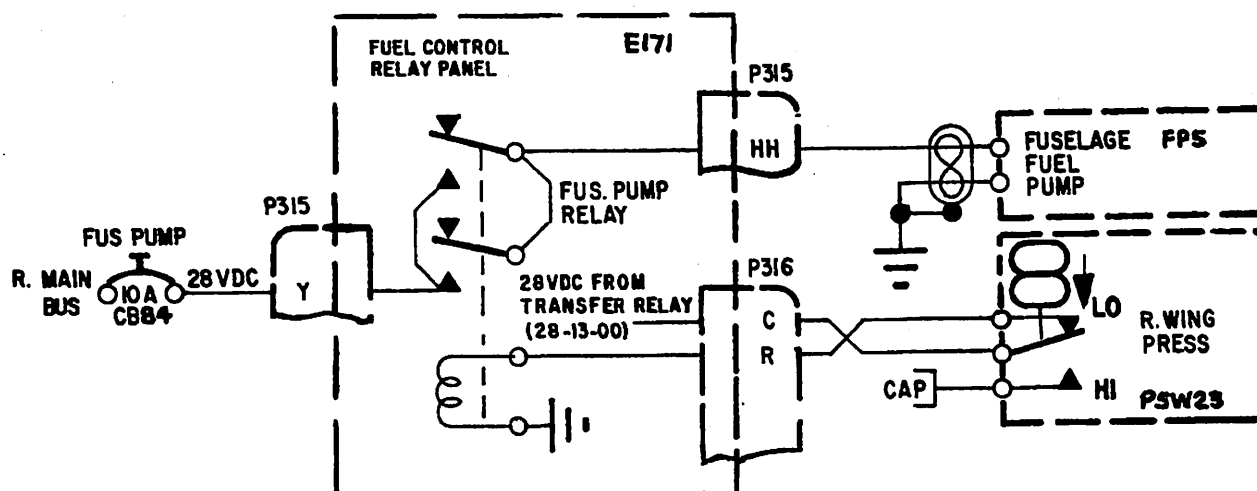
WING FUEL PRESSURE SWITCH - DESCRIPTION AND OPERATION

1. General

A. The wing fuel pressure switch is installed to prevent overpressurization of the wing during transfer of fuselage fuel. The switch is located on the keel beam in the RH wheel well and is plumbed into the inboard access cover on wing spar 5 in the RH wheel well.

2. Operation (See figure 1)

A. The switch controls 28 vdc to the fuselage fuel pump via the fuel control relay panel. The switch senses pressure from the right wing during fuselage fuel transfer and deenergizes the fuselage fuel pump when wing fuel pressure reaches 5 psi. The switch will again energize the fuselage pump when wing fuel pressure drops below 2.5 psi.



Wing Fuel Pressure Switch Electrical Control Schematic
Figure 1

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WING FUEL PRESSURE SWITCH - MAINTENANCE PRACTICES

1. General

A. Maintenance practices on the pressure switch consist of an operational check of the pressure switch and replacement of defective component.

2. Removal/Installation

A. Remove Pressure Switch (See figure 201)

- (1) Defuel the aircraft. (Refer to 12-10-10)
- (2) Lower RH inboard gear door. (Refer to 32-12-00)
- (3) Disconnect pressure switch wiring at splices. Tag wiring.
- (4) Disconnect tube from pressure switch.
- (5) Remove attaching parts, clamps and pressure switch from aircraft.

B. Install Pressure Switch (See figure 201)

- (1) Install pressure switch and secure with clamps and attaching parts.
- (2) Connect pressure switch wiring at splices.
- (3) Connect tube to pressure switch.
- (4) Check pressure switch for proper operation.
- (5) Fuel the aircraft. (Refer to 12-10-10)

3. Inspection/Check

A. Check Pressure Switch (See figure 201)

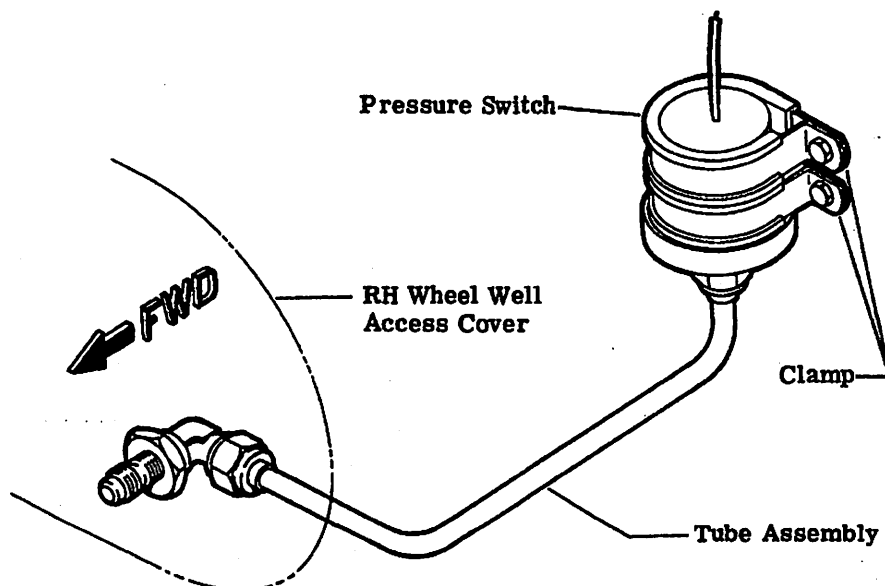
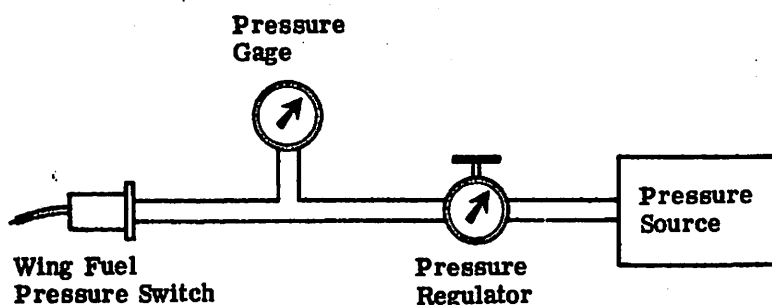
- (1) Test equipment required:
 - (a) Pressure source; minimum 10 psi output.
 - (b) Pressure regulator; 0 to 10 psi.
 - (c) Pressure gage; 0 to 10 psi.
- (2) Disconnect and cap the plumbing which leads from the wing to the wing fuel pressure switch. Use a container to catch fuel which runs out during capping operation. The switch plumbing is located on the inboard access cover on wing spar No. 5 in the RH wheel well.
- (3) Connect pressure gage, pressure regulator and pressure source to the wing fuel pressure switch as shown in the test schematic.
- (4) Set the Fuselage Tank switch to XFER.

CAUTION:

IF THE FUSELAGE PUMP FAILS TO DEENERGIZE AT A PRESSURE OF 5.5 PSI IMMEDIATELY SET THE FUSELAGE TANK SWITCH TO FILL OR WING OVERPRESSURIZATION AND FUEL OVERFLOW MAY OCCUR.

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- (5) Apply pressure to the switch and monitor gage. Fuselage pump should deenergize at 5.0 (± 0.5) psi. If it does not, replace switch and repeat test.
- (6) Bleed pressure off while monitoring gage. Fuselage pump should reenergize at 2.5 (± 0.5) psi. If it does not, replace switch and repeat test.
- (7) Remove pressure source, pressure regulator and pressure gage.
- (8) Uncap and install plumbing.



Wing Fuel Pressure Switch Installation
Figure 201



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DRAIN VALVES - MAINTENANCE PRACTICES

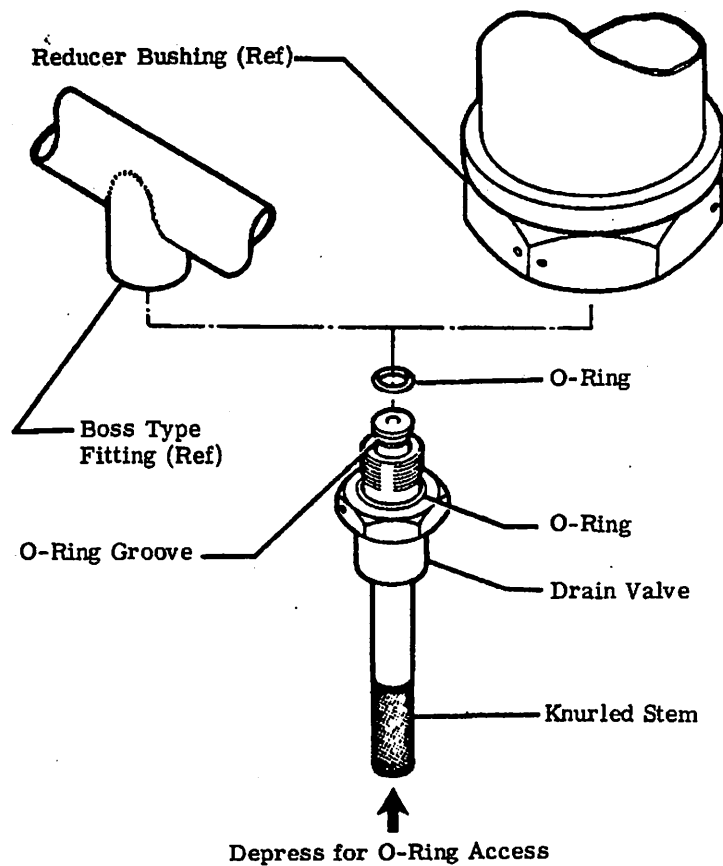
1. General

- A.** A drain valve is incorporated in each inboard fuel probe access panel and in the crossover tube. The valves are used to drain moisture accumulation and sediment from the fuel.
- B.** The valves are spring loaded to the closed position and sealed with an internal O-ring. The valves have a knurled stem which can be depressed with the fingers. The valves may be unscrewed from the boss or reducer bushing in which it is installed. If a drain valve should develop a leak, the valve should be fully depressed to clear any sediment which could be preventing the valve from completely closing. If a valve has developed a leak not caused from sediment holding the valve open, then the O-ring seal should be replaced.

2. Approved Repairs

- A. Drain Valve O-Ring Replacement (See figure 201)**
 - (1) Place a suitable container for catching fuel beneath drain valve to be repaired.
 - (2) Gain access to and loosen drain valve.
 - (3) Quickly unscrew the drain valve and replace with a suitable plug with 7/16-20NF threads.
 - (4) Depress the knurled stem on the drain valve to expose the O-ring.
 - (5) Using a suitable tool, remove the old O-ring.
 - (6) Inspect the O-ring groove and seat for scratches, burrs or dirt.
 - (7) Again with the knurled stem depressed, install a new O-ring.
 - (8) Install a new O-ring at the base of the drain valve threads.
 - (9) Quickly unscrew the plug in the drain valve fitting and install the drain valve.
 - (10) Assure that there are no leaks.
 - (11) Secure any accesses which were removed.

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Drain Valve O-Ring Replacement
Figure 201

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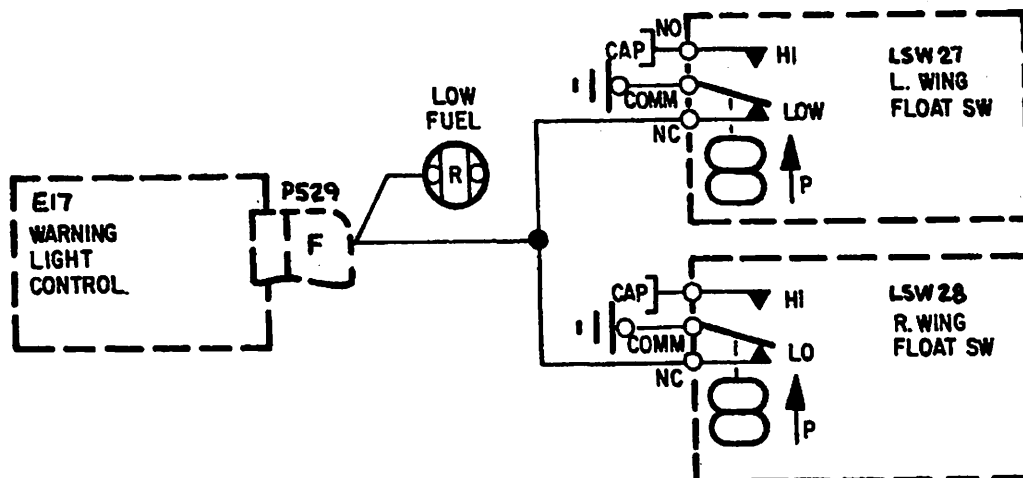
WING FLOAT SWITCH - DESCRIPTION AND OPERATION

1. General

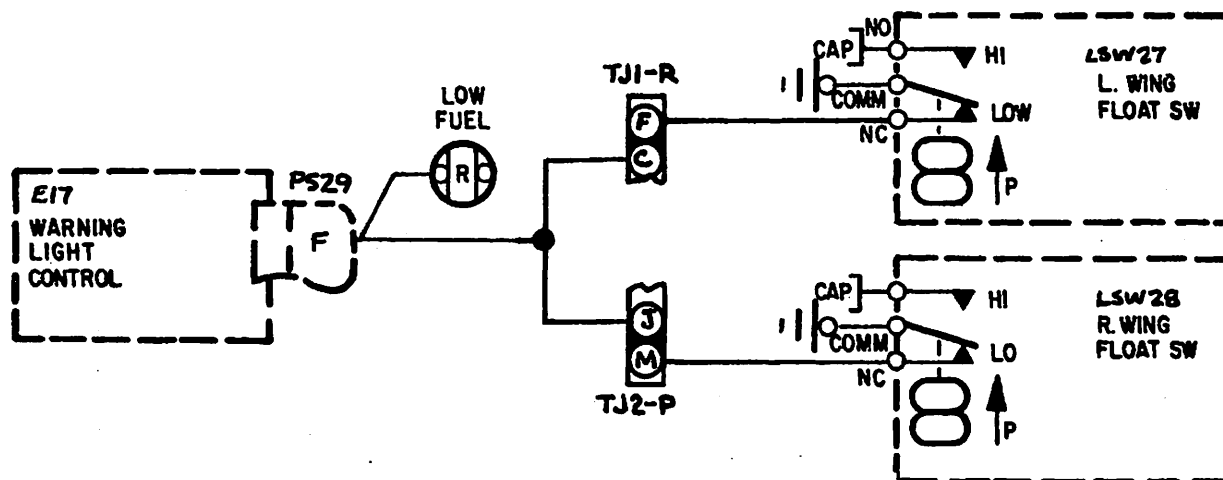
A. The wing float switch is located in the LH and RH wheel well areas on an access panel secured to spar 5.

2. Operation (See figure 1)

A. The float completes a ground circuit to light the LOW FUEL warning light when wing fuel reaches approximately 400 to 500 pounds.



Aircraft 25-061, -070 thru -224



Aircraft 25-225 and Subsequent

Wing Float Switch Electrical Control Schematic
Figure 1

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WING FLOAT SWITCH - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: The following procedures are applicable to either float switch.

A. Remove Float Switch (See figure 201)

- (1) Defuel the aircraft. (Refer to Chapter 12.)
- (2) Depressurize the hydraulic system. (Refer to Chapter 29.)
- (3) Disconnect anti-skid hydraulic plumbing as required to allow removal of the access panel.
- (4) Disconnect float switch electrical wiring at splice.
- (5) Remove attaching parts and access panel from spar 5.
- (6) Remove sealant from wiring and remove nut, washers, and float switch from access panel.

B. Install Float Switch (See figure 201)

- (1) Apply faying surface seal to access panel (fuel side) where float switch seats.
- (2) Install float switch and secure with attaching parts. Assure that float is positioned on a vertical axis.
- (3) Apply sealant (Pro-Seal 890, Class B-2) to switch wiring as shown.
- (4) Apply a fillet seal to switch and access panel.
- (5) Install access panel on spar 5 and secure with attaching parts.
- (6) Connect anti-skid plumbing.
- (7) Connect electrical wiring at splice.
- (8) Bleed the anti-skid brake system. (Refer to Chapter 32.)

2. Inspection/Check

- #### A. Perform operational check in accordance with the current inspection intervals specified in Chapter 5.

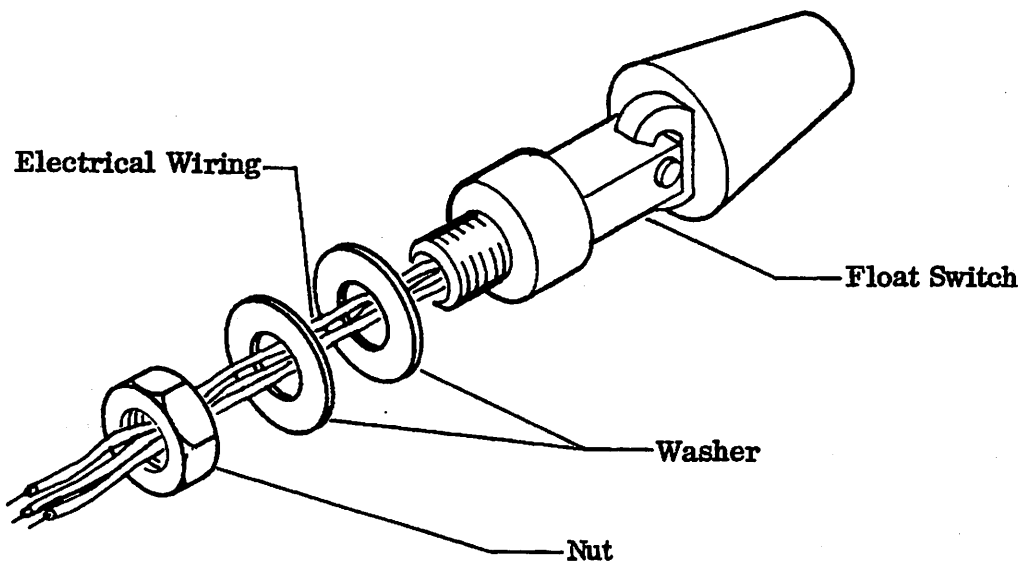
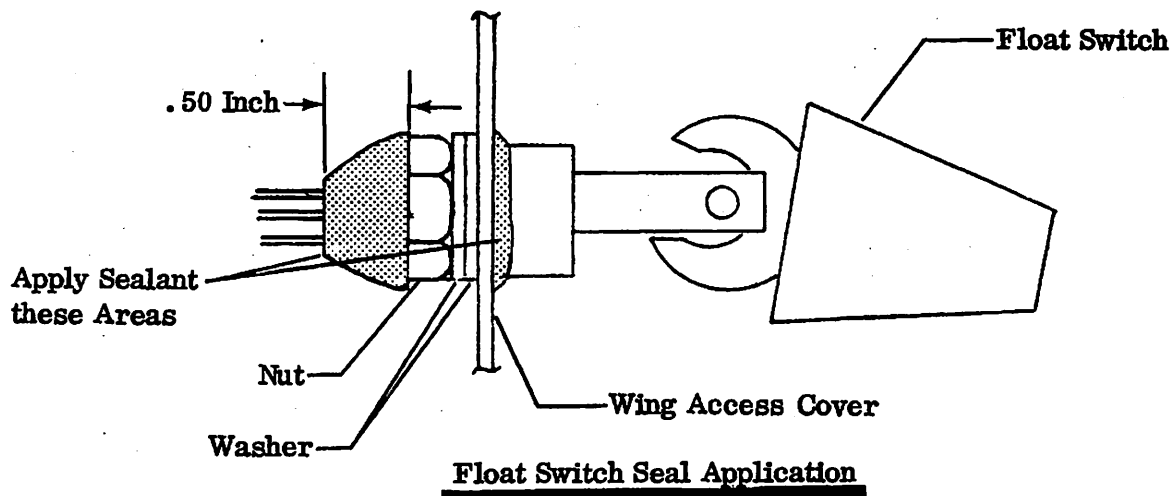
NOTE: • The following operational check must be performed with 600 to 650 pounds of fuel in each wing tank. Larger quantities of fuel will cause the standby pumps to work against the wing flapper valves.

- This operational check is performed by transferring fuel from one wing tank to the other utilizing the crossflow valve and the standby fuel pumps.

- (1) Set Battery Switch(es) to BAT 1 and BAT 2 or to on.
- (2) Set Crossflow Valve Switch to OPEN.
- (3) Set Left Standby Pump Switch to L.
- (4) Monitor LOW FUEL warning light on glareshield.

EFFECTIVITY: ALL

GATES LEARJET CORPORATION
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Wing Float Switch Installation
Figure 201

EFFECTIVITY: ALL

MM-98

28-12-03
Page 202
Dec 2/77

Gates Learjet Corporation
maintenance manual

- (5) When LOW FUEL warning light illuminates set Left Standby Pump Switch off.
- (6) Set Right Standby Pump Switch to R.
- (7) Monitor LOW FUEL warning light on glareshield.

NOTE: The LOW FUEL warning light will extinguish as the left wing tank float switch is actuated (L wing tank fuel level raises); then illuminates as the right wing tank float switch is actuated (R wing tank fuel level lowers).

- (8) When LOW FUEL warning light illuminates set Right Standby Pump Switch off.
- (9) Equalize wing fuel tank loading.
- (10) Set Battery Switch(es) to OFF.

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CHECK VALVE - MAINTENANCE PRACTICES

1. General

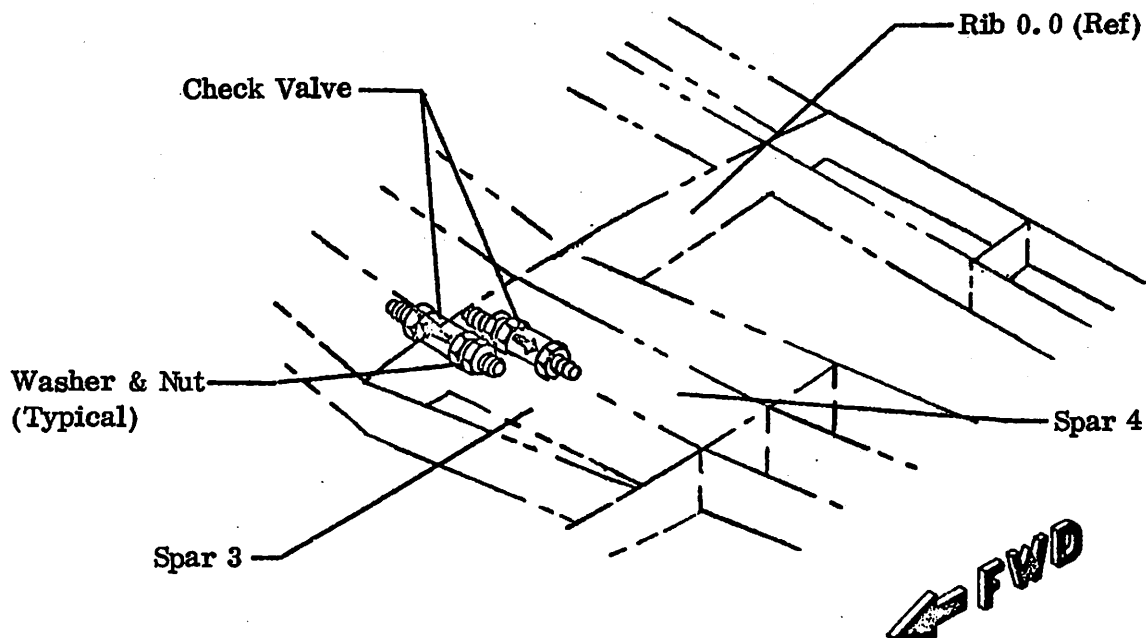
- A. Two check valves are installed in wing rib 0.0 between spar 3 and spar 4.
- B. The valves are installed to allow fuel flow in either direction, this allowing the fuel level to be equal in both wings.

2. Removal/Installation

- A. Remove Check Valve (See figure 201)
 - (1) Defuel the aircraft. (Refer to Chapter 12.)
 - (2) Remove inboard wing fuel probe. (Refer to 28-40-02.)
 - (3) Loosen and remove nut, washer and check valve from wing.
- B. Install Check Valve (See figure 201)

NOTE: Inspect check valve for clogging or contamination. Clean out all contamination by soaking or flushing the valve with kerosene.

- (1) Install check valve and secure with nut and washer.
- (2) Install fuel probe (refer to 28-40-02).



Check Valve Installation
Figure 201

EFFECTIVITY: ALL

MM-98

28-12-04
Page 201
Dec 18/81

Gates Learjet Corporation

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CROSSFLOW VALVE - MAINTENANCE PRACTICES

1. General

- A. The crossflow valve is installed to provide the capability to transfer fuel from one wing to the other, from the fuselage tank to the wings, and from the wings to the fuselage tank.
- B. The crossflow valve is located to the left of the aircraft centerline, forward of the LH jet pump.
- C. Replace the crossflow valve O-rings when fuel is found to be leaking from the valve.

2. Removal/Installation

A. Remove Crossflow Valve

- (1) Defuel the aircraft. (Refer to Chapter 12.)
- (2) Gain access to the crossflow valve by removing the lower wing access panels.
- (3) Disconnect electrical wiring.
- (4) Remove attaching parts and remove connector assembly, forward of the crossflow valve, and remove fuel plumbing from the crossflow valve.
- (5) Remove attaching parts and remove crossflow valve from the aircraft.

B. Install Crossflow Valve

- (1) Position the crossflow valve in place and secure with attaching parts.
- (2) Connect fuel plumbing to the aft side of the crossflow valve.
- (3) Install fuel plumbing connector assembly forward of the crossflow valve. Install connector assembly bolts with the bolt heads forward and torque to 20-25 inch-pounds.
- (4) Connect electrical wires to the crossflow valve.
- (5) Install lower wing access panels and seal in accordance with 28-12-00.
- (6) Fuel the aircraft (Refer to Chapter 12) and check for leaks.

3. Approved Repairs

A. O-Ring Replacement (See figure 201)

NOTE: It is important to note the position of each O-ring to ensure that each new O-ring is installed in its proper position.

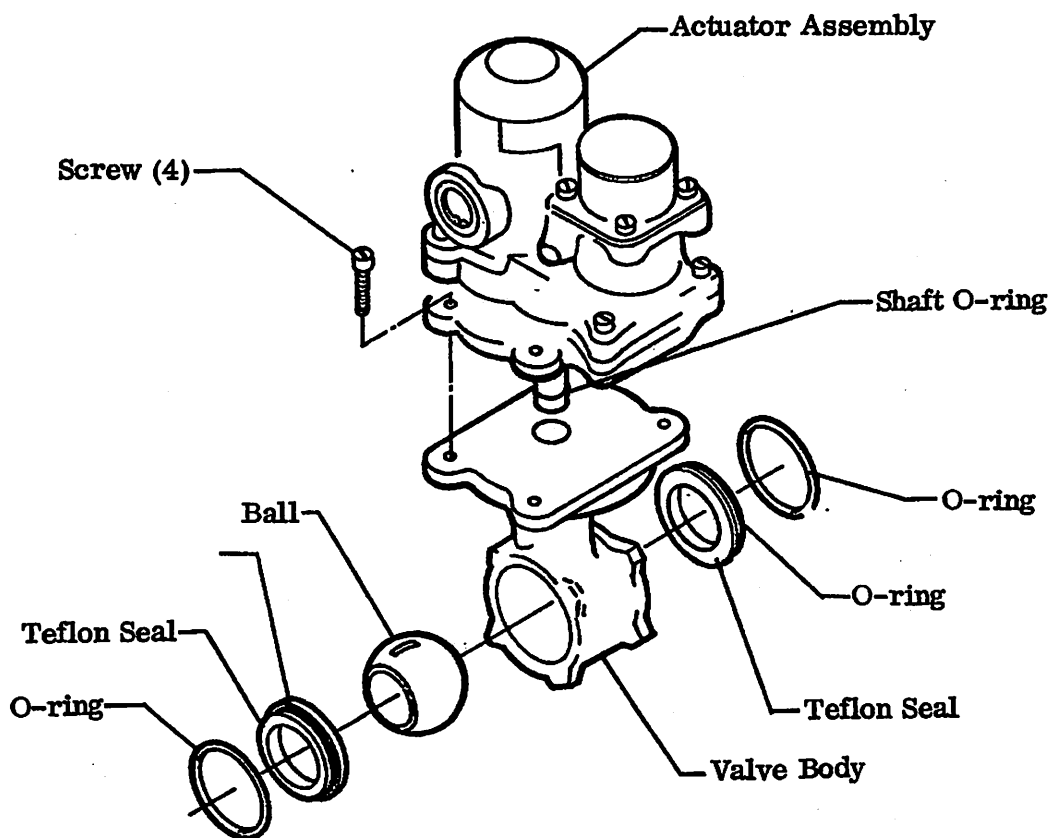
- (1) Remove the fuel crossflow valve from the aircraft as described in paragraph 1. A.
- (2) Remove the outer O-rings, teflon seals, and ball.
- (3) Remove the safety wire and remove the screws which attach the valve body to the actuator assembly. Remove the actuator assembly from the valve body to expose the shaft O-ring.

EFFECTIVITY: ALL

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- (4) Remove the old O-ring from the actuator shaft. Lubricate a new O-ring with petroleum jelly and install the O-ring on the shaft. Ensure that the O-ring is not twisted and is properly installed on the shaft.
- (5) Apply a light coating of petroleum jelly to the shaft O-ring and insert the shaft into the valve body. Secure the valve body to the actuator assembly with the attaching screws and safety wire.
- (6) Align the groove of the ball with the actuating shaft which protrudes into the valve body and install the ball.
- (7) Remove the old O-rings from the teflon seals. Lubricate the new O-rings with petroleum jelly and install the O-rings on the teflon seals.
- (8) Apply a light coating of petroleum jelly to the O-rings installed on the teflon seals and install the teflon seals in the valve body.
- (9) Lubricate the outer O-rings with petroleum jelly and install in the valve body.
- (10) Install the crossflow valve as described in paragraph 2. B.



Crossflow Valve O-Ring Replacement
Figure 201

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FUSELAGE TANK - DESCRIPTION AND OPERATION

1. Description

- A. On 25B/D aircraft the fuselage tank consists of two bladder-type cells located in the cavity between frames 22 and 25. The cavity incorporates two drain ports located in the aft area of the cavity. The drain ports are connected to individual drain lines. The fuselage tank components consist of a fuel transfer pump, float switch, fuel probe, drain valve, vent float valve, and vacuum relief valve. The crossover tube and three interconnect tubes connect the two cells to form the tank.
- B. On 25C/F aircraft the fuselage tank consists of four bladder-type cells located in cavities between frames 18 and 25. The cavity incorporates four drain ports. Two are located in the forward cavity area and two in the aft cavity area. The two forward drain ports are connected to a common drain line. The aft ports are connected to the individual drain lines.

2. Operation

- A. On 25B/D aircraft (see figure 1), fueling of the fuselage fuel tank is accomplished by pumping fuel from the wing tanks through a fuel and transfer line common to both cells. A motorized fuel transfer valve, located in the fuel and transfer line, shuts off fuel flow to the tank when the float switch attains the full position. Additionally, the float switch in the tank full position, deenergizes the wing standby boost pumps and lights the TANK FUEL indicator light. A fuselage tank low pressure switch is installed to alert the pilot when fuselage fuel is depleted. Fuel is transferred from the fuselage tank by a boost pump that provides fuel flow through the LH transfer valve to the wing tanks. In case of fuselage fuel pump failure, the transfer valve will allow gravity transfer to the wing tanks. The tank is vented by a vent tube with an opening at the forward end of the tank and a float valve connected to the vent tube at the aft end of the tank. The vent tube is connected to the external vent system. The vacuum relief valve has a free flow directional arrow imprinted on the valve body. This free flow arrow must point forward when installed. A drain valve, located in the fuel crossover tube, is used to drain moisture accumulation.
- B. On 25C/F aircraft (see figure 2), fueling of the fuselage fuel tank is accomplished by pumping fuel from the wing tanks through two refuel and transfer lines common to the fuel cells. Motorized fuel shutoff valves, located in both refuel and transfer lines, shut off fuel flow to the cells when the float switch, located in the right forward bladder is activated to the full position. The float switch also deenergizes the wing tank standby boost pumps and lights the TANK FULL indicator. Fuel is transferred from the fuselage tank by a boost pump that provides fuel flow through the LH transfer valve to the wing tanks. In case of normal fuel transfer failure, the RH fuselage transfer valve will allow gravity transfer to the wing tanks. A fuselage tank low pressure switch is installed to alert the pilot when fuselage fuel is depleted.

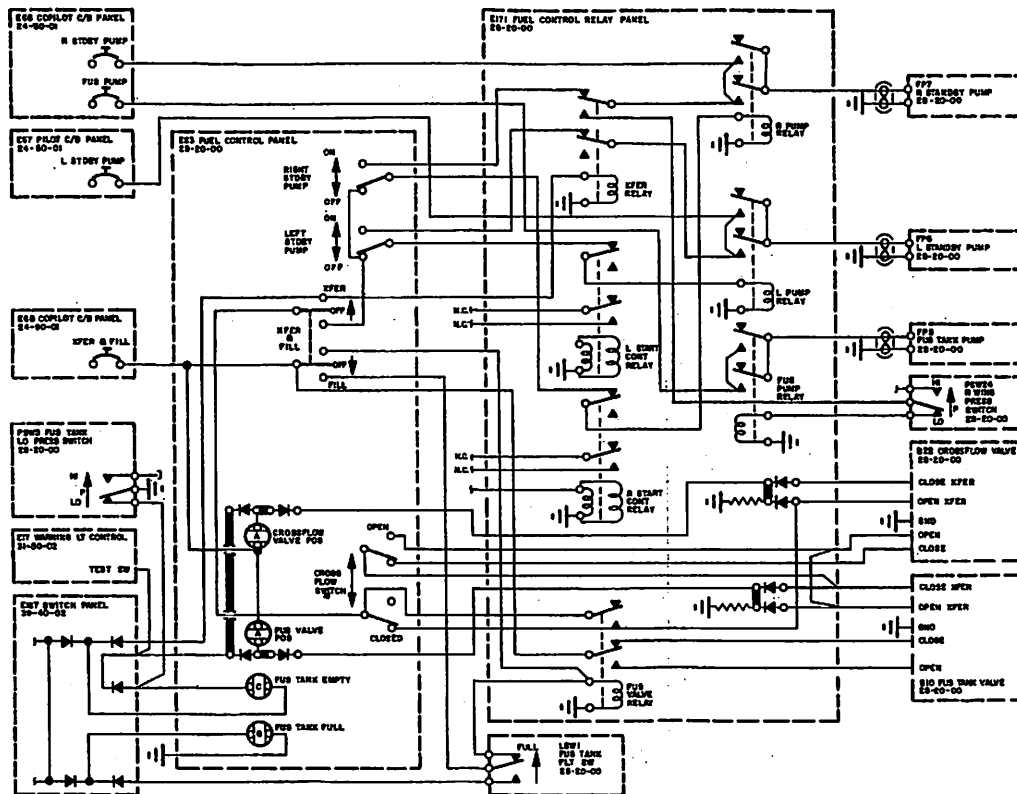
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The tank is vented by a vent tube which opens in the forward bladders. Connected to the forward vent tube is a venting arrangement which prevents siphoning of fuel into the vent system at extreme flight attitudes.

EFFECTIVITY: ALL

MM-98

**28-13-00
Page 2
Dec 18/81**



Fuel Transfer, Crossflow and Fill Electrical Control Schematic
Figure 1

EFFECTIVITY: 25B/D AIRCRAFT

MCN-98

28-13-00
Page 3
Dec 18/81

Gates Learjet Corporation



nd FY11 E
Figure 2

28-13-00
Page 4
Dec 18/81

Gates Learjet Corporation 

maintenance manual

FUSELAGE TANK - MAINTENANCE PRACTICES

1. **General**

- A. Maintenance practices consist of replacement of defective components, inspection and repairing of sealed areas if deterioration or leak occurs.
- B. For maintenance practices on the inspection and repairing of sealed areas in the fuselage fuel cell cavity. (Refer to Chapter 20.)

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FUEL BLADDER - MAINTENANCE PRACTICES

1. Removal/Installation

WARNING: OBSERVE ALL SAFETY PRACTICES DESCRIBED IN 28-00-00. SPECIAL CARE SHOULD BE TAKEN FOR PROPERLY VENTILATING THE FUEL CELLS PRIOR TO PERFORMING MAINTENANCE PROCEDURES.

A. Remove Fuel Bladder (See figure 201)

- (1) Assure that the fuselage tank is empty.
- (2) Lower the tailcone access door and remove batteries. (Refer to 24-32-05.)
- (3) Remove LH and RH pylon access panels, located on lower side of engine pylon, and disconnect main fuel and motive flow lines from pylon tank fittings. Remove bolts freeing bladder fittings from tank structure.
- (4) Remove fuselage fuel probe access cover and disconnect electrical wiring from fuel probe. Remove attaching parts and fuel probe from aircraft. The lower end of the fuel probe is set in the lower fuel probe fitting. Do not remove bonding jumper from fuselage structure.
- (5) Remove lower fuselage access panels to gain access to fuselage transfer pump, crossover tube, and fuel probe lower fitting.
- (6) Remove fuselage fuel transfer pump. (Refer to 28-13-05.)
- (7) Remove tape and sealant securing bladder nipples to fuselage tank structure.
- (8) Remove attaching parts and fuel probe lower fitting.
- (9) Disconnect fuel vent line at upper RH access cover connection.
- (10) Disconnect main fuel and motive flow lines at frame 24 connection and fuel cell connections. Remove aft fuel cell connections. Remove main fuel and motive flow fittings.
- (11) Disconnect fuel filter drain lines at aircraft skin connection.
- (12) Disconnect pressure relief valve drain lines from main fuel line connection and aircraft skin connection.
- (13) Disconnect wiring from fuel pressure switches.
- (14) Remove attaching parts and large fuel cell access covers from aircraft.
- (15) Disconnect internal fuel vent line and remove clamps.
- (16) Remove attaching parts and RH and LH upper access covers from fuel cell.
- (17) Disconnect and remove main fuel and motive flow lines from internal fuel tank fittings.
- (18) Remove clamp and crossover tubes.
- (19) Unlatch fuel cell bladders from structure and remove through large access openings.

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B. Install Fuel Bladder (See figure 201)

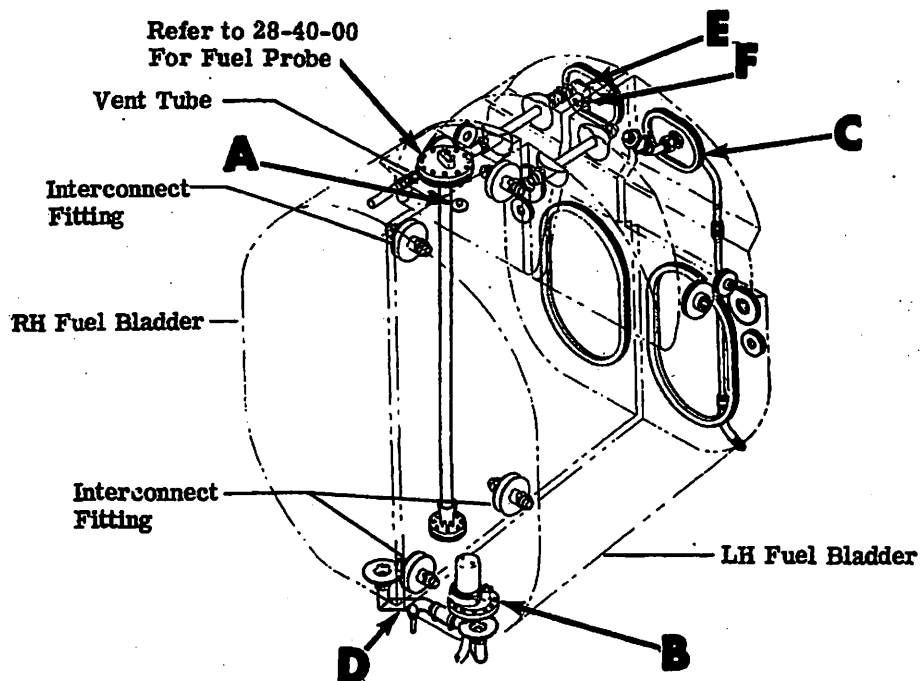
- (1) Clean fuel cell cavities of all burrs and foreign matter. Assure that all rivet heads in cavities are properly sealed.
- (2) Check for fuel cavity drain obstructions.
- (3) Clean fuel bladders and inspect for condition.

NOTE: Prior to installing bladders, cover cell fittings with protective paper and masking tape to avoid possible damage to bladder fittings during installation.

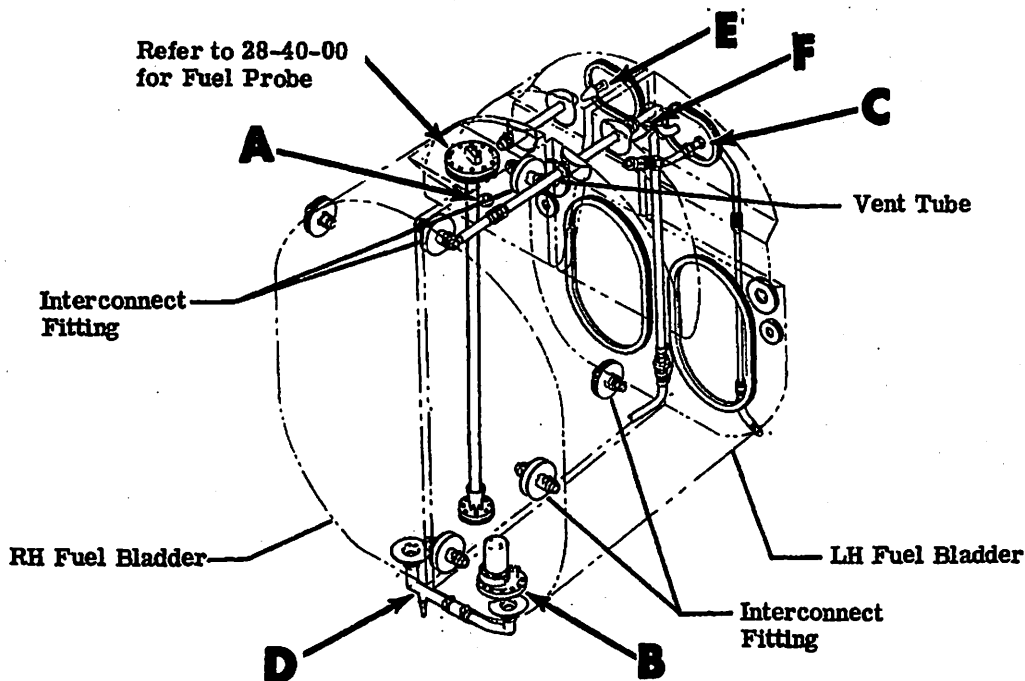
Replace old gaskets with new gaskets.

- (4) Insert LH bladder (bottom end first) through large access opening in aft side of fuel cell. Position bladder in cavity and latch hanger fasteners (11 places). Repeat same procedure for RH cell except bladder has 10 fasteners.
- (5) Remove protective covering from bladder fittings.
- (6) Insert interconnect tubes from LH bladder through structure cell divider into RH bladder. If necessary, dampen interconnect tubes with water to permit tubes to slide into bladder nipples.
- (7) Position interconnect tubes flush with end of internal nipples.
- (8) Secure nipples to tubes with clamps.
- (9) Install fuselage fuel transfer pump. (Refer to 28-13-05)
- (10) Wrap bladder nipple with tape (P/N 853, manufactured by the Minnesota Mining & Mfg. Co.) and apply a fillet seal to tape and tank structure. Allow sealant to dry before installing interconnect tube.
- (11) Fabricate three alignment pins by cutting off the heads of three AN4-30 bolts.
- (12) Install fuel probe and lower fitting as follows:
 - (a) Install alignment pins (equally spaced) around perimeter of bladder metal gasket.
 - (b) Install gasket over alignment pins and position pins through holes in tank structure.
 - (c) With bladder metal gasket held firmly in place, position gasket and upper portion of fuel probe over alignment pins. Install attaching bolts and tighten finger-tight. Remove alignment pins and install remaining bolts. Torque bolts to 50 ($\pm$ 5) inch-pounds.

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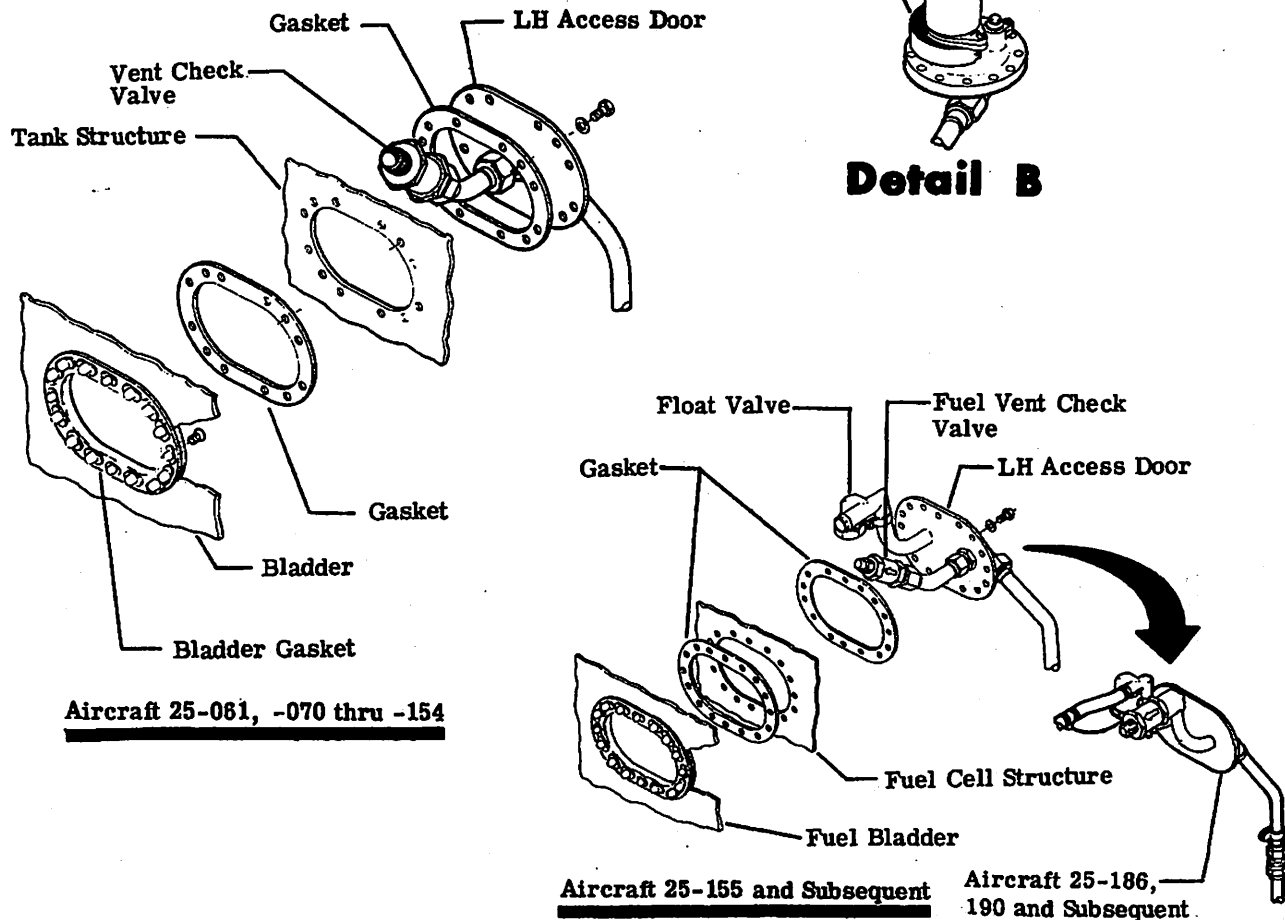
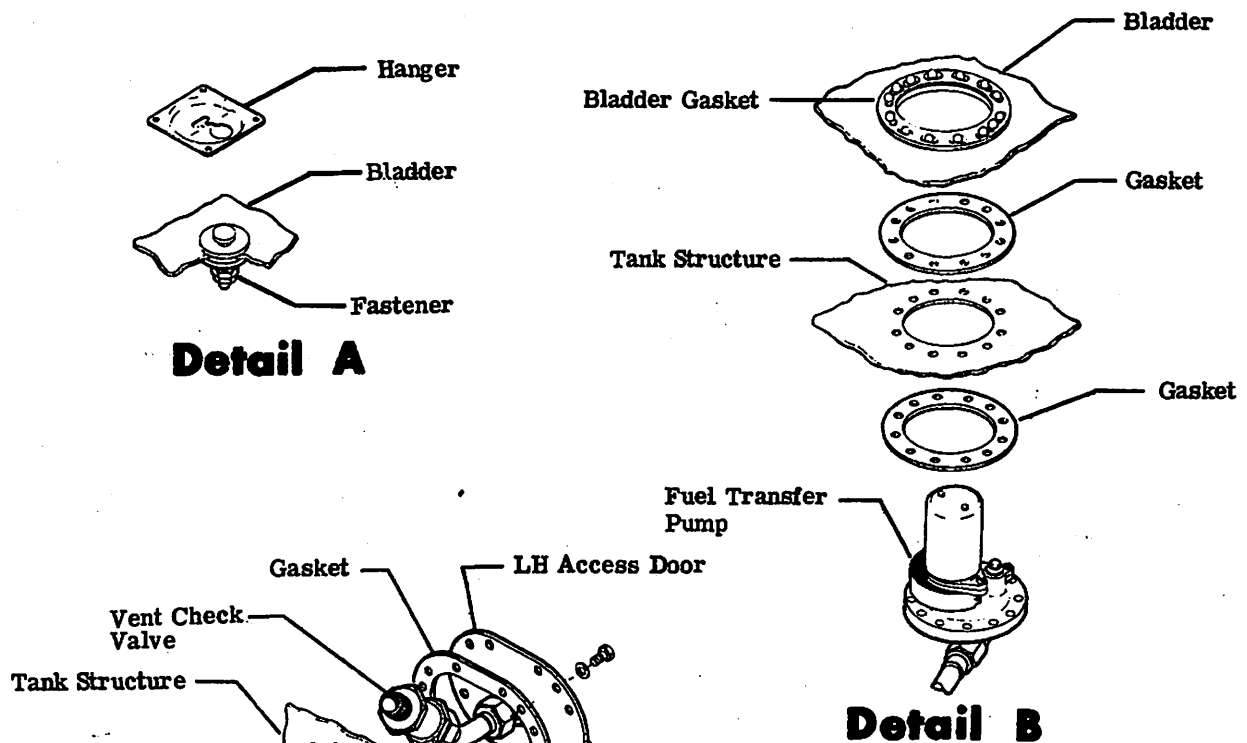
Aircraft 25-081, -070 thru -154



Aircraft 25-155 and Subsequent

Fuselage Fuel Tank Bladder Installation
Figure 201 (Sheet 1 of 3)

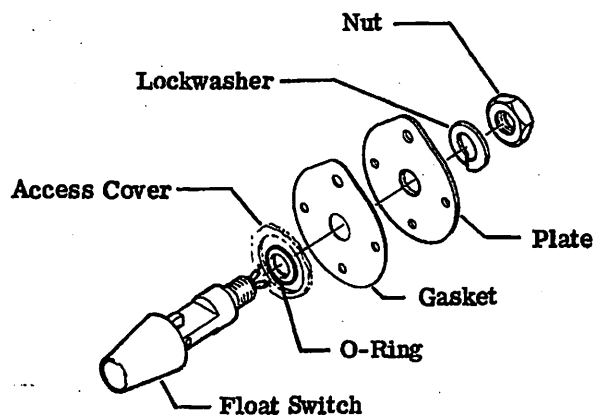
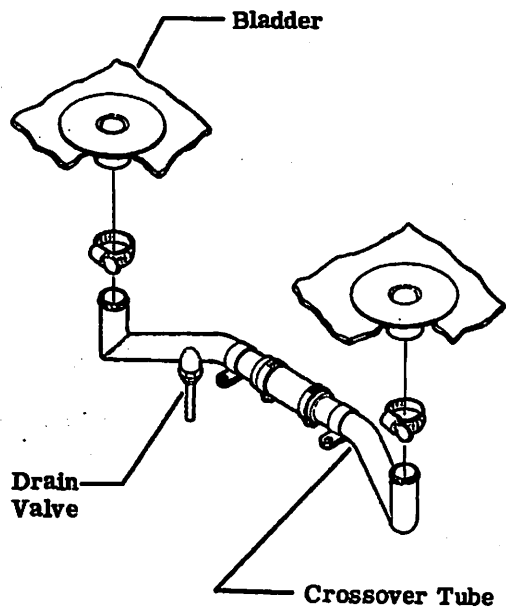
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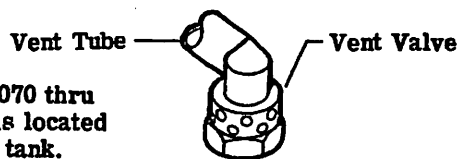
Detail C

Fuselage Fuel Tank Bladder Installation
Figure 201 (Sheet 2 of 3)

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NOTE: On Aircraft 25-081, -070 thru -154, the Vent Valve is located in RH side of fuselage tank.



Fuselage Fuel Tank Bladder Installation
Figure 201 (Sheet 3 of 3)

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- (13) Install fuel cell fittings as follows:
- (a) Install alignment of pins (equally spaced) around perimeter of bladder metal gasket.
 - (b) Install gasket over alignment pins and position pins through holes in tank structure.
 - (c) With bladder metal gasket held firmly in place, position fitting over alignment pins and install bolts. Remove alignment pins and install remaining bolts. Torque bolts in a criss-cross pattern to 50 (± 5) inch-pounds.
 - (d) Repeat steps (a) through (c) on remaining fuel cell fittings.
- (14) Install RH and LH upper access covers as follows:
- (a) Position hose on internal vent line and secure with clamp.
 - (b) Install three alignment pins in perimeter of bladder metal gasket. Position gasket over alignment pins and insert pins into holes around RH access opening.
- NOTE: The LH access cover is installed the same as the RH, except the LH has no vent line connection.
- (c) Position a second gasket over alignment pins. Position RH access cover over alignment pins while seating access cover vent line into hose. Secure hose to vent line with clamp and access cover to tank structure with bolts. Remove alignment pins and install remaining bolts. Torque bolts 50 (± 5) inch-pounds in a crisscross fashion. Safety wire per the double twist method.
- (15) Install large access covers as follows:
- (a) Install alignment pins into bladder metal gaskets equally spaced around perimeter of opening.
 - (b) Install a gasket on alignment pins and position bladder metal gasket and gasket to fuel cell opening.
 - (c) Holding bladder metal gasket in place, install gasket on the aft side of fuel cell structure. Install access covers and secure with three washers and bolts equally spaced around perimeter of access covers. Tighten bolts finger-tight.
 - (d) Remove alignment pins and install remaining washer and bolts.
 - (e) Torque bolts to 50 (± 5) inch-pounds.
 - (f) Repeat steps (a) through (e) on remaining access cover.
- (16) Install all lower access panels.
- (17) Connect all fuel plumbing and electrical wiring previously removed.
- (18) Install aircraft batteries. (Refer to 24-32-05)
- (19) Fill fuselage tank and check for leaks.
- (20) Secure the tailcone access door.

Gates Learjet Corporation
maintenance manual

C. Remove Forward Fuel Bladder (See figure 201)

- (1) Assure that the fuselage tank is empty. (Refer to 12-10-10.)
- (2) Disconnect aircraft batteries.
- (3) Remove interior panels and equipment installed in aft cabin as required to gain access to aft pressure bulkhead.
- (4) Disconnect electrical plug from float switch.
- (5) Remove sealant from coverplate attaching parts. (Refer to Detail A.)
- (6) Remove attaching parts securing coverplate to pressure bulkhead.
- (7) Remove coverplate sufficiently to allow disconnection of electrical connector on aft side of coverplate. Disconnect electrical connector.
- (8) Remove safety wire, attaching parts and float switch from fuel cell.
- (9) Remove attaching parts and pressure bulkhead from aircraft.
- (10) Remove attaching parts and forward cell access covers from fuel cell.
- (11) Remove attaching parts and remove fuel vent tubing, interconnect fittings, tee assemblies, and expansion tubes from forward fuel cells.

NOTE: The interconnect tubes between the forward and aft fuel cells are removed when the aft fuel cell is removed.

- (12) Using slight pressure, push up at bladder fastener locations and release fasteners from hangers.
- (13) Remove forward fuel cell bladders through access opening.

D. Install Forward Fuel Bladder (See figure 202.)

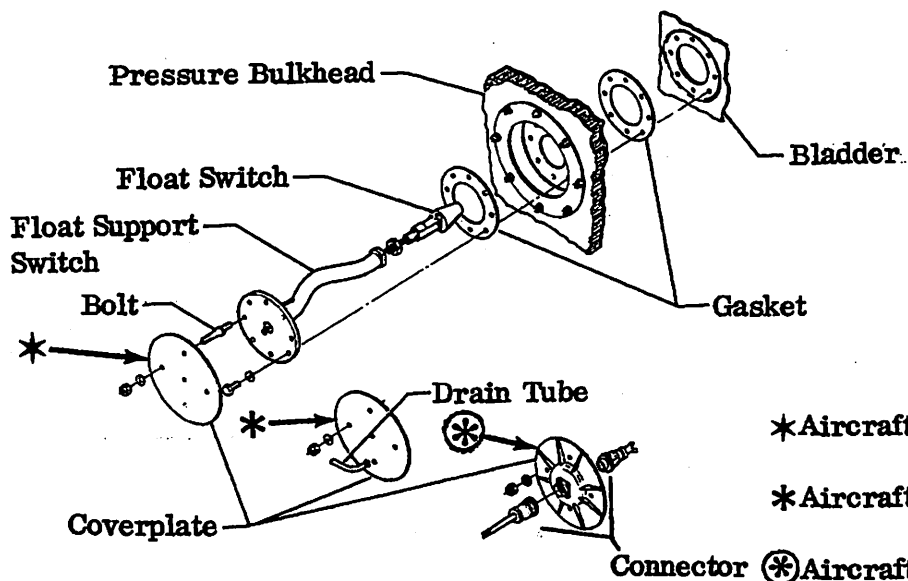
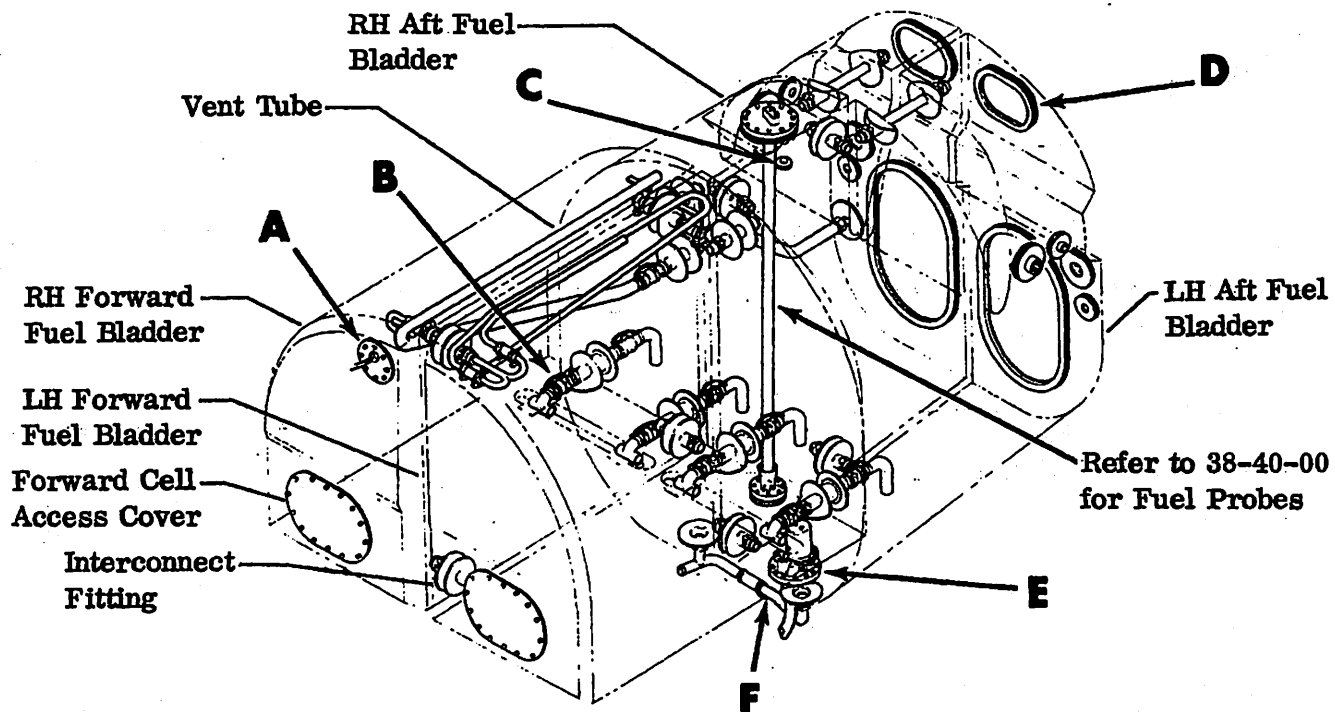
- (1) Clean fuel cavities of all burrs and foreign matter and check for fuel cavity drain obstructions. Assure that all rivet heads in cavities are properly sealed.
- (2) Clean fuel tank bladders and inspect for condition.

NOTE: • Prior to installing bladders, cover fittings with protective paper and masking tape to avoid possible damage during installation.

- Do not reinstall gaskets that have been removed. Replace with new gaskets.

- (3) Insert tank bladders into cavities through openings above cell access supports.
- (4) Remove protective covering from bladder fittings.
- (5) Position tank bladders into cavities and latch hanger fasteners (12 places for each cell).
- (6) Install interconnect fittings. If necessary, dampen tube with water to permit tubes to slide into bladder nipples.
- (7) Install interconnect tube clamps.
- (8) Install the four lower bladder interconnect tubes and insert tee assemblies under fabric retainers.

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*Aircraft 25-272 thru 25-159

*Aircraft 25-070 thru 25-071

Connector (*)Aircraft 25-160 and Subsequent

Detail A

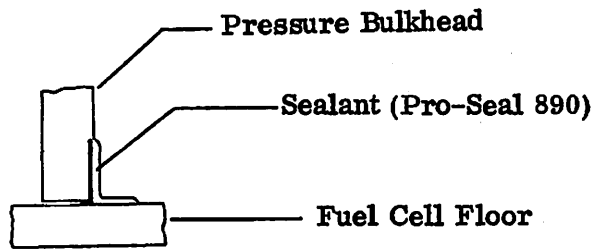
Fuselage Fuel Tank Bladder Installation
Figure 202 (Sheet 1 of 3)

EFFECTIVITY: 25C/F AIRCRAFT

MM-98

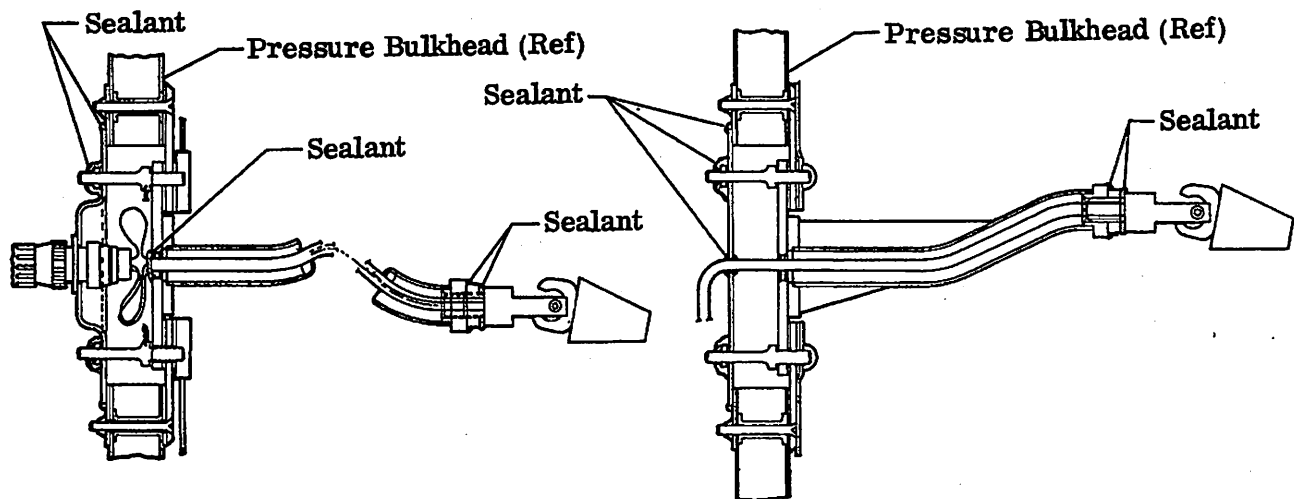
28-13-01
Page 208
Dec 18/81

 GATES LEARJET CORPORATION
maintenance manual



Apply faying surface seal on fuel cell floor. Apply a coat of paste wax around the edge and down the center of the pressure bulkhead.

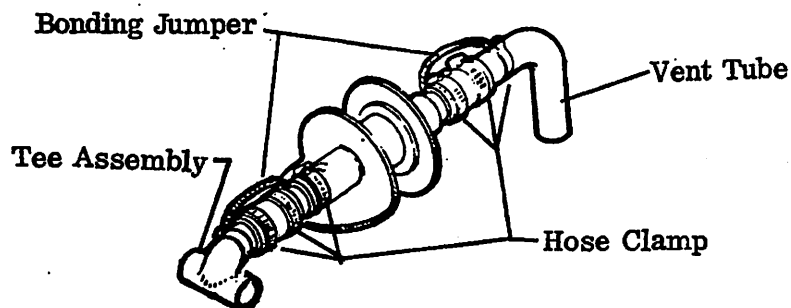
PRESSURE BULKHEAD SEALING



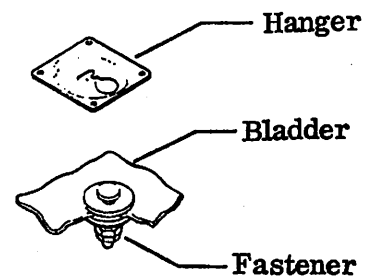
Aircraft 25-160 and Subsequent

Aircraft 25-070 thru 25-159

FLOAT SWITCH SEALING



Detail B



Detail C

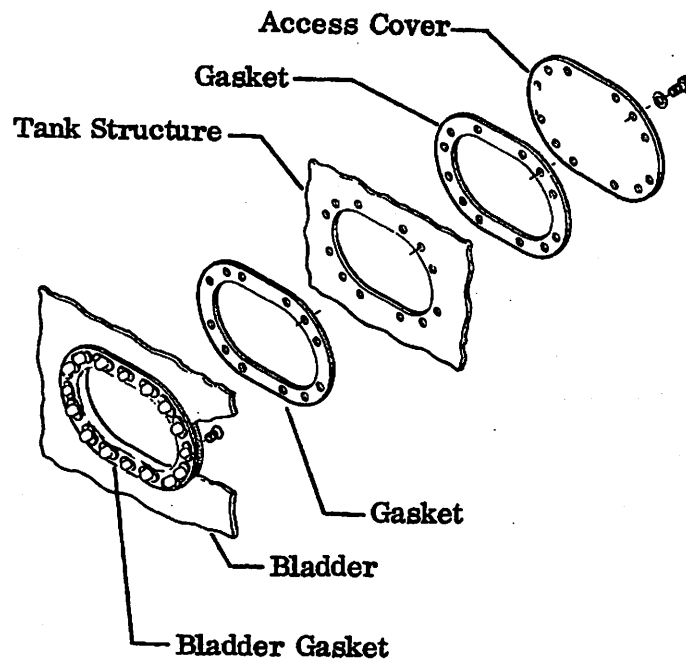
**Fuselage Fuel Tank Bladder Installation
Figure 202 (Sheet 2 of 3)**

EFFECTIVITY: 25C/F AIRCRAFT

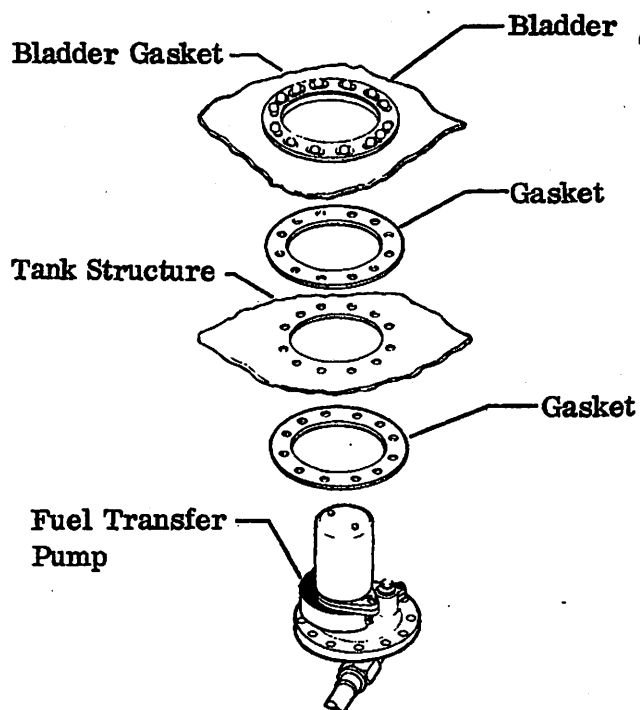
MM-98

28-13-01
Page 209
Dec 18/81

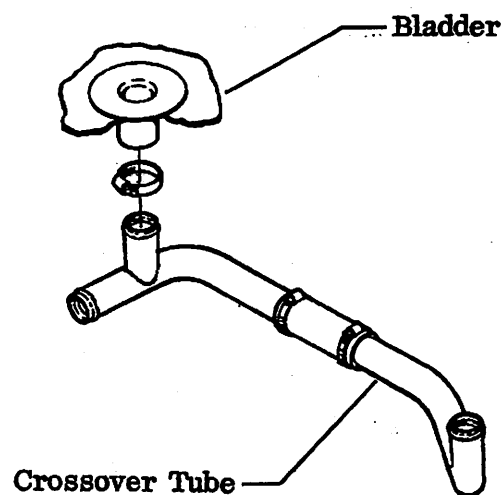
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maintenance manual



Detail D



Detail E



Detail F

Fuselage Fuel Tank Bladder Installation
Figure 202 (Sheet 3 of 3)

EFFECTIVITY: 25C/F AIRCRAFT

MM-98

28-13-01
 Page 210
 Dec 2/77

Gates Learjet Corporation

maintenance manual

- (9) Secure interconnect tubes and tee assemblies with clamps and install bonding jumpers.
- (10) Connect vent tubes to tank vent connection and secure with clamps.
- (11) Install vent tube support clamps.
- (12) Install vent tube bonding jumper.
- (13) Fabricate alignment pins by cutting off heads of three AN4-30 bolts.
- (14) Install tank bladder access covers as follows:
 - (a) Install alignment pins (equally spaced around perimeter of bladder fitting).
 - (b) Install gasket over alignment pins and position pins through holes of bladder support.
 - (c) Install gasket on forward side of bladder support and install access covers and secure with washers and bolts equally spaced around perimeter of access cover. Tighten bolts finger-tight.
 - (d) Remove alignment pins and install remaining washers and bolts.
 - (e) Torque bolts in a crisscross pattern to 50 (± 5) inch-pounds.
 - (f) Repeat steps (a) thru (e) on remaining access cover.
- (15) Install four alignment pins into upper RH access bladder fittings. Install gasket on alignment pins. Position alignment pins at 12, 3, 6 and 9 o'clock around perimeter of bladder fitting.
- (16) Install pressure bulkhead as follows:
 - (a) Inspect each bulkhead nutplate for condition of threads and security of mounting.
 - (b) Clean sealing surface.
 - (c) Apply a generous coat of paste wax to the perimeter and down the center of the bulkhead.
 - (d) Apply faying surface seal to the pressure bulkhead. (See figure 202.)
 - (e) Insert float switch fitting alignment pins into fittings of bulkhead.
 - (f) Install washer and bolts around perimeter and through the center of the bulkhead. Tighten bolts finger-tight.
 - (g) Tighten bolts in a crisscross pattern and torque. (Refer to Chapter 20.)
 - (h) Inspect for a continuous bead of sealant around the pressure bulkhead.
- (17) Carefully position float switch support on alignment pins. Ensure that float switch float is in the vertical position and gaskets are in place.
- (18) Install gasket and upper access cover over alignment pins. Install bolts in open holes and tighten finger-tight.
- (19) Remove alignment pins and install remaining bolts. Torque bolts in crisscross pattern to 50 (± 5) inch-pounds.
- (20) Install coverplate with rivets (P/N CR2249-4).
- (21) Apply sealant around perimeter of coverplate and over rivets.



GATES LEARJET CORPORATION

maintenance manual

- (22) Install electrical equipment and cabin pressurization components previously removed.
 - (23) Install interior lining.
 - (24) Connect aircraft batteries.
 - (25) Fill fuselage tank and check for leaks.
- E. Remove Aft Fuel Bladder (See figure 202)
- (1) Assure that the fuselage tank is empty.
 - (2) Lower tailcone access door.
 - (3) Remove aircraft batteries. (Refer to 24-32-01)
 - (4) Remove LH and RH pylon access panels, located on lower side of engine pylon, and disconnect main fuel and motive flow lines from pylon tank fittings. Remove bolts and free bladder fittings from tank structure.

NOTE: The lower end of the fuel probe is set in the lower fuel probe fitting.

- (5) Remove fuselage fuel probe access panel and disconnect electrical wiring from fuel probe. Remove attaching parts and fuel probe from aircraft. Do not remove bonding jumper from fuselage structure.
- (6) Remove lower fuselage access panels to gain access to fuselage transfer pump, crossover tube, and fuel probe lower fitting.
- (7) Remove fuselage fuel transfer pump. (Refer to 28-13-05)
- (8) Remove tape and sealant securing bladder nipples to fuselage tank structure.
- (9) Remove attaching parts and fuel probe lower fitting.
- (10) Disconnect main fuel and motive flow lines at frame 24 connection and fuel cell connections. Remove aft fuel cell main fuel and motive flow fittings.
- (11) Disconnect fuel filter drain lines at aircraft skin connection.
- (12) Disconnect pressure relief valve drain lines from engine main fuel line connection and aircraft skin connection. Remove drain lines from aircraft.
- (13) Disconnect wiring from fuel pressure switches.
- (14) Remove attaching parts and large fuel cell access covers from aircraft.
- (15) Disconnect internal fuel vent line and remove clamps.
- (16) Remove attaching parts and RH and LH upper access covers from fuel cell.
- (17) Disconnect and remove main fuel and motive flow lines from internal fuel tank fittings.
- (18) Remove clamp and interconnect tubes.
- (19) Unlatch fuel cell bladders from structure and remove through large access opening.

GATES LEARJET CORPORATION
maintenance manual

F. Install Aft Fuel Bladder (See figure 202)

- (1) Clean fuel cell cavities of all burrs and foreign matter. Assure that all rivet heads in cavities are properly sealed.
- (2) Clean fuel tank bladders and inspect for condition.

NOTE: Prior to installing bladders, cover fittings with protective paper and masking tape to avoid possible damage during installation.

Do not reinstall gaskets that have been removed.
Replace with new gaskets.

- (3) Insert LH bladder (bottom end first) through large access opening in aft side of fuel cell. Position bladder in cavity and latch hanger fasteners (11 places). Repeat same procedure for RH cell except bladder has 10 fasteners.
- (4) Install interconnect tubes. If necessary, dampen tube with water to permit tubes to slide into bladder nipples.
- (5) Install interconnect tube clamps.
- (6) Secure nipples to tubes with clamps.
- (7) Install fuselage fuel transfer pump. (Refer to 28-13-05).
- (8) Wrap bladder nipple with tape (P/N 853, manufactured by the Minnesota Mining & Mfg. Co.) and apply a fillet seal to tape and tank structure. (Refer to 28-10-00) Allow sealant to dry before installing crossover tube.
- (9) Fabricate three alignment pins by cutting off the heads of three 1/4 x 3 inch bolts.
- (10) Install fuel probe and lower fitting as follows:
 - (a) Install alignment pins (equally spaced) around perimeter of bladder metal gasket.
 - (b) Install gasket over alignment pins and position pins through holes in tank structure.
 - (c) With bladder metal gasket held firmly in place, position gasket and upper portion of fuel probe over alignment pins. Install attaching bolts and tighten finger-tight. Remove alignment pins and install remaining bolts. Torque bolts in a crisscross pattern to 50 (±5) inch-pounds.
- (11) Install fuel cell fittings as follows:
 - (a) Install alignment pins (equally spaced) around perimeter of bladder metal gasket.
 - (b) Install gasket over alignment pins and position pins through holes in tank structure.
 - (c) With bladder metal gasket held firmly in place, position fitting over alignment pins and install bolts. Remove alignment pins and install remaining bolts. Torque bolts in a crisscross pattern to 50 (±5) inch-pounds.



GATES LEARJET CORPORATION

maintenance manual

- (d) Repeat steps (a) thru (c) on remaining fuel cell fittings.
- (e) Position hoses no internal vent line and secure with clamps.
- (12) Install RH and LH upper access covers as follows:
 - (a) Install three alignment pins in perimeter of bladder metal gasket. Position gasket over alignment pins and insert pins into holes around RH access opening.
 - (b) Position a second gasket over alignment pins. Position RH access cover over alignment pins and install washers and bolts.
 - (c) Remove alignment pins and install remaining washers and bolts. Torque bolts 25 (± 5) inch-pounds in a crisscross pattern. Safety wire per the double twist method.
- (13) Install large access covers as follows:
 - (a) Install alignment pins into bladder metal gaskets equally spaced around perimeter of opening.
 - (b) Install a gasket on alignment pins and position metal gasket and gasket to fuel cell opening.
 - (c) Holding metal bladder gasket in place, install gasket on the aft side of fuel cell structure. Install access cover and secure with three washers and bolts equally spaced around perimeter of cover panel. Tighten bolts finger-tight.
 - (d) Remove alignment pins and install remaining washer and bolts.
 - (e) Torque bolts in a crisscross pattern to 50 (± 5) inch-pounds.
 - (f) Repeat steps (a) thru (e) on remaining access cover.
- (14) Install all lower access panels.
- (15) Connect all fuel plumbing and electrical wiring previously removed.
- (16) Install aircraft batteries. (Refer to 24-32-01)
- (17) Fill fuselage tank and check for leaks.
- (18) Secure tailcone access door.

GATES LEARJET CORPORATION
maintenance manual

FUEL TRANSFER VALVE - MAINTENANCE PRACTICES

1. Description

- A. Motorized fuel transfer valves are two-position (open and closed) ball-and-seat type and are electrically operated by 28 vdc.

2. Operation

- A. On 25B/D Aircraft the fuselage fuel tank transfer valve is located in the refuel and transfer line from the fuselage to the right wing tank. The fuselage fuel transfer valve is controlled by the Fuselage Tank Transfer-Fill Switch on the fuel control panel.
- B. On 25C/F Aircraft there is a fuel transfer valve in the RH and LH refuel and transfer line from the fuselage tank to the wing tanks. The fuselage fuel tank transfer valves are controlled by the Fuselage Tank Transfer-Fill Switch, also the RH fuel transfer valve can be controlled by the FUS Valve Switch. The Fuselage Tank Transfer-Fill Switch and the FUS Valve Switch are located on the fuel control panel.

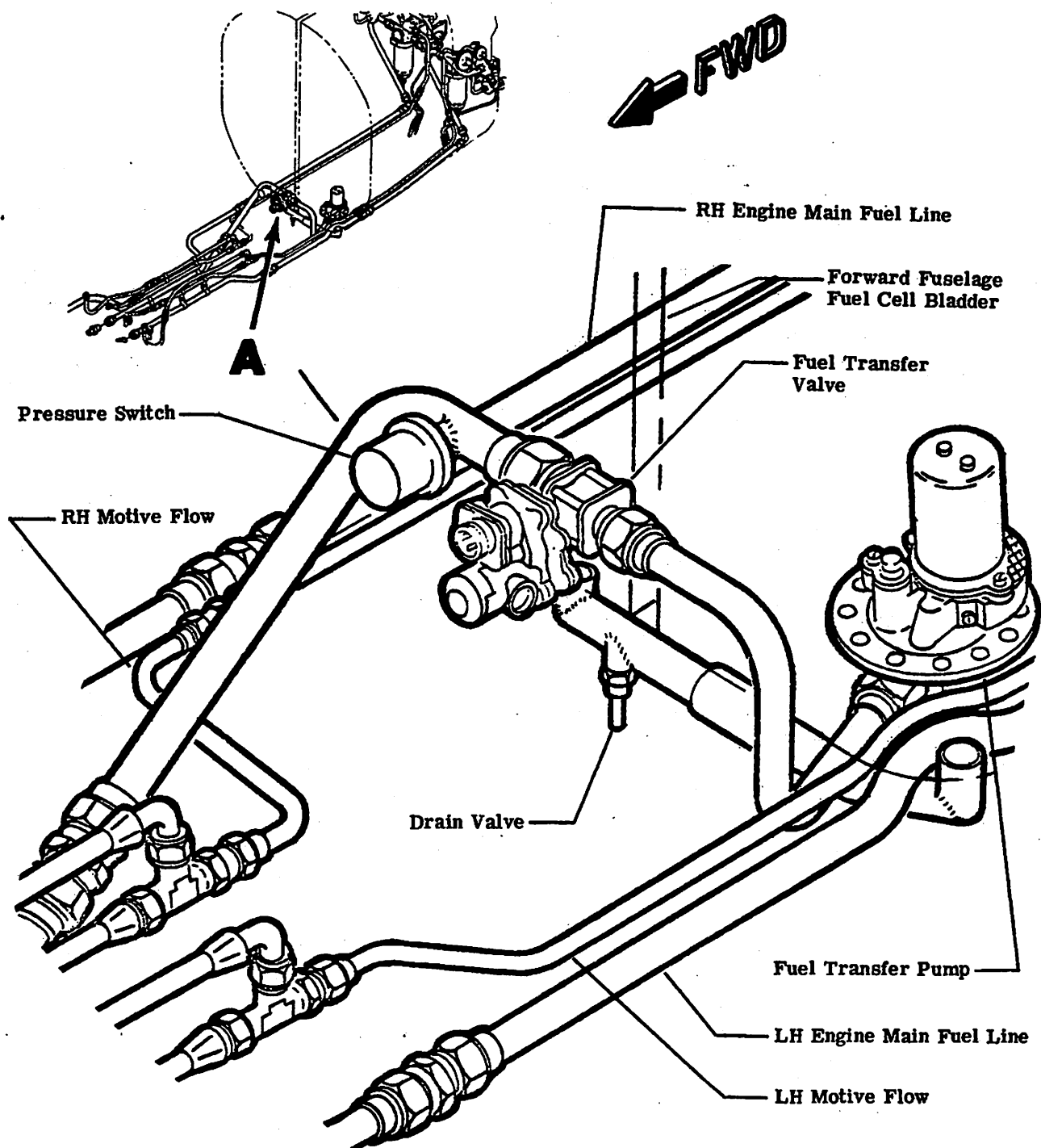
3. Removal/Installation

- A. Remove Transfer Valves (Typical) (See figures 201 and 202)
- (1) Defuel the aircraft. (Refer to 12-00-00)
 - (2) Remove lower fuselage access covers.
 - (3) Provide a suitable container to catch fuel when plumbing is disconnected from transfer valves.
 - (4) Disconnect electrical plug from transfer valve.
 - (5) Disconnect plumbing and remove transfer valve from aircraft.

CAUTION: THE FUEL TRANSFER VALVES INCORPORATE TWO TEFLON SEALING GASKETS, ONE BLACK AND ONE WHITE, AGAINST THE BALL VALVE. THE BLACK GASKET INCORPORATES A SMALL ORIFICE TO RELIEVE SYSTEM PRESSURE. THEREFORE, WHEN INSTALLING THE FUEL TRANSFER VALVES, ALWAYS POSITION THE VALVE SO THAT THE BLACK GASKET IS NEAREST THE PUMP (GREATEST PRESSURE).

- B. Install Transfer Valves (Typical) (See figures 201 and 202)
- (1) Install transfer valve and connect plumbing. Assure that Teflon gasket is properly installed.
 - (2) Connect electrical plug to valve.
 - (3) Fuel the aircraft, refer to 12-00-00, and check for leaks.
 - (4) Install access covers previously removed.

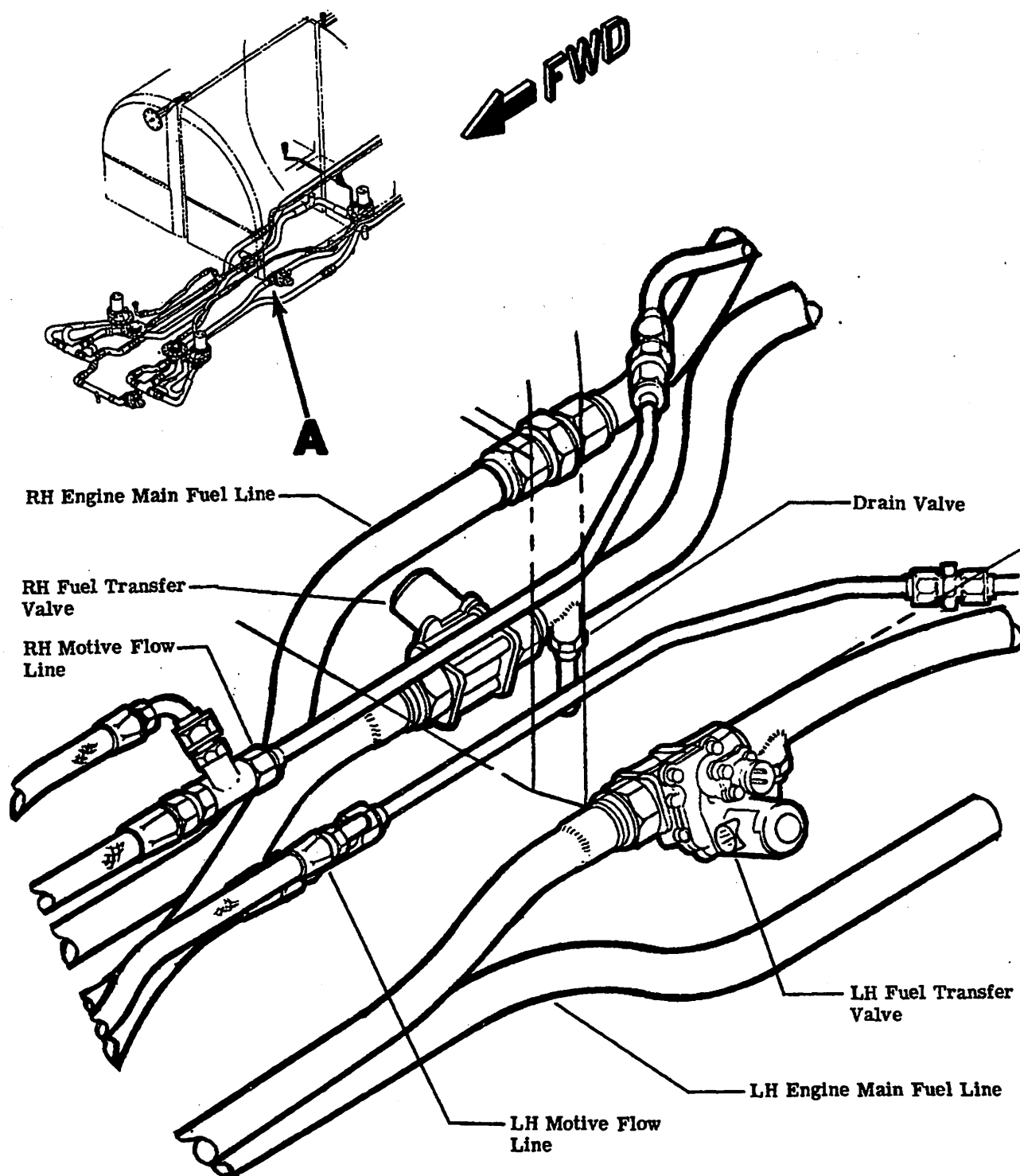
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maintenance manual



Detail A

Fuel Transfer Valve Installation
Figure 201

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maintenance manual



Detail A

Fuel Transfer Valve Installation
Figure 202

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maintenance manual

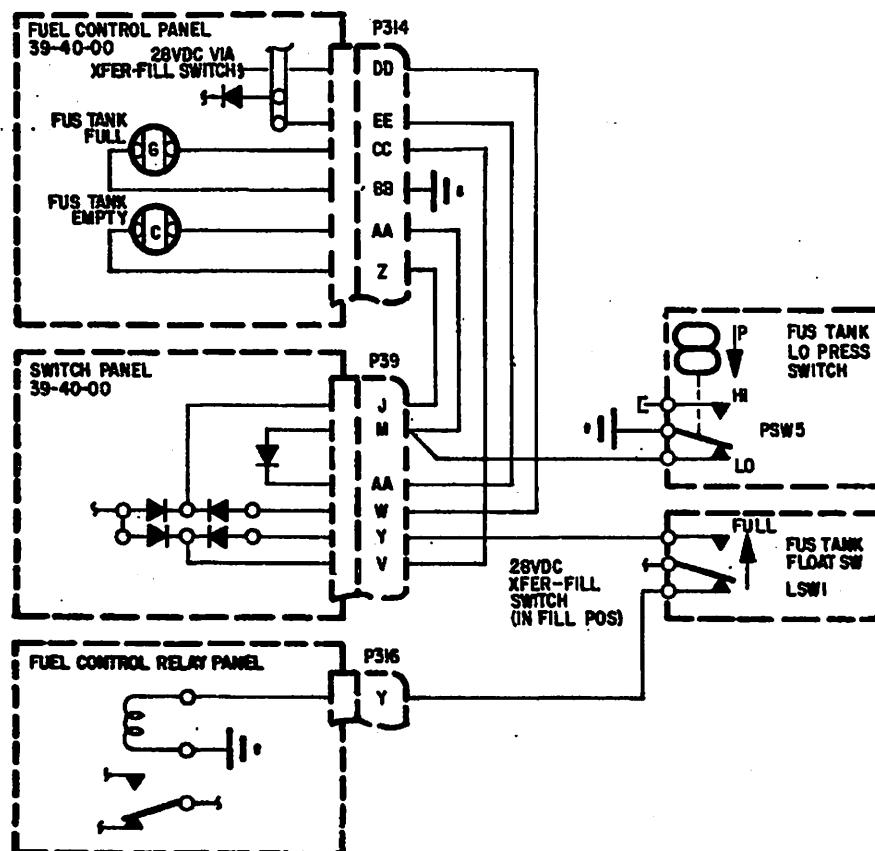
FUSELAGE PRESSURE SWITCH - DESCRIPTION AND OPERATION

1. General

- A. The fuselage tank low pressure switch is installed to alert the pilot when fuselage fuel is depleted.
- B. On 25B/D Aircraft the pressure switch is located forward of frame 22 directly above the keel beam. Access to the switch is gained through the center access cover aft of the right wheel well.
- C. On 25C/F Aircraft the pressure switch is located in the LH transfer line. Access to the switch is gained through the lower fuselage access panels.

2. Operation (See figure 1)

- A. The function of the switch is to sense pressure from the fuselage transfer-fill line and energize the FUS Tank Empty indicator light on the fuel control panel. The switch closes and energizes the light when pressure drops below 2.75 psi. The switch reopens at a pressure of 3.75 psi.



Fuselage Pressure Switch Electrical Control Schematic
Figure 1

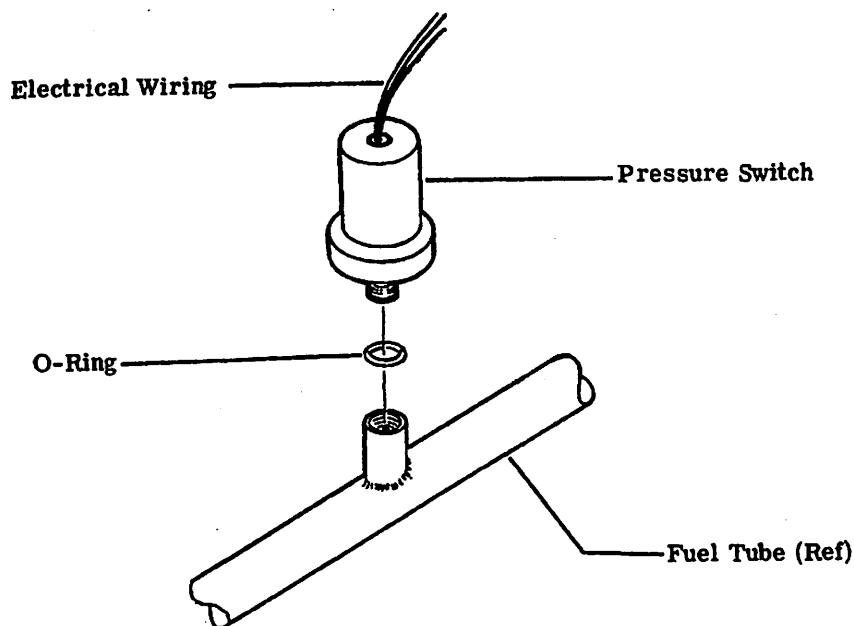
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FUSELAGE PRESSURE SWITCH - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: The following removal and installation procedures are typical.

- A. Remove Pressure Switch (See figure 201)**
- (1) Defuel the fuselage tank. (Refer to 12-00-00)
 - (2) Gain access to pressure switch installation.
 - (3) Disconnect switch wiring at splice. Tag wiring.
 - (4) Remove pressure switch from fuel transfer tube.
- B. Install Pressure Switch (See figure 201)**
- (1) Install pressure switch in fuel transfer tube.
 - (2) Connect wiring at splice.
 - (3) Fuel the aircraft. (Refer to 12-00-00)
 - (4) Install removed access covers.



**Fuselage Pressure Switch Installation
Figure 201**

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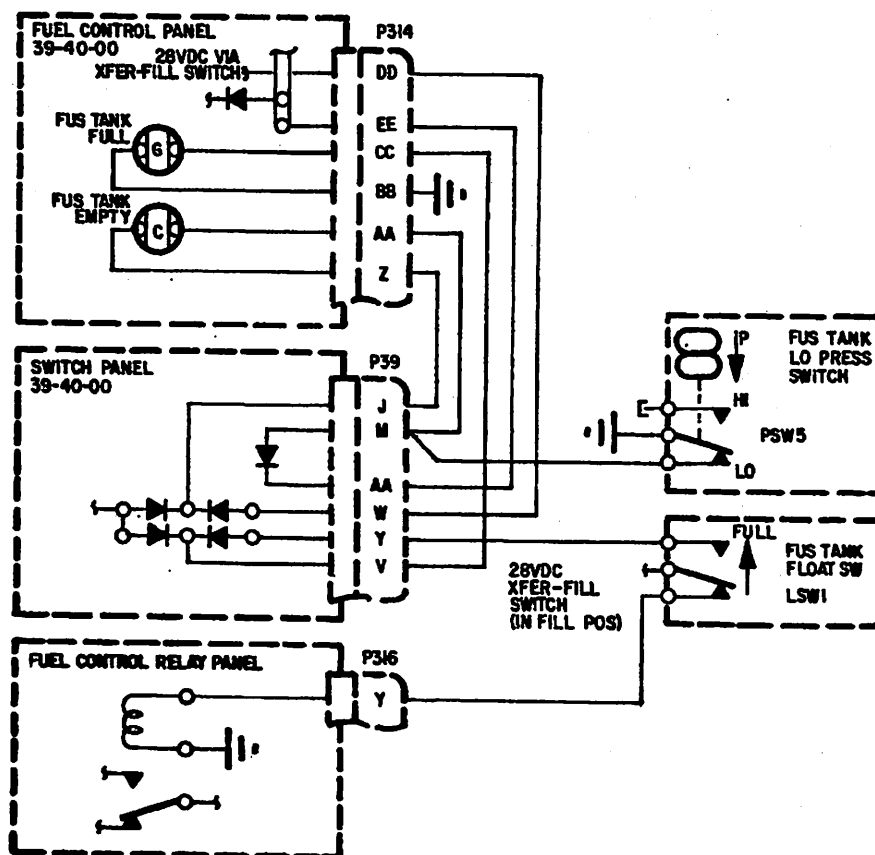
FUSELAGE FLOAT SWITCH - DESCRIPTION AND OPERATION

1. Description

- A. On 25B/D Aircraft, a float switch is installed in the aft area of the fuselage tank.
- B. On 25C/F Aircraft, a float switch is installed in the forward upper RH side of the fuselage tank.

2. Operation

- A. On 25B/D Aircraft, the float switch, when actuated, closes the transfer valve located in the fill and transfer line and the crossflow valve. The float switch also deenergizes the wing tank standby boost pumps and lights the TANK FULL indicator.
- B. On 25C/F Aircraft, the float switch, when actuated, closes the transfer valves located in both refuel and transfer lines. The float switch also deenergizes the wing tank standby boost pumps and lights the TANK FULL indicator.



Fuselage Float Switch Electrical Control Schematic
Figure 1

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FUSELAGE FLOAT SWITCH - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Float Switch (25B/D Aircraft) (See figure 201)

WARNING: OBSERVE ALL SAFETY PRACTICES DESCRIBED IN 28-00-00. SPECIAL CARE SHOULD BE TAKEN FOR PROPERLY VENTILATING THE FUEL CELLS PRIOR TO PERFORMING MAINTENANCE PROCEDURES.

- (1) Defuel fuselage tank.
- (2) Lower tailcone access door.
- (3) Disconnect float switch electrical wiring at splice. Tag wiring.
- (4) Remove attaching parts, plate, gasket and float switch from access cover.

B. Install Float Switch (25B/D Aircraft) (See figure 201)

- (1) Assemble float switch, plate, and gasket. Install assembly and secure with attaching parts.
- (2) Connect float switch electrical wiring at splices.
- (3) Refuel fuselage tank and check for leaks.
- (4) Secure tailcone access door.

C. Remove Float Switch (25C/F Aircraft) (See figure 201)

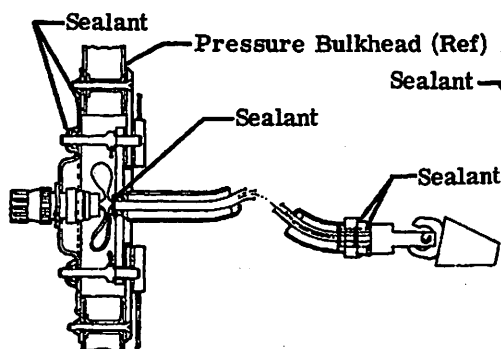
WARNING: OBSERVE ALL SAFETY PRACTICES DESCRIBED IN 28-00-00. SPECIAL CARE SHOULD BE TAKEN FOR PROPERLY VENTILATING THE FUEL CELLS PRIOR TO PERFORMING MAINTENANCE PROCEDURES.

- (1) Defuel the fuselage tank.
- (2) Remove interior lining forward of frame 18. Remove cabin air conditioning components and electrical equipment.
- (3) Remove sealant from float switch wiring that goes thru cover plate. On Aircraft 25-160 and Subsequent, disconnect electrical connector.
- (4) Remove sealant, attaching parts, and float switch coverplate from aft pressure bulkhead.
- (5) Remove attaching parts and float switch support from aft pressure bulkhead.
- (6) Remove sealant from nut securing float switch to support tube.
- (7) Disconnect wiring at splices or from electrical connector and remove float switch from support.

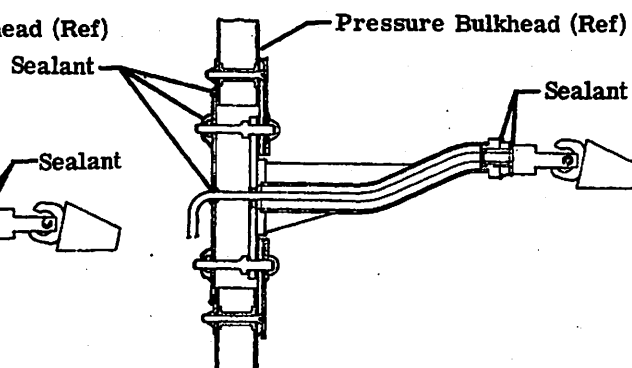
D. Install Float Switch (25C/F Aircraft) (See figure 201)

- (1) Position float switch on support as shown and secure with nut.
- (2) Apply sealant, in accordance with Chapter 20, and seal around perimeter of nut and float switch.

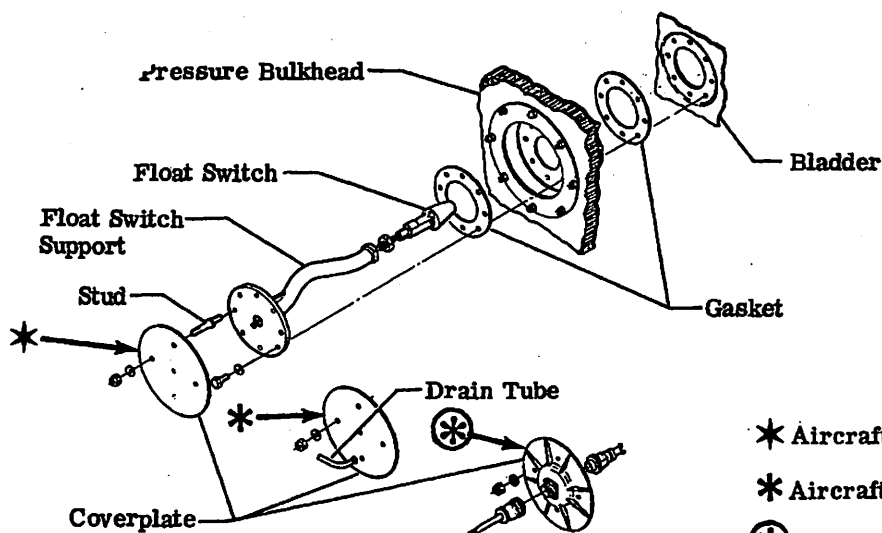
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Aircraft 25-160 and Subsequent



Aircraft 25-070 thru 25-159

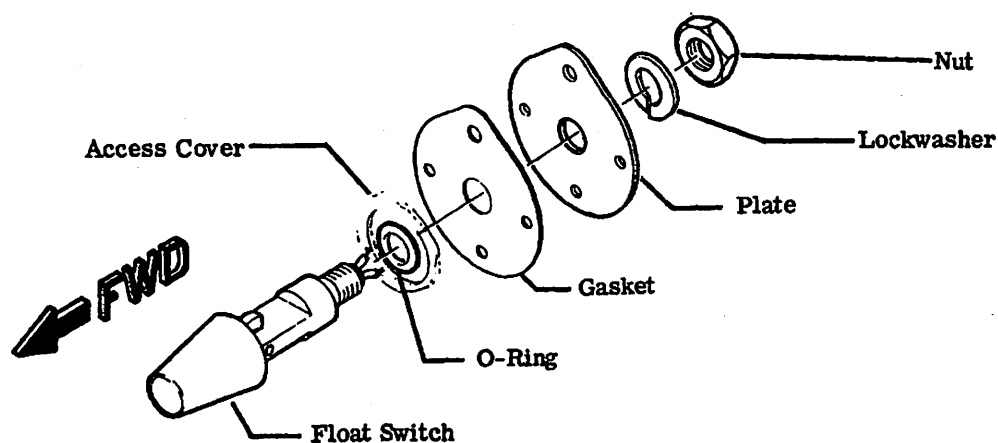


* Aircraft 25-072 thru 25-159

* Aircraft 25-070 thru 25-071

(*) Aircraft 25-160 and Subsequent

25C/F AIRCRAFT



25B/D AIRCRAFT

**Float Switch Installation
Figure 201**

EFFECTIVITY: ALL

MM-98

**28-13-04
Page 202
Dec 2/77**

Gates Learjet Corporation
maintenance manual

- (3) Connect electrical wiring at splices or to electrical connector.
- (4) Install float switch support and secure with attaching parts.
- (5) Install coverplate and secure with attaching parts. Apply sealant around perimeter of coverplate, nuts, and electrical wiring (if applicable). On Aircraft 25-160 and Subsequent, connect electrical connector.
- (6) Install previously removed equipment.

EFFECTIVITY: ALL

MM-98

28-13-04
Page 203
Dec 18/81



GATES LEARJET CORPORATION

maintenance manual

FUEL BOOST PUMP - DESCRIPTION AND OPERATION

1. Description

- A.** An electric fuel boost pump is installed in the left fuselage tank. The fuel pump is a submerged-type with a maximum output of 18 psi. The pump is used to transfer fuel from the fuselage tank to the wing tanks. The pump, powered by 28 vdc, is controlled by the Fuselage Transfer Switch on the fuel control panel.

2. Operation

- A.** During the fill cycle the boost pump is deenergized. When the Fuselage Transfer Switch is set to XFER the pump is energized and transfers fuel to the wing tank. When fuel transfer is completed the Fuselage Transfer Switch must be set to OFF to deenergize the pump.
- B.** A pressure switch (reference 28-12-02) sensing fuel pressure in the right wing, will deenergize the fuselage pump if fuel pressure reaches 5 psi. The pump will reenergize when the pressure drops below 2.50 psi.

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maintenance manual

FUEL BOOST PUMP - MAINTENANCE PRACTICES

1. **General**

A. Maintenance practices consist of removal and installation of fuel pump.

2. **Removal/Installation**

A. Remove Fuel Boost Pump (See figure 201)

- (1) Defuel the fuselage fuel tank.
- (2) Disconnect aircraft batteries.
- (3) Remove lower fuselage access panels to gain access to the fuel pump installation.
- (4) Disconnect electrical wiring from pump.

CAUTION: USE CARE NOT TO DAMAGE THE TAPE AND FILLET SEAL AROUND BLADDER NIPPLES WHEN REMOVING CROSSOVER TUBE.

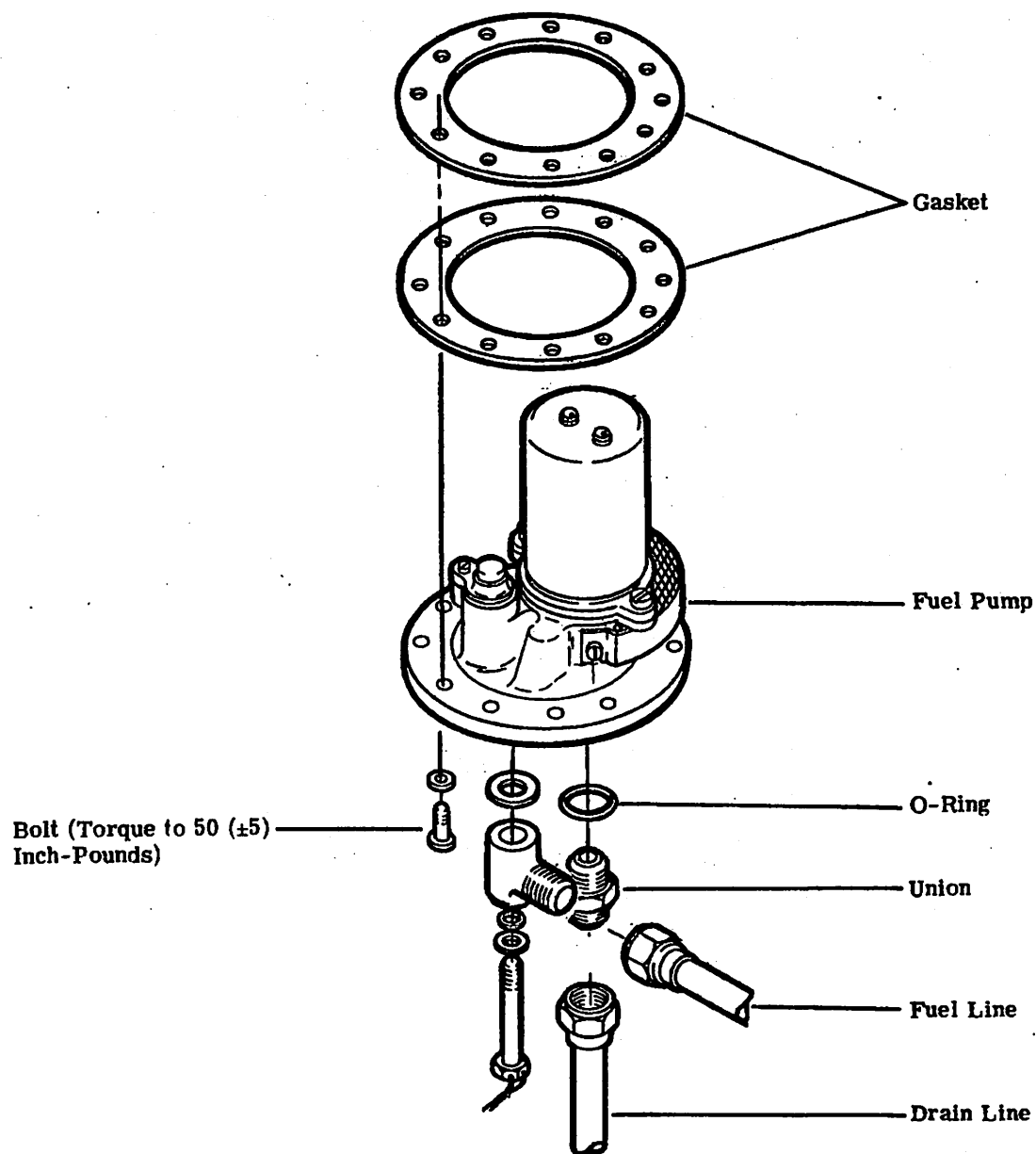
- (5) Loosen clamps securing bladder nipples to crossover tube. Free crossover tube from nipples and remove from aircraft.
- (6) Disconnect and remove fuel line between the pump and transfer valve.
- (7) Remove attaching parts and fuel pump from aircraft.

B. Install Fuel Boost Pump (See figure 201)

NOTE: Replace gaskets at each fuel pump installation.

- (1) Position gasket between fuel cell fitting and fuel cell cavity floor.
- (2) Place remaining gasket over mounting flange of pump.
- (3) Secure with attaching parts. Torque bolts 50 ($\pm$ 5) inch-pounds and safety wire.
- (4) Connect fuel line between the pump and transfer valve.
- (5) Install fuel tank crossover tube and secure with clamps.
- (6) Connect electrical wiring and battery quick-disconnects.
- (7) Fuel the fuselage tank and inspect for leaks.
- (8) Install lower access panels.

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Fuel Boost Pump Installation
Figure 201

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FUEL VENT SYSTEM - DESCRIPTION AND OPERATION

1. General

- A. The fuel vent system provides continuous ram air pressure to the tip tanks, wing tanks and the fuselage tank while the aircraft is in flight. Components of the vent system are: float valves, pressure relief valves, sump assembly, drain valve, vacuum relief valve, ram air-scoops, flame arrestors and vent tubing.
- B. For removal and installation of pressure relief valves, refer to 28-11-04.
- C. For removal and installation of vacuum relief valves, refer to 28-11-05.

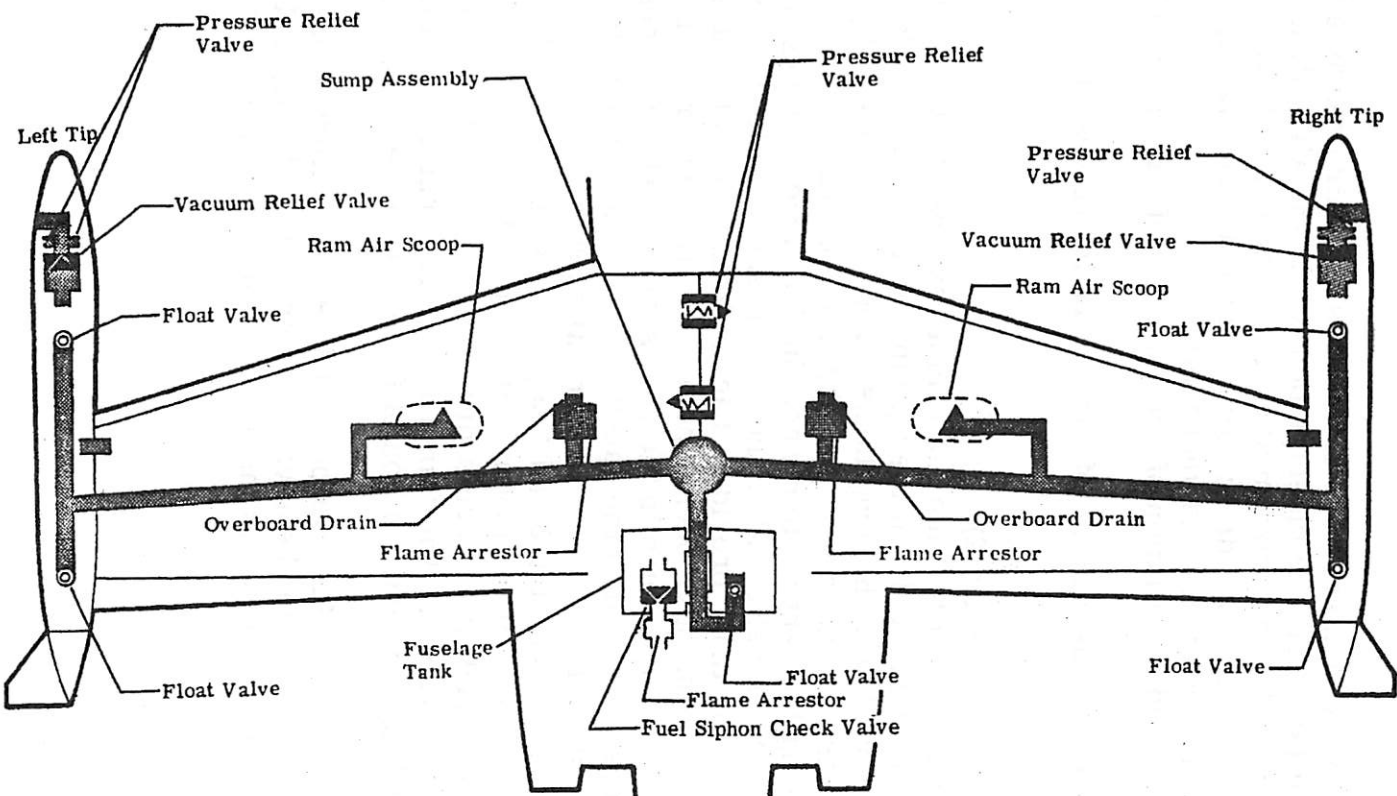
2. Operation (See figures 1 and 2)

- A. In flight, air is admitted into the ram air-scoops located under each wing through tubing and into the tip tanks and fuselage tank. Ram air pressure is vented to the wing tanks through the wing-to-tip tank interconnects. Float valves, located in the tip tanks, prevent fuel from entering the vent system when the fuel level reaches the vent ports. A vacuum relief valve and two pressure relief valves, located in each tip tank, provide a backup to the normal vent system. The vacuum relief valve opens and allows air to enter the tank when a vacuum condition occurs in the vent system. The two pressure relief valves, attached to the vacuum relief line, protect the tanks from excessive pressure due to thermal expansion. The pressure relief valves are set at 1.0 and 1.5 psi. A sump assembly and a drain valve located at the lower centerline of the fuselage and also drain tubes in each wing permit draining of moisture and/or fuel from the vent system. The flame arrestors, installed in the over-board vent line, prevent ignition of fumes in the vent system due to lightning strike.
- B. On 25B/D Aircraft a fuel siphon vent line is installed in the tailcone section. The vent line check valve is located on the upper LH access cover. A flame arrestor is installed in the fuel siphon vent line.



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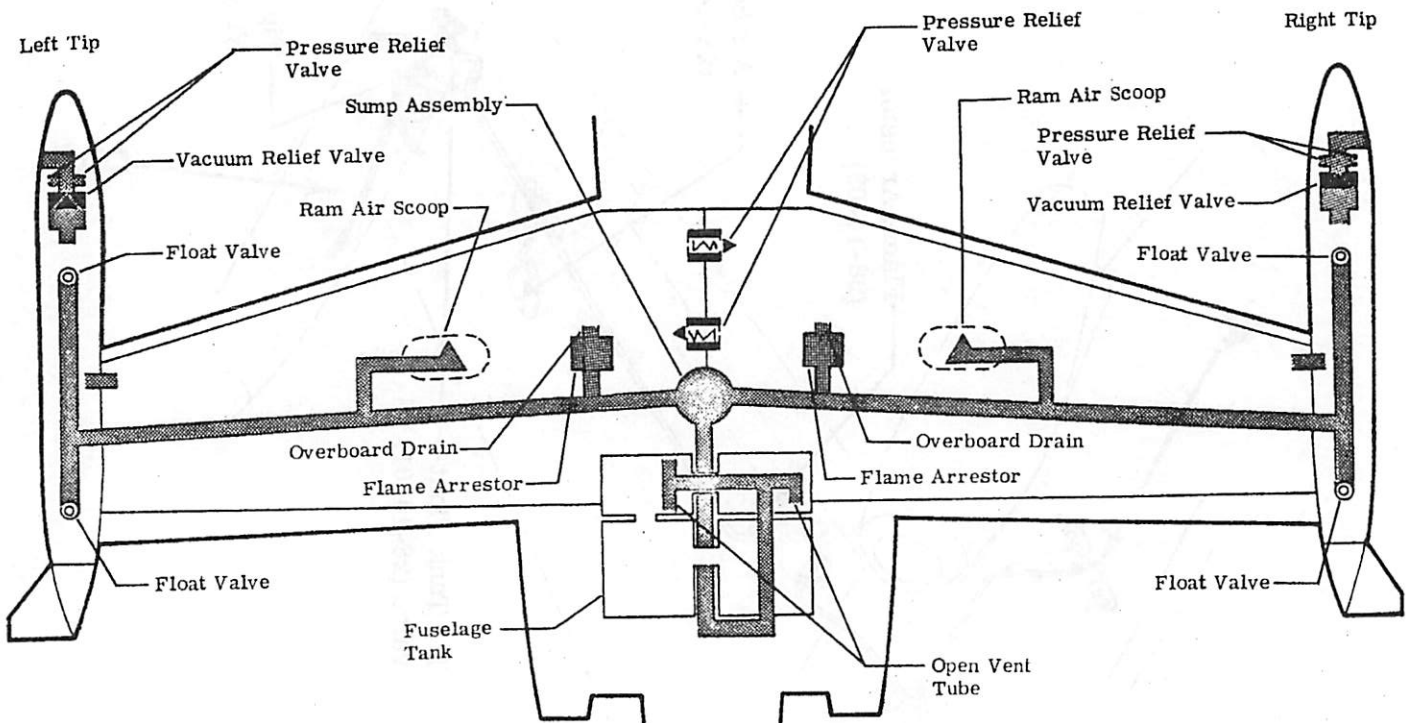
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Fuel Vent System Schematic
Figure 1 (Sheet 1 of 2)

EFFECTIVITY: 25B/D AIRCRAFT

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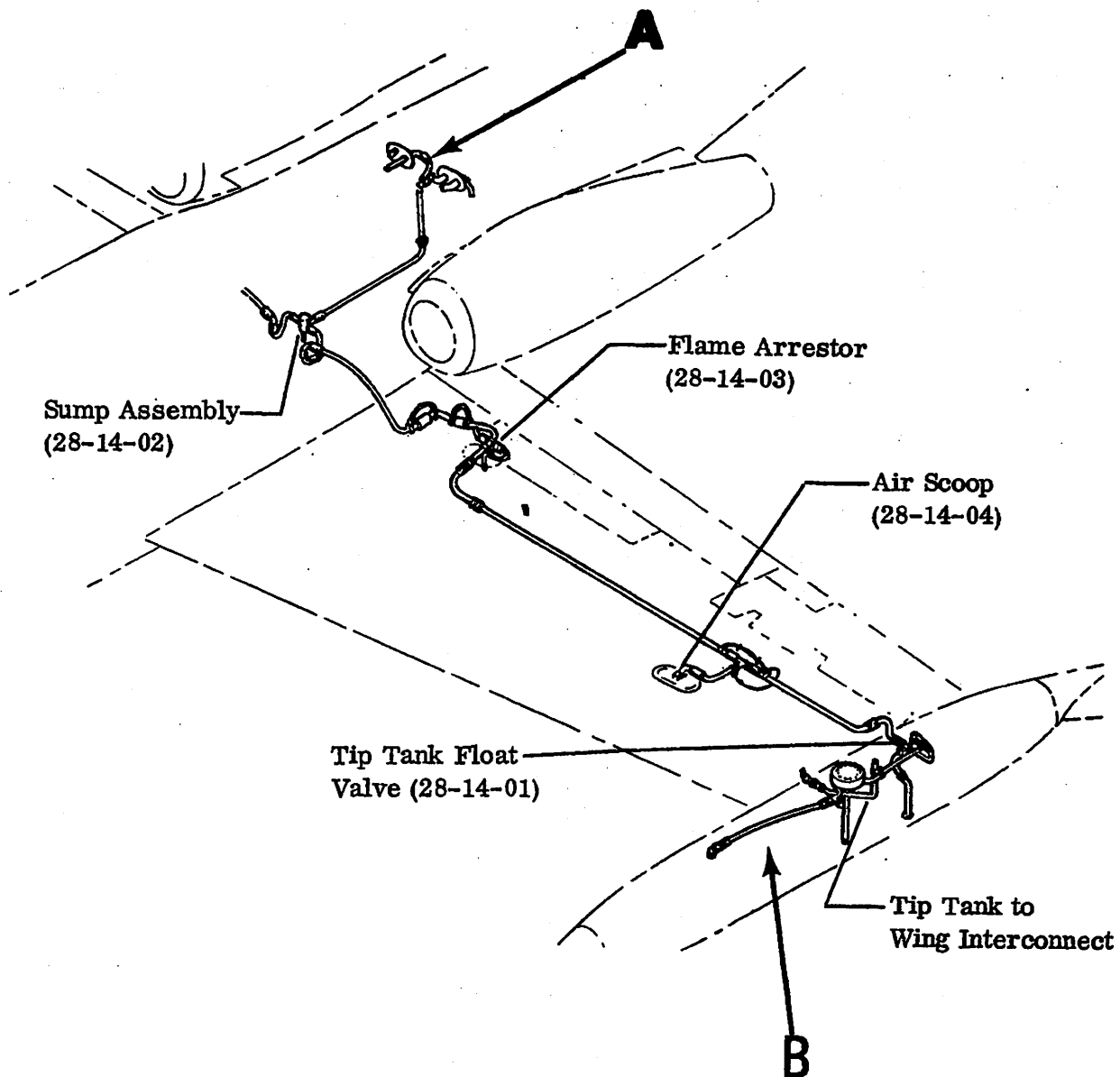


Fuel Vent System Schematic
Figure 1 (Sheet 2 of 2)

EFFECTIVITY: 25C/F AIRCRAFT

MM-98

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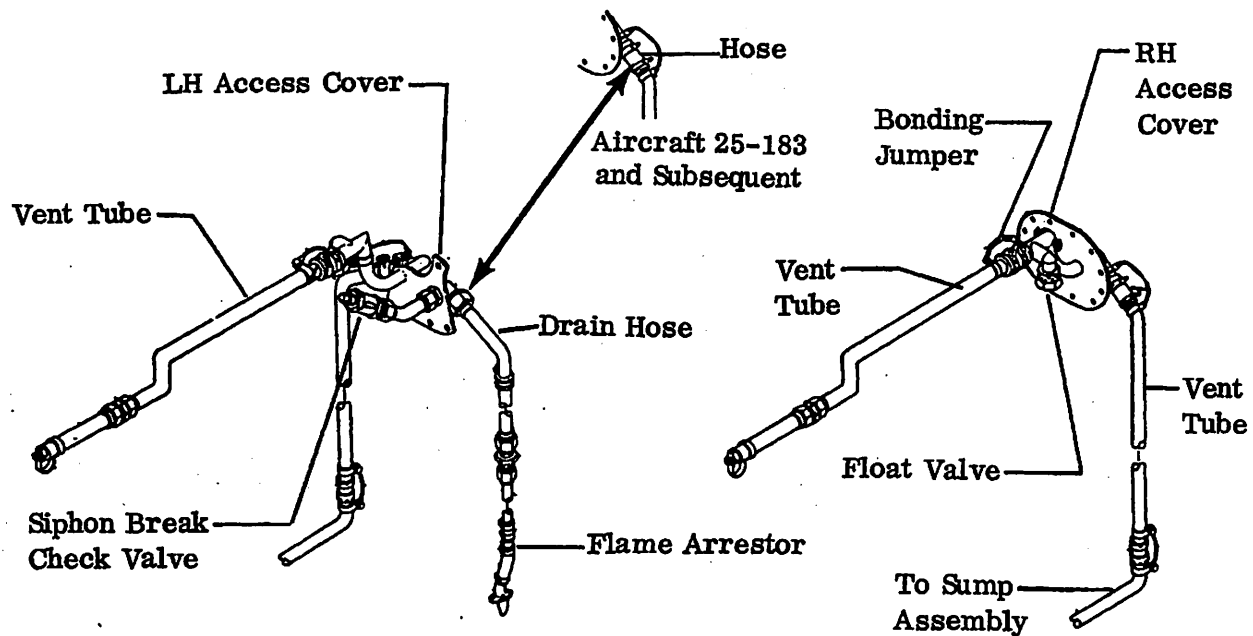
Fuel Vent System Installation
Figure 2 (Sheet 1 of 3)

EFFECTIVITY: ALL

MM-98

28-14-00
Page 4
Dec 18/81

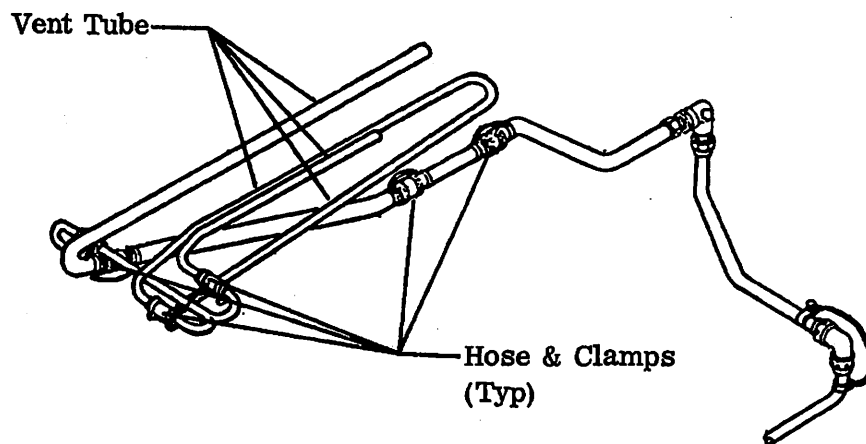
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maintenance manual



Aircraft 25-155 and Subsequent

25-061, 25-070 thru 25-154

25B/D Aircraft



25C/F Aircraft

Detail A

Fuel Vent System Installation
Figure 2 (Sheet 2 of 3)

EFFECTIVITY: ALL

MM-98

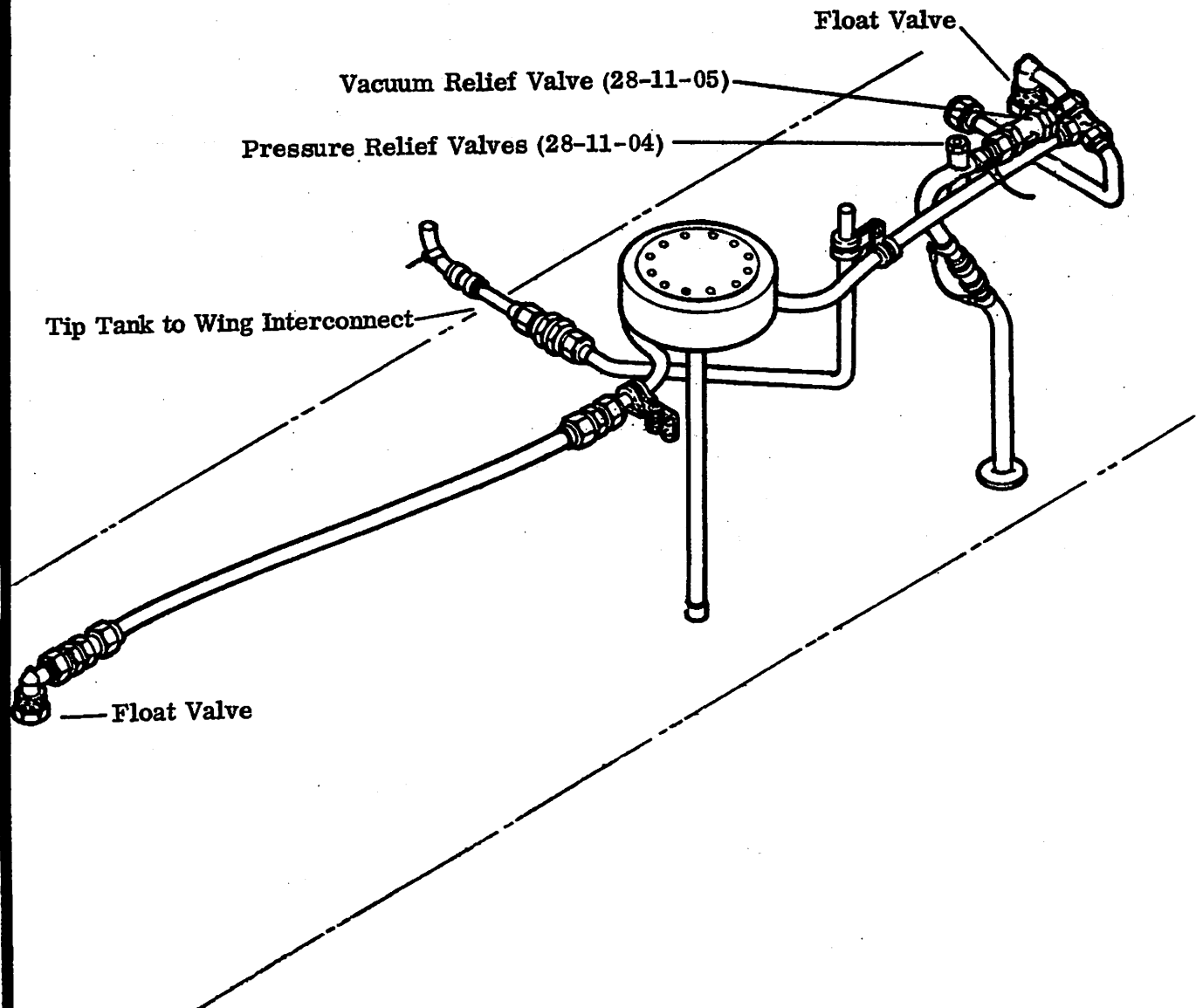
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Page 5

Dec 18/81

Gates Learjet Corporation

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Detail B

Fuel Vent System Installation
Figure 2 (Sheet 3 of 3)

A-12-13C-3

EFFECTIVITY: ALL

MM-98

28-14-00
Page 6
Dec 18/81

GATES LEARJET CORPORATION
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FUEL VENT SYSTEM - MAINTENANCE PRACTICES

1. General

- A. Maintenance practices consist of removal and installation of the system components.**
- B. The removal and installation procedures are outlined in the applicable components section.**
- C. Prior to performing maintenance practices on the aircraft fuel vent system, maintenance personnel should read, thoroughly understand and carefully adhere to the safety practices as outlined in 28-00-00.**

2. Adjustment/Test

A. Functional Test Fuel Vent System Check Valves

- (1) Blow through fuselage tank and tip tank vacuum relief valves, assure that vent lines and vacuum relief valves are not plugged. This indicates that vacuum relief valves are opening properly.**
- (2) Apply suction to fuselage tank and tip tank vacuum relief valves, assure that airflow is stopped. This indicates that vacuum relief valves are closing properly.**



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FLOAT VALVE - MAINTENANCE PRACTICES

1. General

- A. On 25B/D Aircraft two float valves are located in each tip tank and one in the fuselage tank.
- B. On 25C/F Aircraft two float valves are located in each tip tank.

2. Description

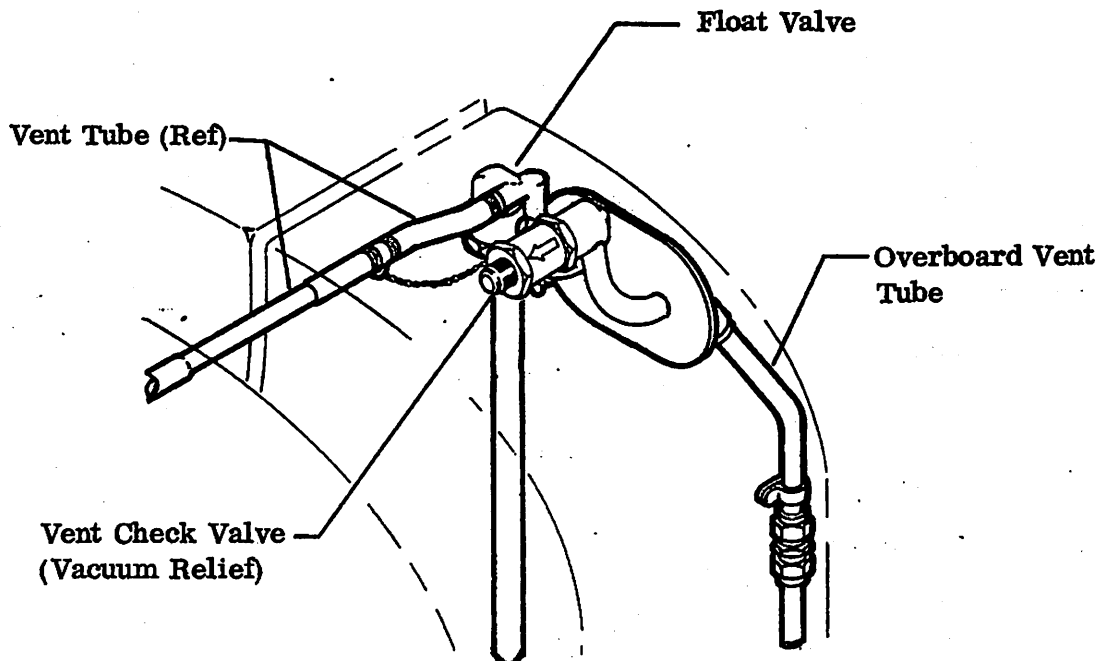
The float valve consists of a housing and a plunger and float assembly incorporating an O-ring. The plunger and float are movable and as fuel level rises the plunger and float move up sealing the vent tube outlet. This prevents fuel from collecting in the vent system.

3. Removal/Installation

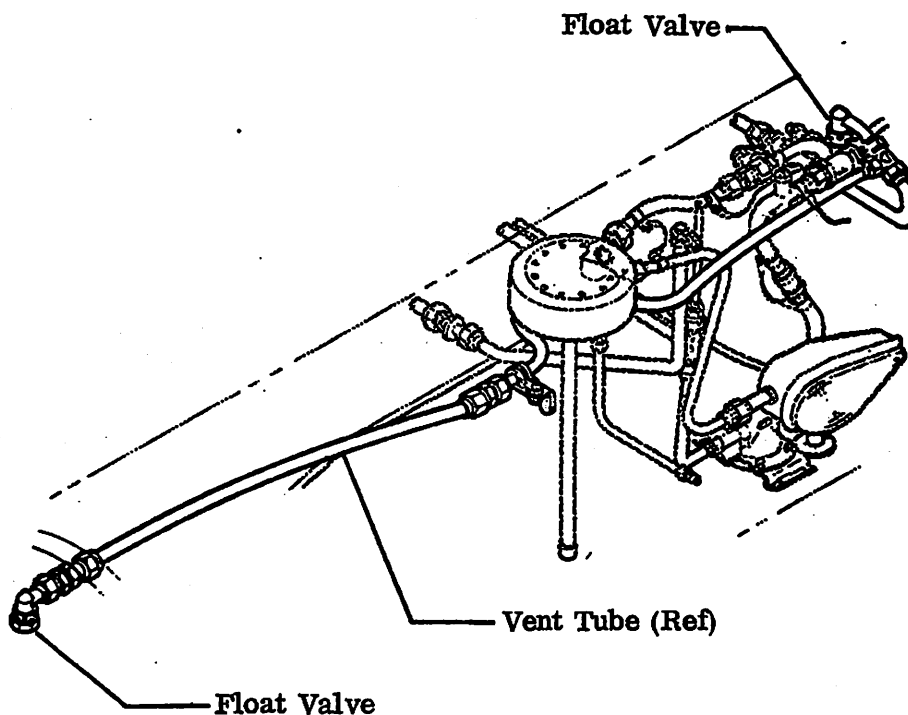
NOTE: Steps A and B apply to either model. Steps C and D apply only to 25B/D Aircraft.

- A. Remove Tip Tank Float Valves (See figure 201)
 - (1) Defuel the tip tank. (Refer to 12-00-00)
 - (2) Remove access panels to gain access to float installation.
 - (3) Loosen and remove float assembly.
- B. Install Tip Tank Float Valve (See figure 201)
 - (1) Install float assembly.
 - (2) Apply a thin coat of anti-corrosion grease to access covers (refer to 28-11-00, figure 202)
 - (3) Install gaskets and access covers and secure with attaching parts.
- C. Remove Fuselage Tank Float Valve (See figure 201)
 - (1) Defuel the fuselage fuel tank. (Refer to 12-00-00)
 - (2) Lower tailcone access door.
 - (3) Disconnect vent tube from upper RH access panel.
 - (4) Disconnect fuel plumbing from equipment installed on large RH access panel.
 - (5) Remove attaching parts and large access panel.
 - (6) Loosen and remove float valve from vent line.
- D. Install Fuselage Tank Float Valve (See figure 201)
 - (1) Install float valve assembly on vent line.
 - (2) Install access cover.
 - (3) Connect plumbing to equipment previously removed.
 - (4) Connect vent tube to upper RH access cover.
 - (5) Secure tailcone access door.

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maintenance manual



Fuselage Float Valve Installation (25B/D Aircraft)



Tip Tank Float Valve Installation

**Float Valve Installation
Figure 201**

EFFECTIVITY: ALL.

**28-14-01
Page 202
Dec 2/77**

Gates Learjet Corporation
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4. Approved Repairs

A. Float Valve Flapper Replacement

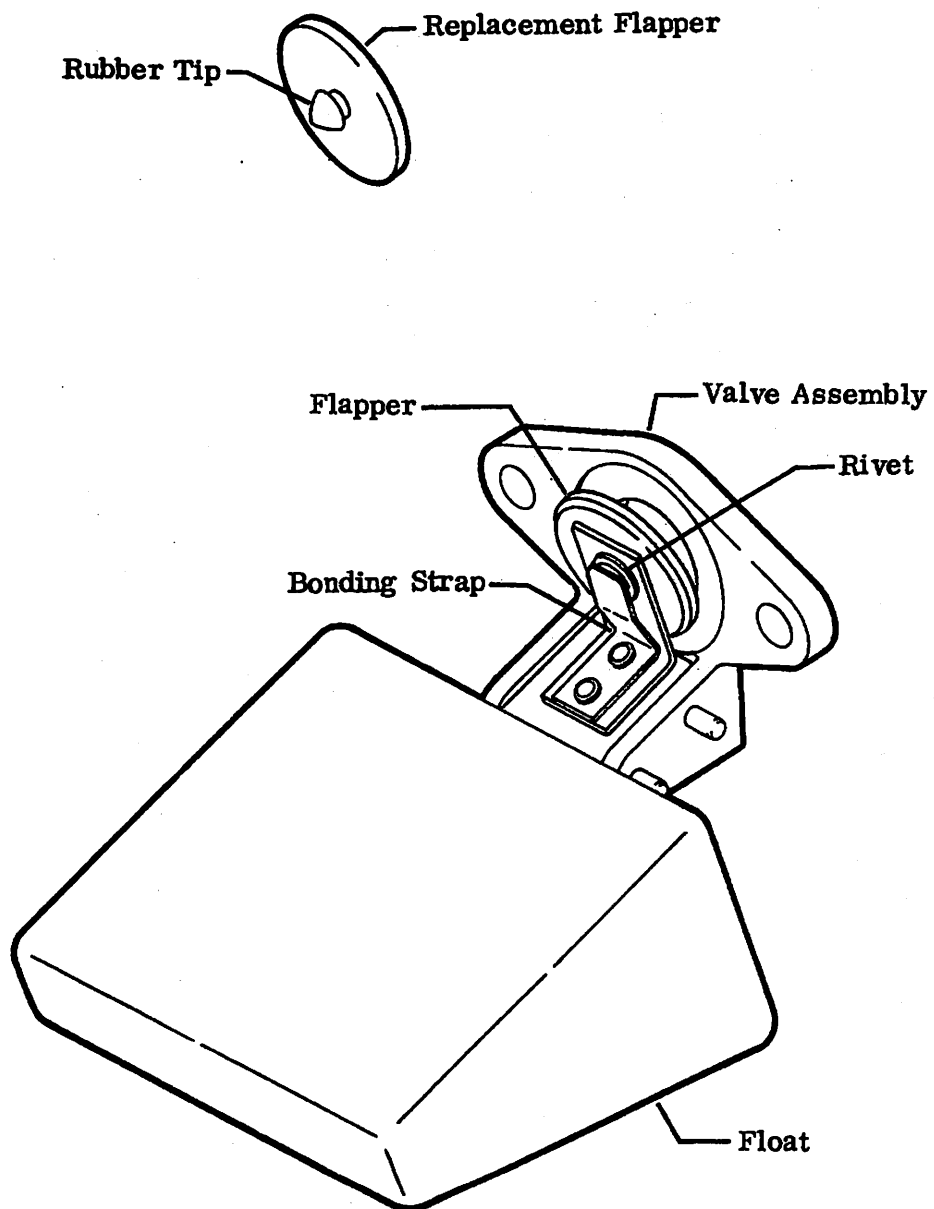
NOTE: Applies to aircraft with 6600346-1 or -2 valves installed.

- (1) Remove applicable float valve as described in paragraph 3.A. or 3.C. (See figure 202.)
- (2) Drill out rivet attaching flapper to valve assembly and discard flapper.
- (3) Cut bonding strap away from rivets attaching it to valve assembly and discard.
- (4) Insert rubber tip on replacement flapper through rivet hole in valve assembly.

NOTE: Make certain rubber tip on flapper is completely through valve assembly and seated flush.

- (5) Install float valve as described in applicable paragraph 3.B or 3.D.

Gates Learjet Corporation
maintenance manual



12-106A

Fuel Vent Valve Flapper Replacement
Figure 202

EFFECTIVITY: ALL

MM-98

28-14-01
Page 204
May 25/84

GATES LEARJET CORPORATION
maintenance manual

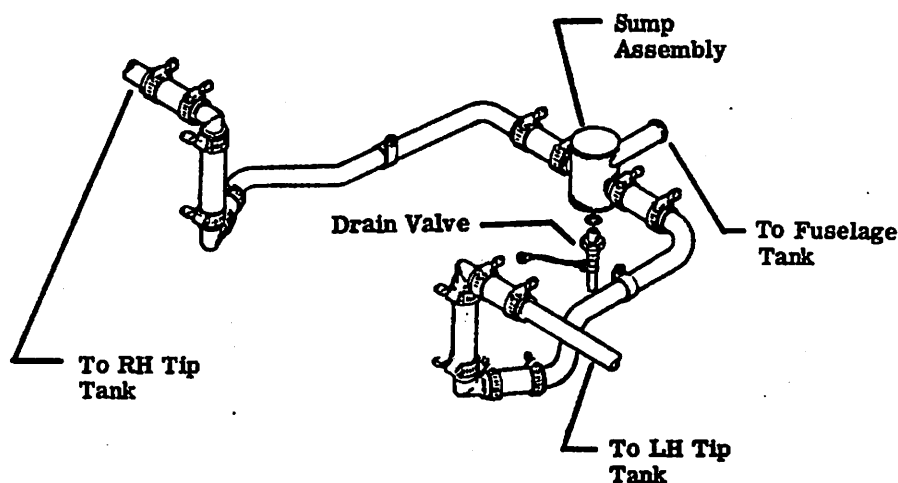
SUMP ASSEMBLY - MAINTENANCE PRACTICES

1. General

- A. The sump assembly is installed in the aft portion of the keel beam. The sump is the lowest point in the vent system and collects any accumulation of moisture or fuel. The sump incorporates a manual drain valve to drain any accumulation of moisture or fuel.

2. Removal/Installation

- A. Remove Sump Assembly (See figure 201)
(1) Remove lower fuselage access panel.
(2) Loosen clamps securing bonding jumpers and hoses.
(3) Remove sump assembly from aircraft.
- B. Install Sump Assembly (See figure 201)
(1) Install sump assembly.
(2) Connect hoses and bonding jumpers and secure with clamps.
(3) Install access cover.



Sump Assembly Installation
Figure 201

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FLAME ARRESTOR - MAINTENANCE PRACTICES

1. General

- A.** The flame arrestor is installed in the fuel vent system at approximately wing station 53.04 between spars 7 and 8. Access to the flame arrestor is through an access panel at the same location. The flame arrestor prevents ignition of fumes in the vent due to lightning following the overboard drain.

2. Removal/Installation

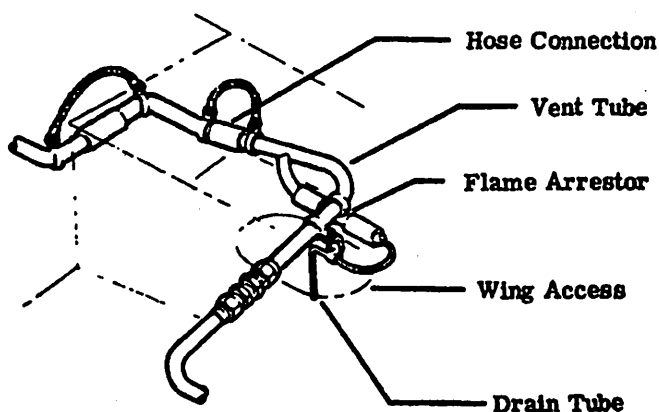
NOTE: The following procedure is applicable to either flame arrestor.

A. Remove Flame Arrestor (See figure 201)

- (1) Remove access panel.
- (2) Loosen and remove clamps, hoses, overboard drain line and flame arrestor from aircraft.

B. Install Flame Arrestor (See figure 201)

- (1) Install flame arrestor and overboard drain line and secure with hoses and clamps. Assure the drain line protrudes past lower wing surface 1.00 (± 0.06) inch.
- (2) Install access cover.



**Flame Arrestor Installation
Figure 201**

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AIRSCOOP - MAINTENANCE PRACTICES

1. General

- A. The air scoop is installed on the wing lower skin at wing station 139.57 between spars 4 and 5. The air scoop is secured to an access cover at the same location.

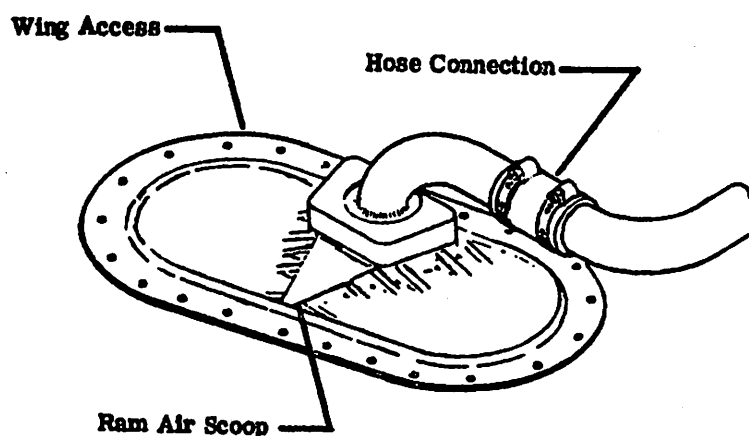
2. Removal/Installation

A. **Remove Airscoop** (See figure 201)

- (1) Defuel the aircraft. (Refer to 12-00-00)
- (2) Remove screws securing air scoop to access cover.
- (3) Remove attaching parts and access covers from wing.

B. **Install Airscoop** (See figure 201)

- (1) Apply lightning strike protective seal to access cover. (Refer to 28-12-00)
- (2) Secure gasket and air scoop to access panel.
- (3) Install access cover and secure with attaching parts.
- (4) Fuel the aircraft and check for leaks.



Airscoop Installation
Figure 201



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DISTRIBUTION - DESCRIPTION AND OPERATION

1. General

- A. The fuel distribution system consists of two independent systems, one for each engine. Each distribution system consists of a jet pump, an electric boost pump, a fuel filter, a shutoff valve, a motive flow valve, a pressure switch and a relief valve.

2. Operation

- A. Fuel System Operation (25B/D Aircraft) (See figures 1 thru 4)

- (1) During normal mode of operation, the Jet Pump switches are set to ON, the Crossflow switch is set to CLOSE, and the Fuselage Transfer switch is set to OFF. When the Starter Generator switch is set to START, the motive flow control valve closes and the standby boost pumps are energized, supplying fuel for engine starting. When the Starter Generator switch is set to GEN position, the standby pumps are deenergized and the motive flow valves open, allowing high pressure fuel for jet pump operation. Under these conditions, the tip tank jet pumps are transferring fuel into their respective wing tanks and the wing tank jet pumps are feeding fuel to their respective engine.

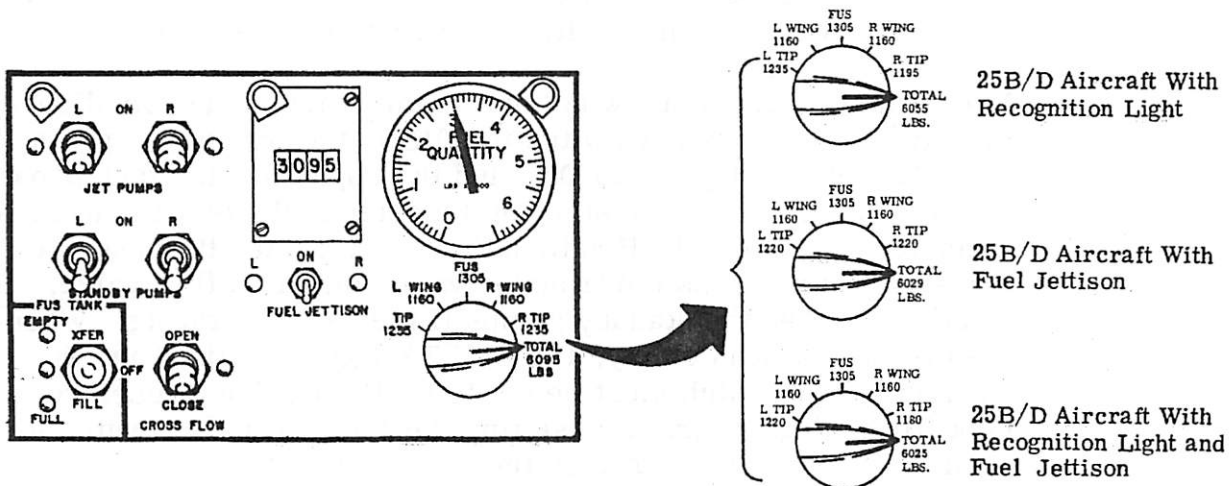
NOTE: During a fuselage fill, fuselage transfer, or fuel crossflow operation, if the crossflow valve or fuselage tank valve position lights are lighted, the valves are not in the position selected by the switches.

Fuel transfer, from one wing tank to the other, is accomplished by setting the Crossflow switch to OPEN, the transferring wing tank standby boost pump to ON, and the opposite wing tank boost pump to OFF. When transferring fuselage fuel, setting the Fuel Transfer switch to XFER automatically energizes the fuselage transfer pump, opens the transfer valve and crossflow valve, and deenergizes the standby pumps if they are operating. When the fuselage tank is empty, the EMPTY light will light and the Fuel Transfer switch must be set to OFF. During a fuselage tank fill mode of operation, setting the Fuel Transfer switch to FILL automatically energizes the standby boost pumps and opens the crossflow valve and fuselage tank valve. When the fuselage tank is full, the FULL light will be lighted and the fill operation will automatically stop. The Fuel Transfer switch must then be set to OFF. In case of jet pump failure, the pressure drop in the main fuel line will cause a pressure sensing switch to light the applicable indicator light (L FUEL PRESS or R FUEL PRESS). The applicable Standby Pump switch

EFFECTIVITY: ALL

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maintenance manual

should immediately be set to ON to restore normal fuel system pressure.



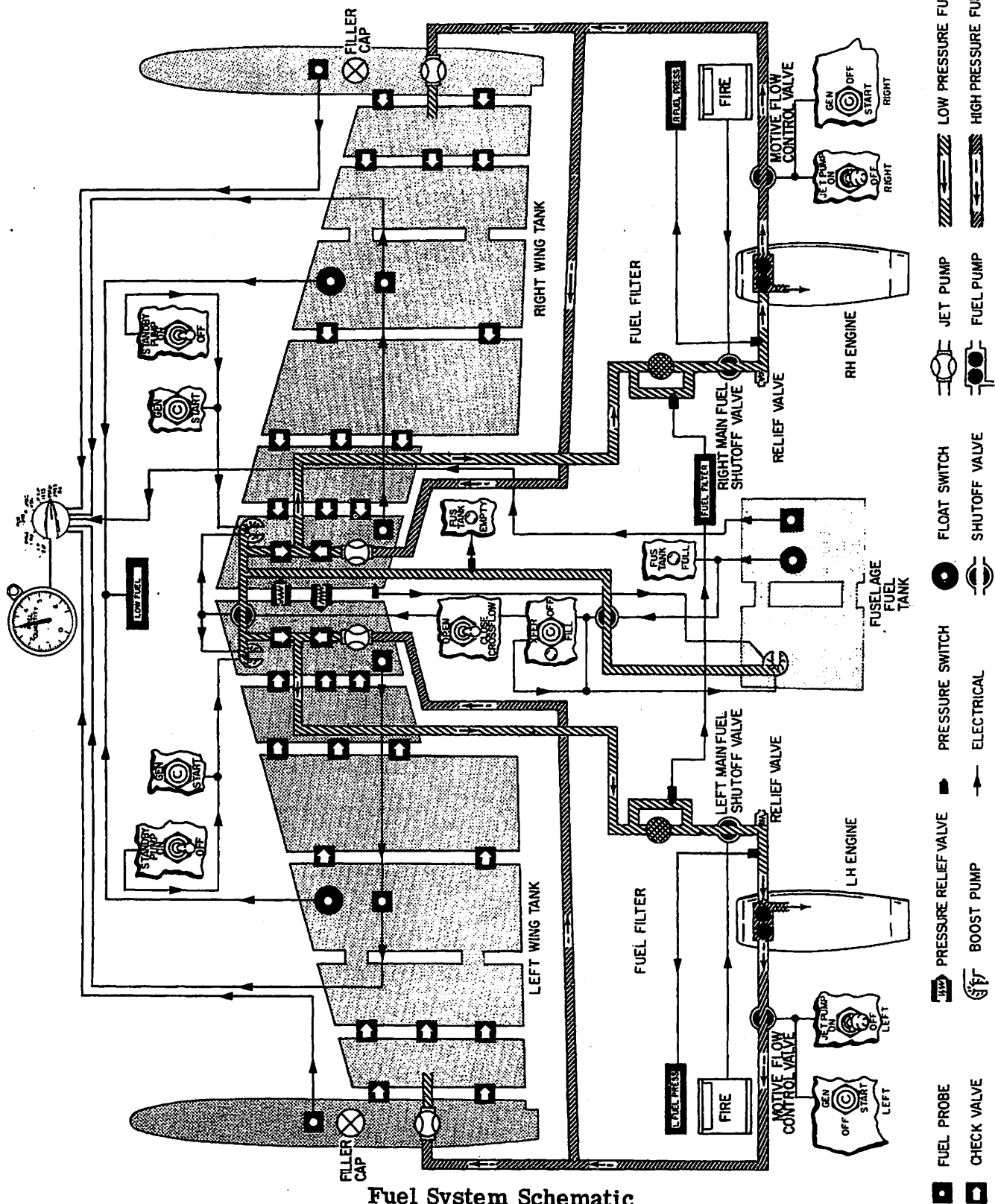
25B/D Aircraft
Fuel Control Panel
Figure 1

EFFECTIVITY: ALL

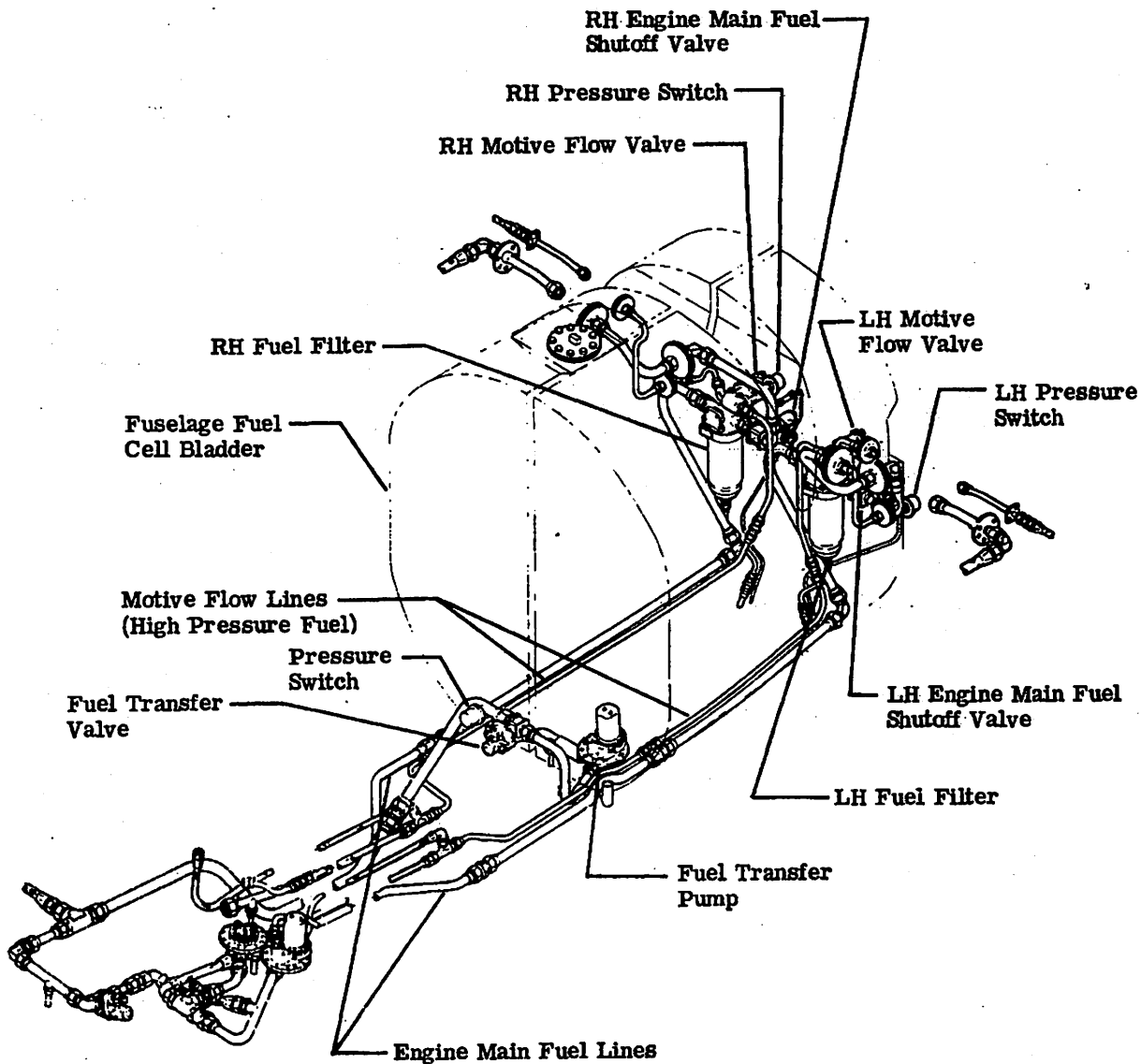
MM-98

28-20-00
 Page 2
 Dec 18/81

 GATES LEARJET CORPORATION
maintenance manual



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maintenance manual



Distribution System Component Locator
 Figure 3

EFFECTIVITY: 25B/D AIRCRAFT

MM-98

28-20-00

Page 4

Dec 18/81



GATES LEARJET CORPORATION

maintenance manual

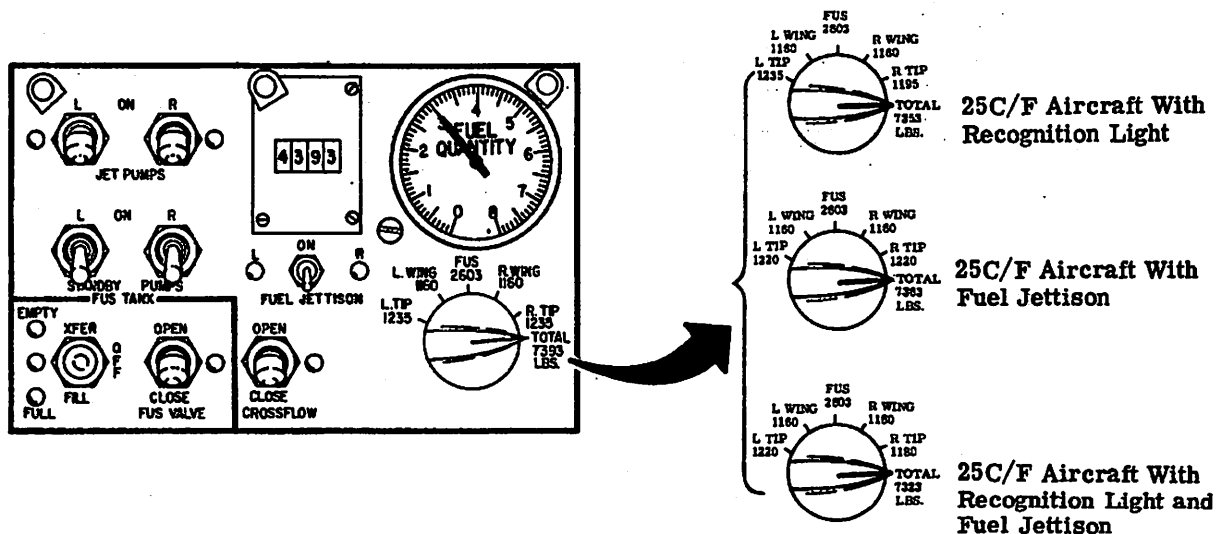
B. Fuel System Operation (25C/F Aircraft) (See figures 5 thru 8)

- (1) During a normal mode of operation, the Jet Pump switches are set to ON, the Crossflow switch is set to CLOSE, and the Fuselage Transfer switch is set to OFF. When the Starter-Generator switch is set to START, the motive flow valve closes and the applicable standby boost pump is energized, supplying fuel for engine starting. When the Starter-Generator switch is set to GEN, the standby pump is deenergized and the motive flow valve opens, allowing high pressure fuel to operate the jet pumps. Under these conditions, the tip tank jet pumps are transferring fuel into their respective wing tanks and the wing tank jet pumps are feeding fuel to their respective engines.

NOTE: During a fill, fuselage transfer, or fuel crossflow operation, if the crossflow valve or fuselage tank valve indicator lights are lighted, the valves are not in the position selected by the switches. In addition, if the jet pump indicator lights are lighted, the motive flow valves are not in the position selected by the switches.

Fuel transfer, from one wing tank to the other, is accomplished by setting the Crossflow switch to OPEN, the transferring wing tank standby boost pump to ON, and the opposite wing tank boost pump to OFF. During a fuselage fuel transfer mode of operation, setting the fuel Transfer switch to XFER automatically energizes the fuselage transfer pump, opens the LH fuselage transfer valve and crossflow valve, and deenergizes the standby boost pumps if they were operating. When the fuselage tank is empty, a pressure switch in the LH fuselage tank transfer line will cause the fuselage tank EMPTY light to light. The Transfer switch must be set to OFF. In case of normal fuel transfer failure, the RH transfer valve can be opened by setting the Fus Valve switch on the Fuel Control panel to OPEN allowing fuel to gravity flow from the fuselage tank to the wing tank. Gravity flow operation is stopped by setting the Fus Valve switch to CLOSE. During a fuselage tank fill mode of operation, setting the fuel transfer switch to FILL automatically energizes the standby boost pumps and opens the crossflow valve and the RH and LH fuselage tank transfer valves. When the fuselage tank is full, the FULL light will be lighted and the fill operation will automatically stop. The transfer switch should then be set to OFF. In case of jet pump failure, the pressure drop in the main fuel line to the engines will cause a pressure switch to energize the applicable indicator light (L FUEL PRESS or R FUEL PRESS). The applicable standby boost pump switch should immediately be set to ON to restore normal fuel system pressure.

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Fuel Control Panel
Figure 5

EFFECTIVITY: 25C/F AIRCRAFT

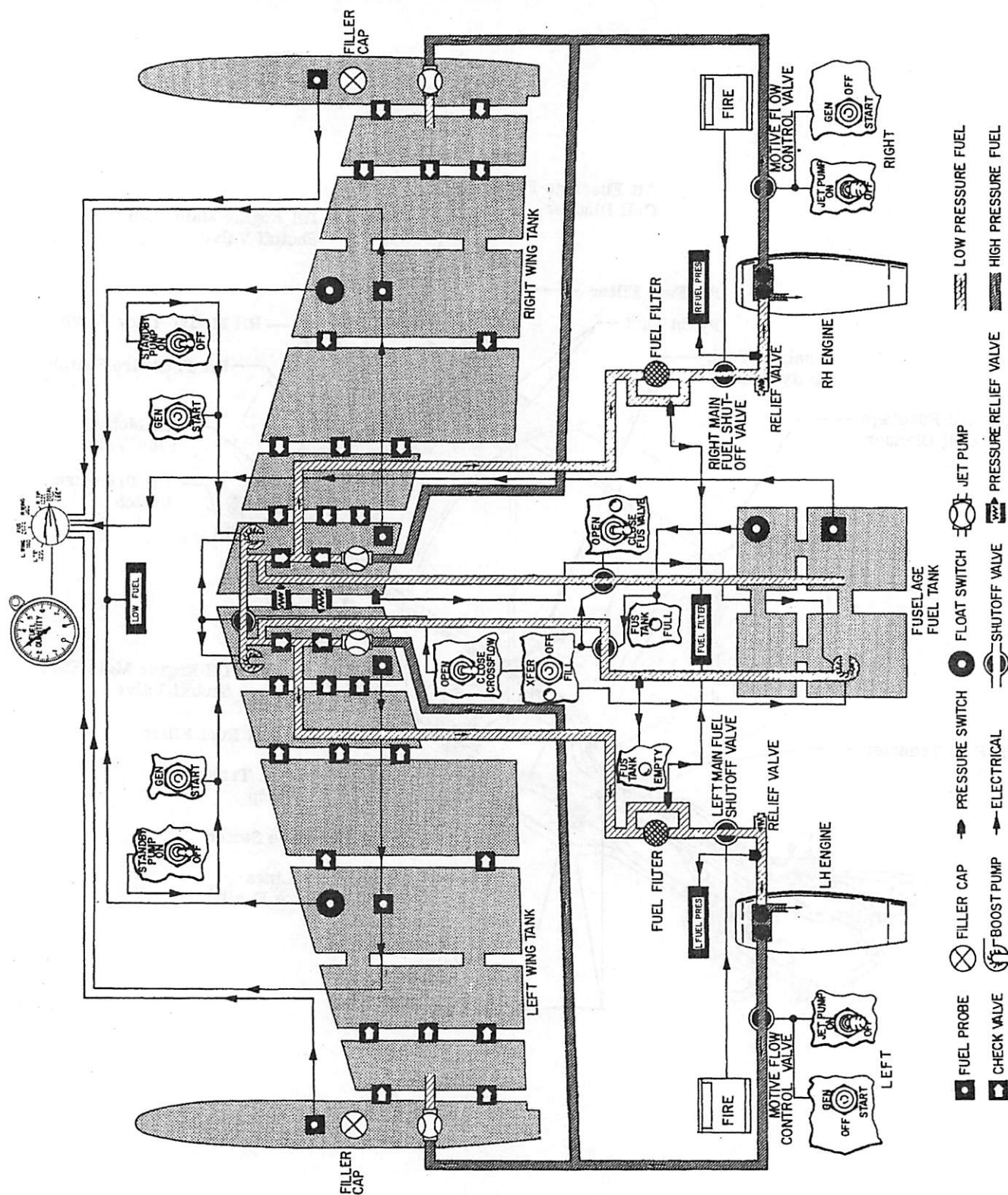
MM-98

28-20-00

Page 8

Dec 18/81

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maintenance manual



Fuel System Schematic
Figure 6

EFFECTIVITY: 25C/F AIRCRAFT

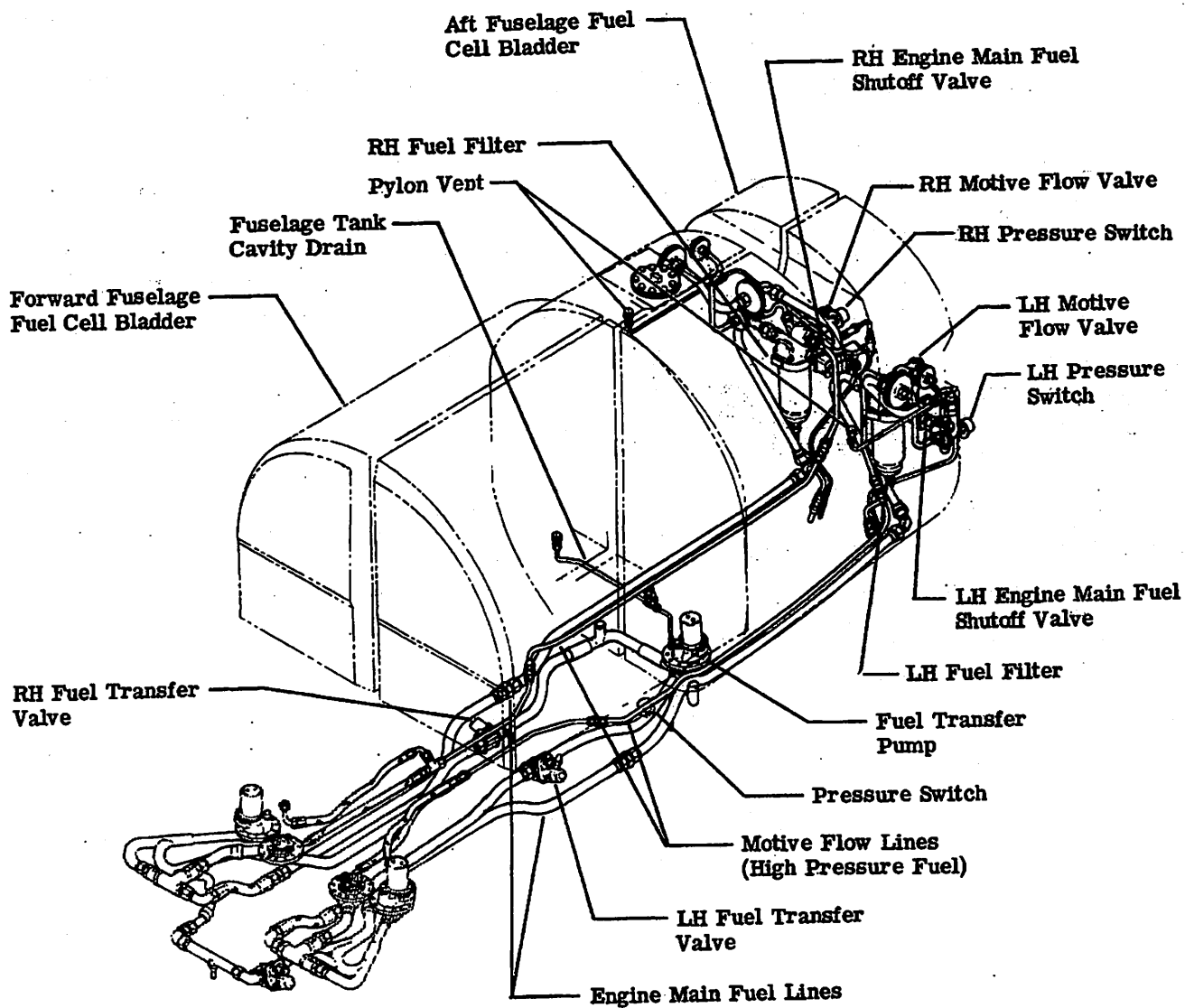
MM-98

28-20-00

Page 9

Dec 18/81

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Distribution System Component Locator
Figure 7

GATEWAY LEARNJET CORPORATION



281 DC  CBI48

1. STAY BUMP

28V DC 0.125A CB147

Figure 8

M24-90

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3. Adjustment/Test

A. Test Engine Fuel Flow System

- (1) Set Battery Switches to ON.
- (2) Set Crossflow Switch to OPEN.
- (3) Set L Standby Pump Switch to ON. L & R FUEL PRESS lights will go out.
- (4) Set L Jet Pump Switch to OFF; then to ON. Indicator light will momentarily illuminate then go out after each setting of the switch. This indicates the motive flow valve has moved to correspond with the switch setting.
- (5) Set Crossflow Switch to CLOSED. R FUEL PRESS light will illuminate and L FUEL PRESS light will remain out.
- (6) Set L Standby Pump Switch to OFF and Crossflow Switch to OPEN.
- (7) Set R Standby Pump Switch to ON. L & R FUEL PRESS lights will go out.
- (8) Set R Jet Pump Switch to OFF; then to ON. Indicator light will momentarily illuminate then go out after each switch setting. This indicates the motive flow valve has moved to correspond with the switch setting.
- (9) Set Crossflow Switch to CLOSED. L FUEL PRESS light will illuminate and R FUEL PRESS light will remain out.
- (10) Set Crossflow Switch to OPEN, R Standby Pump Switch to OFF and Battery Switches to OFF.

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JET PUMP - MAINTENANCE PRACTICES

1. Removal/Installation

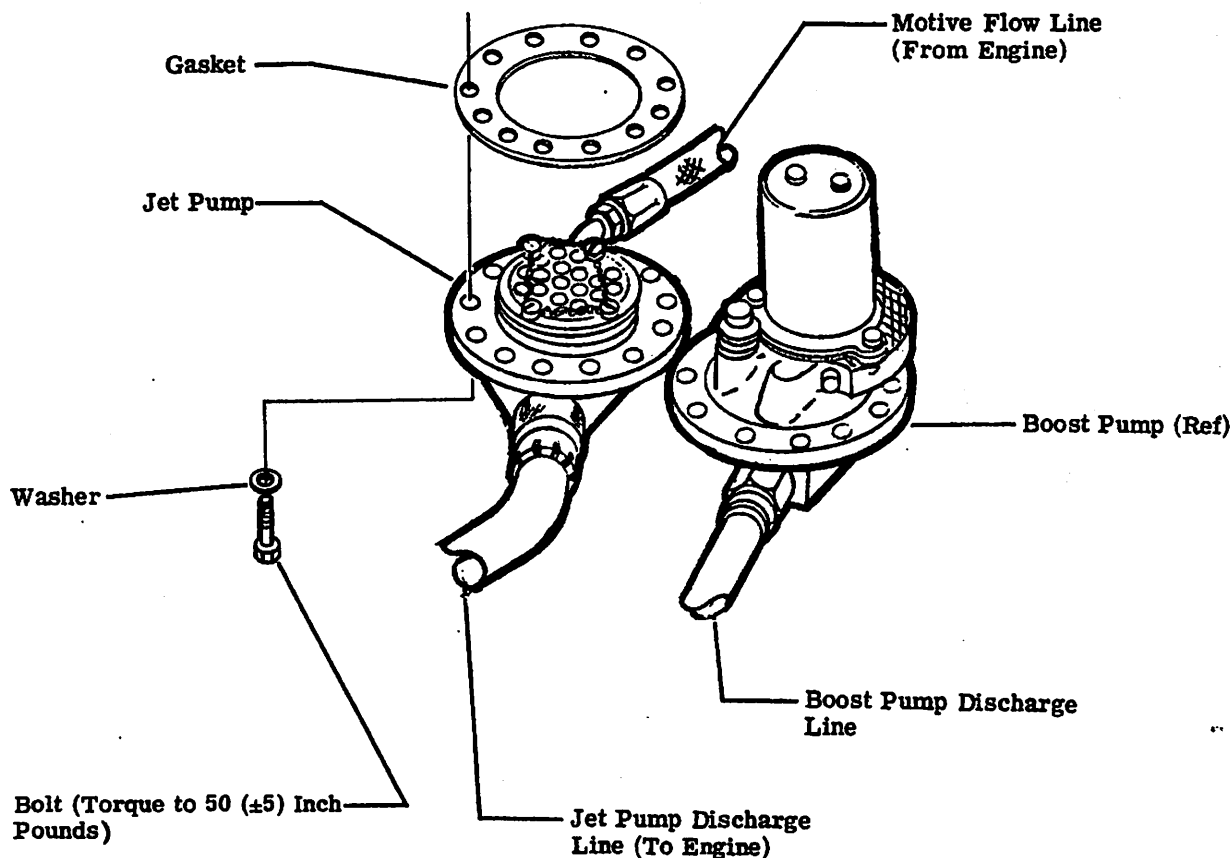
NOTE: The following procedures are applicable for either the LH or RH jet pumps.

A. Remove Jet Pump (See figure 201)

- (1) Defuel the aircraft. (Refer to 12-00-00)
- (2) Remove lower fuselage access panels to gain access to the fuel pump installation.
- (3) Disconnect fuel plumbing lines from pump.
- (4) Remove attaching parts and pump from aircraft.

B. Install Jet Pump (See figure 201)

- (1) Install gasket and jet pump and secure with attaching parts.
- (2) Torque bolts to 50 (± 5) inch-pounds and safety-wire.
- (3) Connect fuel plumbing to pump.
- (4) Fuel the aircraft. (Refer to 12-00-00) Inspect for leaks.
- (5) Install lower fuselage access panels.



**Jet Pump Installation
Figure 201**



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FUEL BOOST PUMP - MAINTENANCE PRACTICES

1. General

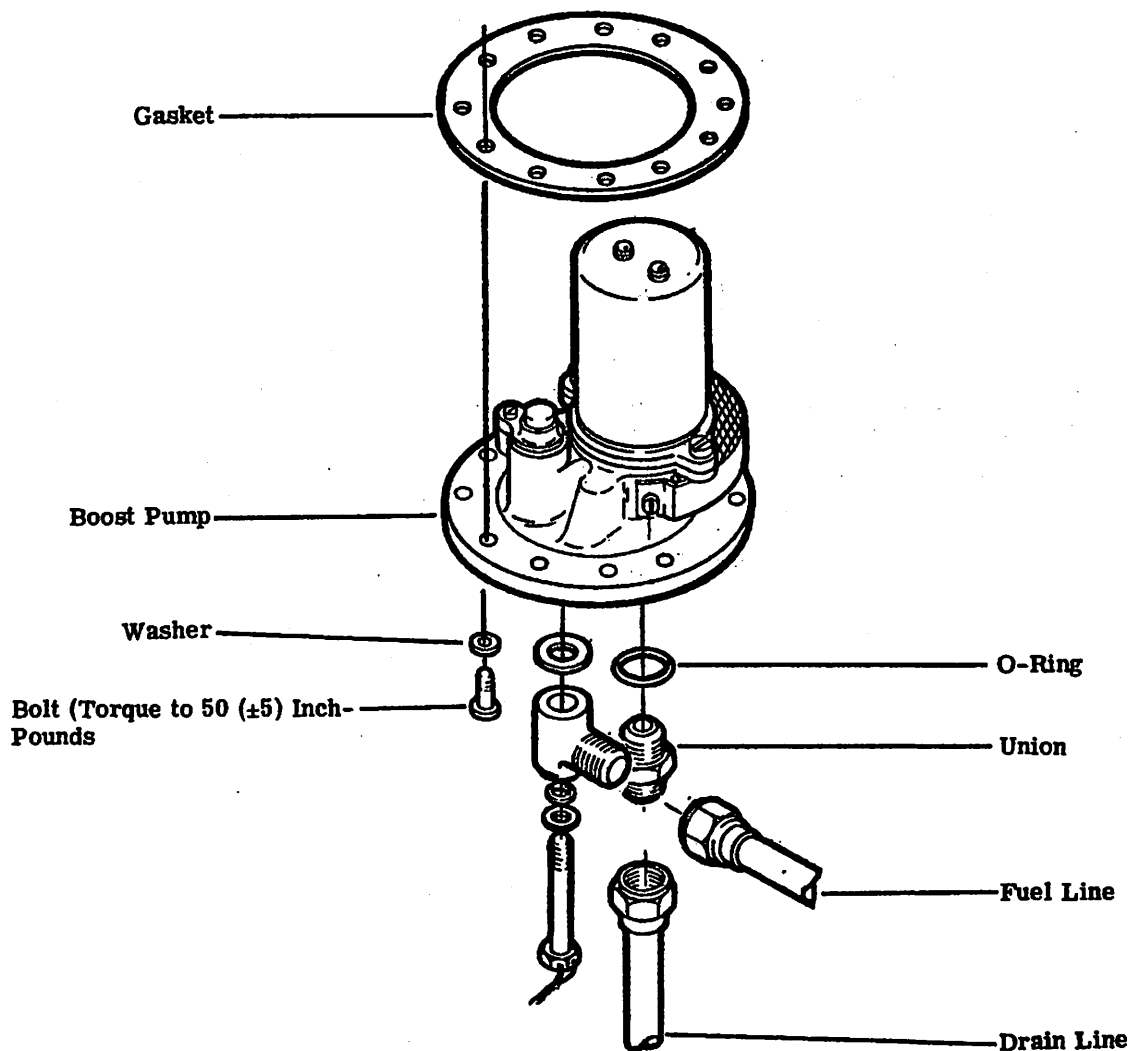
- A. An electric boost pump is installed in each distribution system. The pump is located adjacent to the jet pump. The pump is a submerged-type with a maximum output of 18 psi. The pump is used for fuel crossflow, one wing to the other, fuselage fill operation, for engine starting and as a standby should a jet pump failure occur.

2. Removal/Installation

NOTE: The following procedures are applicable for either the LH or RH boost pump.

- A. Remove Boost Pump (See figure 201)
- (1) Defuel the aircraft. (Refer to 12-00-00)
 - (2) Remove lower fuselage access covers to gain access to the fuel pump installation.
 - (3) Disconnect electrical wiring from pump.
 - (4) Disconnect fuel plumbing from pump.
 - (5) Remove attaching parts and pump from aircraft.
- B. Install Boost Pump (See figure 201)
- (1) Install gasket and boost pump and secure with attaching parts.
 - (2) Torque bolts 50 ($\pm$ 5) inch-pounds and safety wire bolts.
 - (3) Connect fuel plumbing to pump.
 - (4) Fuel the aircraft and check for leaks. (Refer to 12-00-00)
 - (5) Install lower fuselage access covers.

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Fuel Boost Pump Installation
Figure 201

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FUEL FILTER - DESCRIPTION AND OPERATION

1. General

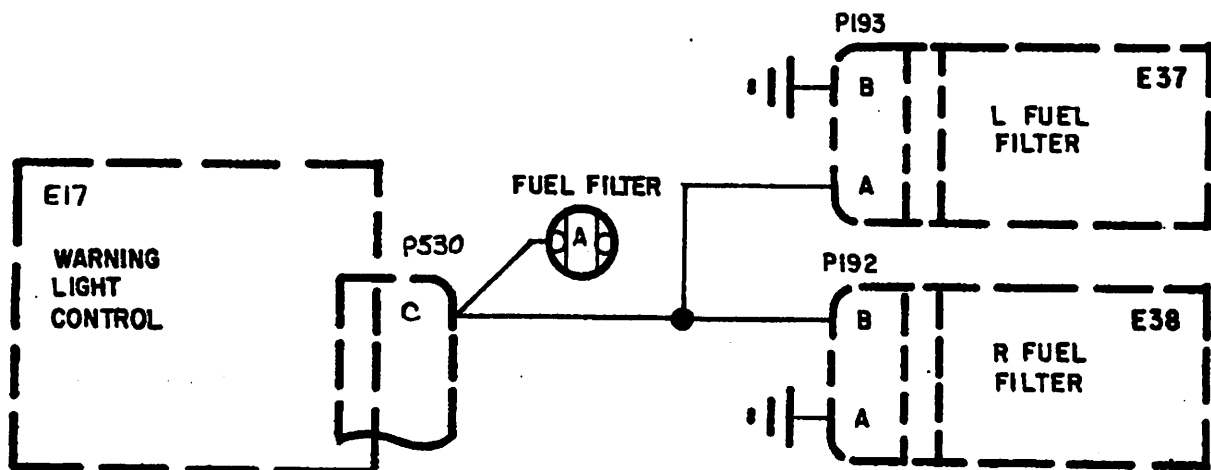
A. A low pressure fuel filter is installed in each main fuel line within the tailcone.

2. Description

A. The filter is equipped with either a disposable, paper element or a cleanable metal element. The paper element consists of a single filter element and bypass valve designed to open at 2.25 psid. The metal element consists of a dual filter element with a primary and secondary bypass valve designed to open at 1.25 and 2.25 psid respectively.

3. Operation (See figure 1)

A. A switch is incorporated in the head of the filter assembly which lights the FUEL FILTER caution light when a pressure differential of 1.25 ($\pm$.25) psi occurs. The caution light, when illuminated, indicates a clogged filter element. Remove, discard and replace paper filter elements after the first 30 operational hours on a new aircraft. Thereafter, replace paper elements after each 100 hours of operation. Remove and clean metal elements after the first 30 operational hours on a new aircraft. Thereafter, clean metal elements after each 100 hours of operation. When installing element and filter bowl, torque T-bolt 45 to 50 inch-pounds.



Fuel Filter Caution Light Electrical Control, Schematic
Figure 1

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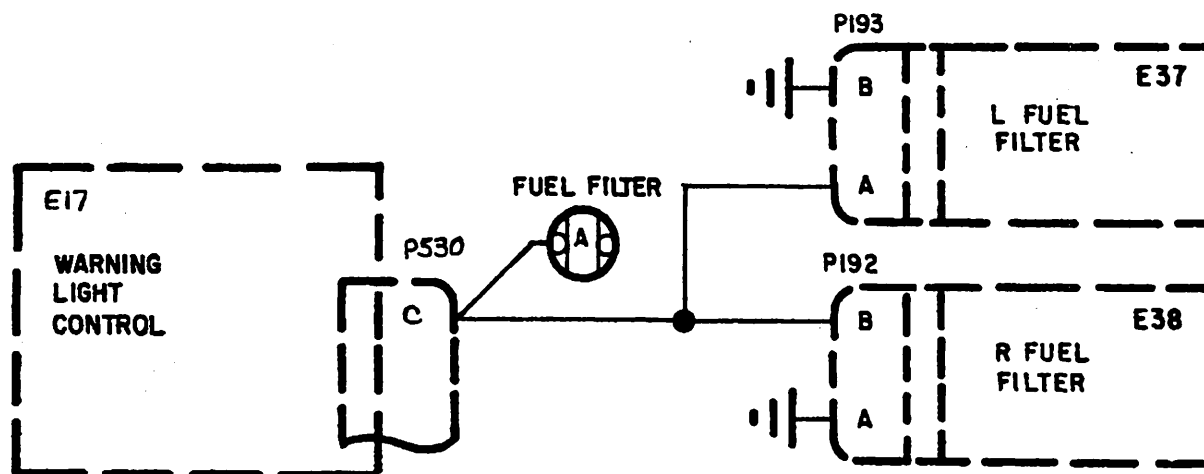
FUEL FILTER - DESCRIPTION AND OPERATION

1. Description

- A. A low pressure fuel filter is installed in each main fuel line within the tail-cone. The filter is equipped with either a disposable paper element or a cleanable metal element. The paper element consists of a single filter element and bypass valve designed to open at 2.25 psid. The metal element consists of a dual filter element with primary and secondary bypass valves designed to open at 1.25 and 2.25 psid respectively.

2. Operation (See figure 1.)

- A. A switch is incorporated in the head of the filter assembly which illuminates the FUEL FILTER caution light when a pressure differential of 1.25 (± 25) psid (Dynamic Filters P/N 101194) or of 0.95 to 1.30 psid (Purlator P/N 7577295) occurs. The caution light, when illuminated, indicates a clogged filter element. Remove, discard, and replace paper filter elements after the first 30 operational hours of a new aircraft. Thereafter, replace paper elements in accordance with current inspection requirements (See Chapter 5). Remove and clean metal elements after the first 30 operational hours of a new aircraft. Thereafter, clean metal elements in accordance with current inspection requirements. (See Chapter 5.) When installing element and filter bowl, torque T-bolt 45 to 50 inch-pounds.



Fuel Filter Caution Light Electrical Control, Schematic
Figure 1

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FUEL FILTER - MAINTENANCE PRACTICES

1. General

- A. Maintenance practices consist of removal and installation of filter assembly and replacement of filter element on aircraft equipped with paper element.

2. Removal/Installation

A. Remove Fuel Filter Assembly (See figure 201)

- (1) Lower tailcone access door.
- (2) Disconnect and remove aircraft batteries. (Refer to 24-32-01.)
- (3) Drain fuel filter using overboard drain valve.
- (4) Disconnect electrical plug from filter head.
- (5) Disconnect fuel plumbing and overboard drain line.
- (6) Remove attaching parts and filter from aircraft.

B. Install Fuel Filter Assembly (See figure 201)

- (1) Install fuel filter and secure with attaching parts.
- (2) Connect fuel plumbing and overboard drain line.
- (3) Connect electrical plug to filter head.
- (4) Install aircraft batteries. (Refer to 24-32-01.)
- (5) Check filters for leaks using either standby boost pump.
- (6) Secure tailcone access door.

C. Remove Filter Element (See figure 201)

- (1) Lower tailcone access door.
- (2) Disconnect and remove aircraft batteries. (Refer to 24-32-01.)
- (3) Drain fuel filter using overboard drain valve.
- (4) Loosen T-bolt clamp and remove filter bowl and filter element from head.
- (5) On aircraft equipped with cleanable elements, the upper gasket is epoxied to the filter head. Do not remove gasket unless gasket is loose or shows signs of deterioration.
- (6) On aircraft equipped with cleanable elements, the lower gasket is epoxied to the primary bypass valve. Do not remove gasket unless gasket is loose or shows signs of deterioration.
- (7) On aircraft equipped with cleanable elements, clean filter element and secondary bypass valve assembly. (Refer to Cleaning/Painting.)

D. Install Filter Element (See figure 201)

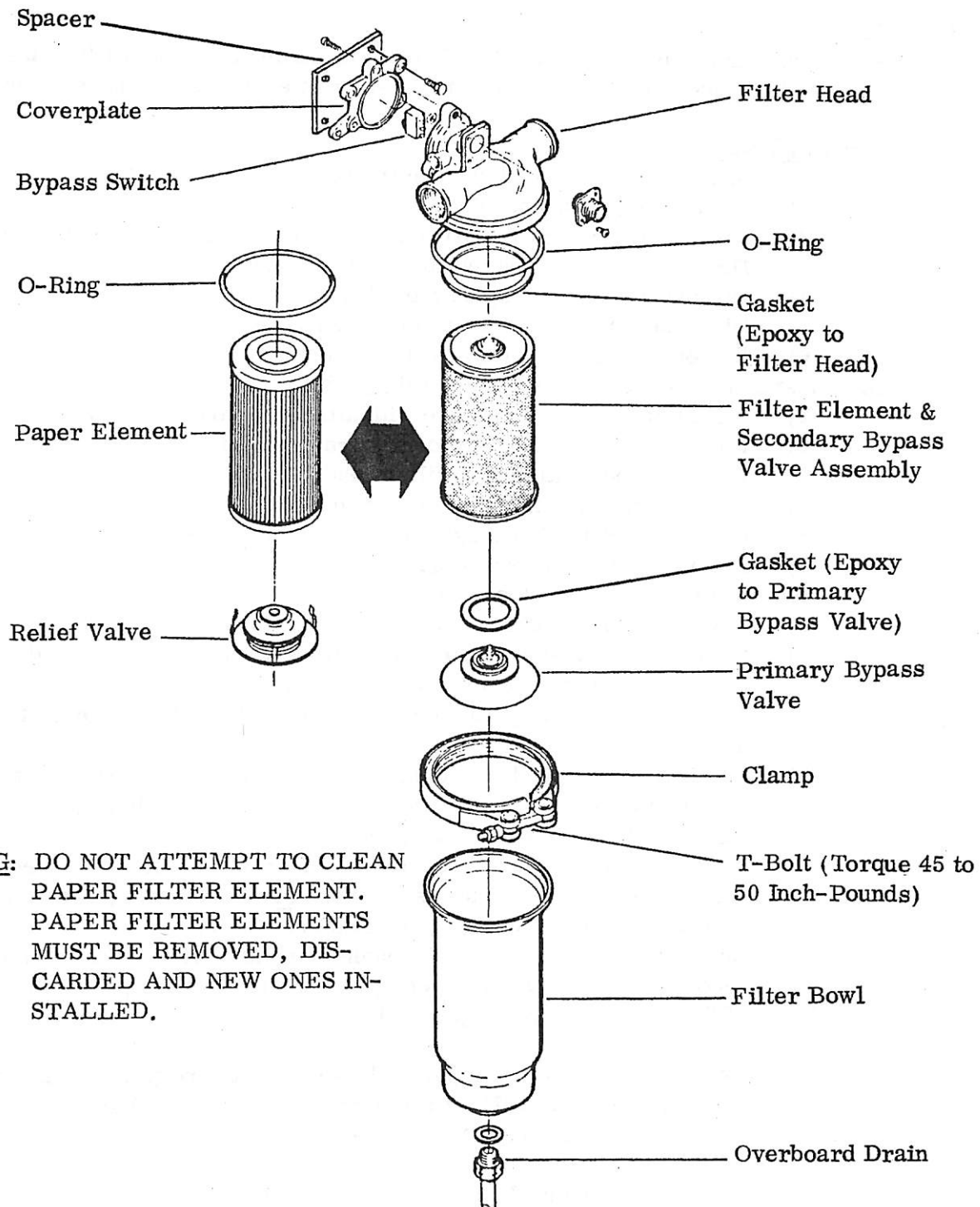
NOTE: • When replacing the upper or lower gasket, epoxy gasket into place using resin CB-1078 with reactor 2100 (John C. Dolph Co., Monmouth Junction, N. J.) or equivalent.

- If O-ring and gaskets are replaced, the date on the assembly date plate must be changed.

- (1) Assemble primary bypass valve, filter element and secondary bypass valve assembly and O-ring in bowl.

EFFECTIVITY: ALL

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maintenance manual



WARNING: DO NOT ATTEMPT TO CLEAN PAPER FILTER ELEMENT. PAPER FILTER ELEMENTS MUST BE REMOVED, DISCARDED AND NEW ONES INSTALLED.

Fuel Filter Assembly
Figure 201

EFFECTIVITY: ALL

MM-98

28-20-03
Page 202
Dec 18/81

Gates Learjet Corporation

maintenance manual

- (2) Install bowl and secure with T-bolt clamp. Torque T-bolt clamp to 45 to 50 inch-pounds.
- (3) Install aircraft batteries. (Refer to 24-32-01.)
- (4) Check filters for leaks using either standby boost pump.
- (5) Secure tailcone access door.

3. Adjustment/Test

A. Test Fuel Filter Bypass Switch (Dynamic Filters)

- (1) Remove filter element. (Refer to "Remove Fuel Filter Element.")
- (2) Install plastic plug where filter element was removed.
- (3) Disconnect fuel plumbing from fuel filter inlet and outlet ports.
- (4) Connect pressure source, pressure regulator, and pressure gauge to filter inlet port.
- (5) Disconnect electrical connector from filter.
- (6) Connect an ohmmeter across pins of filter receptacle.
- (7) Switch should close at 1.25 (± 0.25 psi).
- (8) Disconnect ohmmeter, pressure source, pressure regulator, pressure gauge and plastic plug.
- (9) Install filter element (Refer to "Install Fuel Filter Element") and insert electrical connector.

B. Test Fuel Filter Bypass Switch (Purolator Filter)

- (1) Remove filter element (Refer to "Remove Fuel Filter Element".)
- (2) Install plastic plug.
- (3) Disconnect fuel plumbing from fuel filter inlet and outlet ports.
- (4) Connect pressure sources, pressure regulators, and pressure gauges to filter inlet and outlet ports.
- (5) Remove electrical connector from filter.
- (6) Connect an ohmmeter across pins of filter receptacle.
- (7) Apply 10 psi pressure to inlet and outlet ports.
- (8) Decrease pressure at outlet port to 9.1 psi while maintaining inlet pressure at 10 psi to create a pressure differential of 0.9 psid across bypass switch.
- (9) There should be no electrical continuity across the pins of the filter receptacle.
- (10) Gradually decrease pressure at outlet port. Continuity shall occur between 0.95 to 1.30 psid.
- (11) Release all pressure at inlet and outlet port.
- (12) Disconnect pressure sources, pressure regulators, pressure gauges, plastic plug and ohmmeter.
- (13) Install filter (Refer to Fuel Filter Element") and electrical connector.

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4. Cleaning/Painting

A. Clean Metal Filter Element

WARNING: DO NOT ATTEMPT TO CLEAN PAPER FILTER ELEMENT.
PAPER FILTER ELEMENTS MUST BE REMOVED, DISCARDED
AND NEW ONES INSTALLED.

- (1) Remove bulk of dirt by vapor degreasing filter element and secondary bypass valve assembly in trichloroethane for ten (10) minutes.
- (2) Sonic clean filter element and secondary bypass valve assembly in mild detergent (MIL-D-16791, Type 1) or equivalent for 20 to 30 minutes.
- (3) Rinse in hot water (130 to 200°F) for 20 minutes.

CAUTION: DO NOT BLOW ACROSS SCREEN WITH SHOP AIR.
ELEMENT MUST BE ALLOWED TO AIR DRY.

- (4) Allow to air dry, or dry in oven at 100 to 200°F.
- (5) Examine filter. If any contaminant is still visible, repeat cleaning procedure.

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maintenance manual

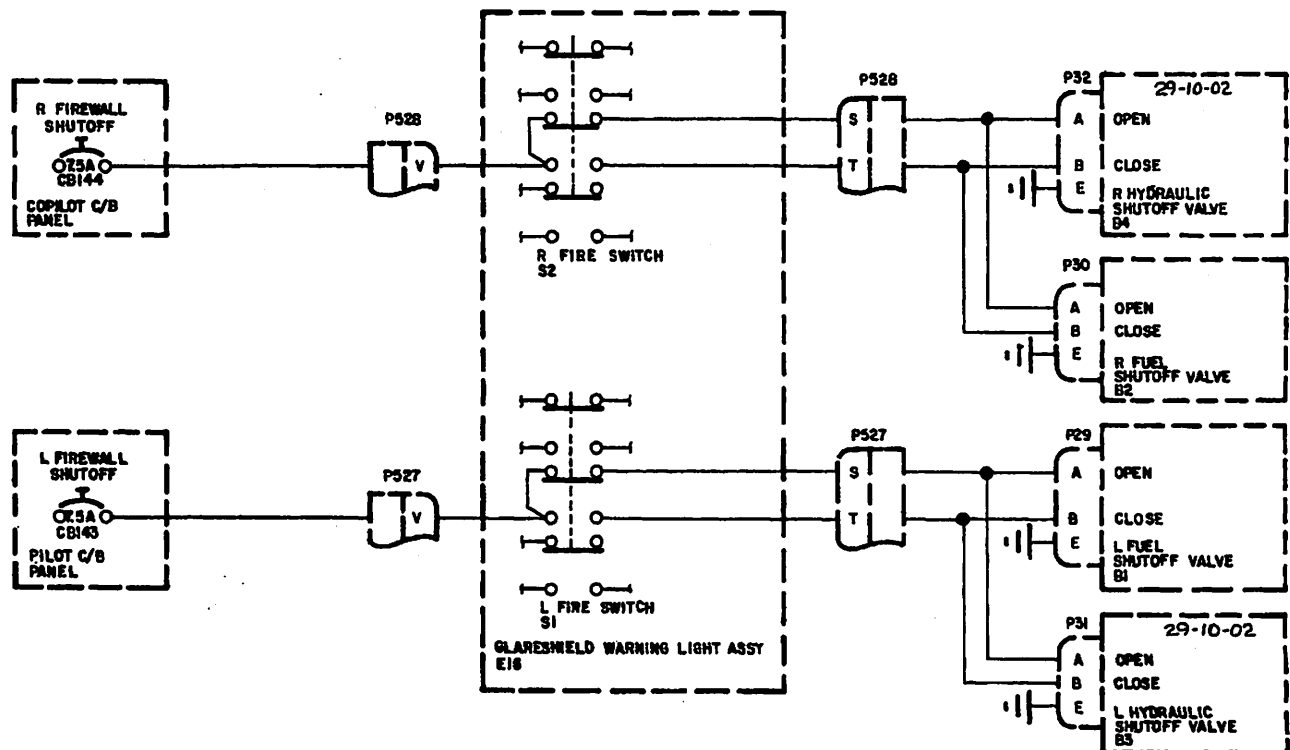
FUEL SUPPLY SHUTOFF VALVE - DESCRIPTION AND OPERATION

1. General

- A. Each distribution system incorporates a fuel shutoff valve. The shutoff valves are located adjacent to the fuel filters. The shutoff valves are two-position (open and closed) ball-and-seat type and are electrically operated.

2. Operation (See figure 1)

- A. The left engine main fuel shutoff valve(s) control fuel flow to the left engine and is closed only in case of fire in the left engine. Operation is controlled by the left FIRE switch on the glareshield. The right engine main fuel shutoff valve(s) controls fuel flow to the right engine and is closed only in case of fire in the right engine. Operation is controlled by the right FIRE switch. The fuel crossflow shutoff valve controls fuel flow between the wing tanks.



Fuel Supply Shutoff Valve Electrical Control Schematic
Figure 1

EFFECTIVITY: ALL

MM-98

28-20-04
Page 1
Dec 18/81

Gates Learjet Corporation ➞

maintenance manual

FUEL SUPPLY SHUTOFF VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Shutoff Valve (See figure 201)

- (1) Open tailcone access door.
- (2) Remove aircraft batteries. (Refer to 24-32-01.)
- (3) Disconnect fuel plumbing and electrical plug from shutoff valve.
- (4) Remove nuts, washers, and bolts installed through shutoff valve fittings. Remove shutoff valve from aircraft.

CAUTION: THE FUEL SUPPLY SHUTOFF VALVE INCORPORATES TWO TEFLON SEALING GASKETS, ONE BLACK AND ONE WHITE, AGAINST THE BALL VALVE. THE BLACK GASKET INCORPORATES A SMALL ORIFICE TO RELIEVE SYSTEM PRESSURE. THEREFORE, WHEN INSTALLING THE FUEL SUPPLY SHUTOFF VALVE, ALWAYS POSITION THE VALVE SO THAT THE BLACK GASKET IS TO THE PRESSURE SIDE (NEAREST THE PUMP).

NOTE:

- On Aircraft 25-061, 25-070 thru 25-344, install fuel supply shutoff valves with the electrical receptacle at lower portion of valve.
- On Aircraft 25-345 and Subsequent, install fuel supply shutoff valve with the electrical receptacle at the upper portion of valve.

B. Install fuel supply shutoff valve (See figure 201)

- (1) Position O-ring and shutoff valve onto fitting still attached to fuel filter.
- (2) Secure with attaching bolts, washers, and nuts. Torque nuts 20 to 25 inch-pounds.
- (3) Connect electrical plug to shutoff valve.
- (4) Install aircraft batteries. (Refer to 24-32-01.)
- (5) Check shutoff valve for leaks using applicable standby boost pump.
- (6) Install tailcone access door.

2. Inspection/Check

A. Functional check of the fuel supply shutoff valves requires two mechanics, one in the cockpit and one in the tailcone access opening.

CAUTION: DO NOT DEPRESS EITHER ARMED LIGHT WHEN PERFORMING THIS CHECK AS THE FIRE EXTINGUISHER SYSTEM IS ARMED AFTER THE FIRE LIGHT IS DEPRESSED.

- (1) Set Battery Switch(es) to ON.
- (2) Close left engine main fuel shutoff valve by depressing the left FIRE

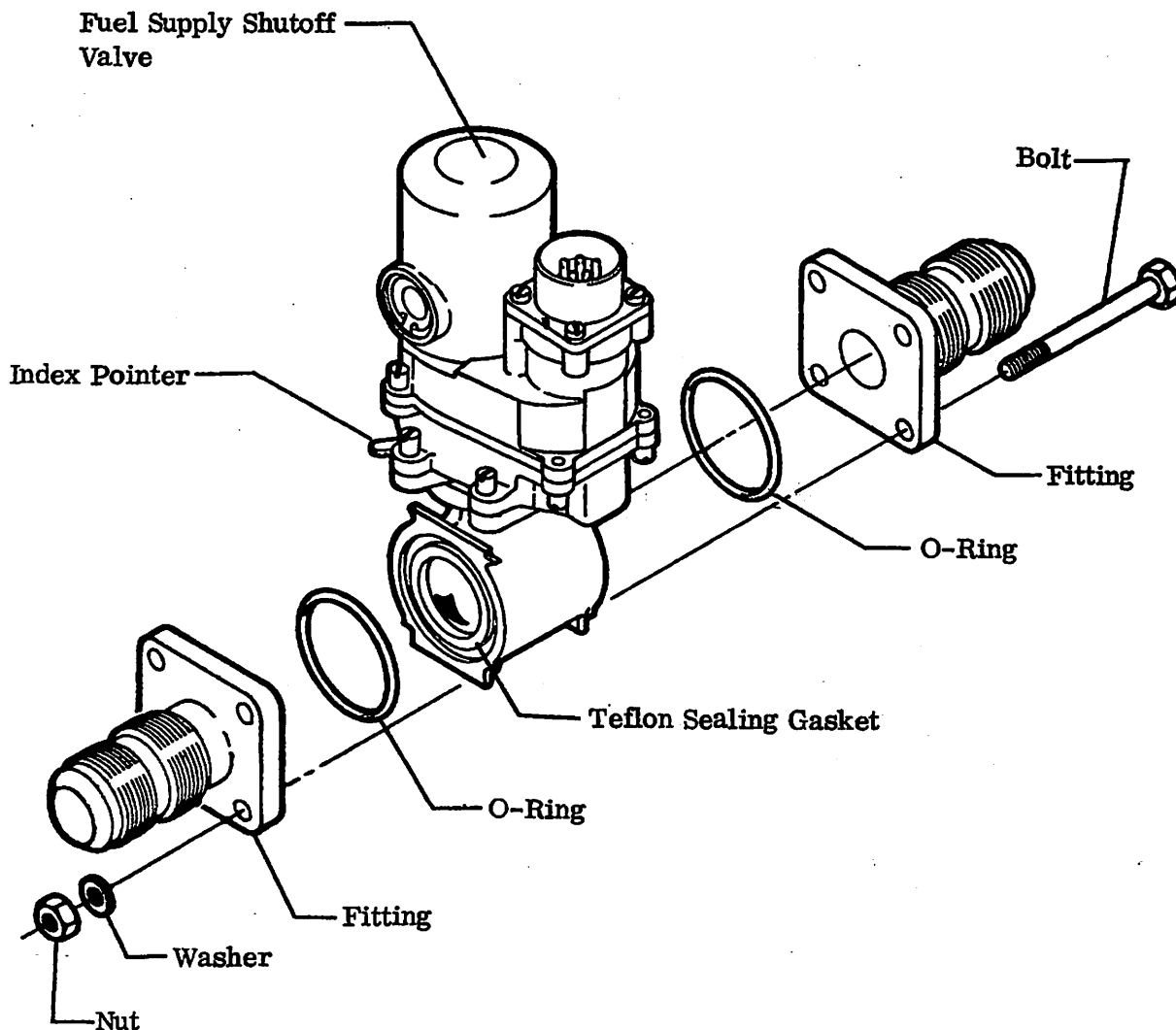
EFFECTIVITY: ALL

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maintenance manual

Switch. The pinhead light, located inboard of the FIRE Switch, will illuminate indicating a completed circuit to the shutoff valves. The left ARMED lights will also illuminate.

- (3) Observe the index pointer on the left main fuel shutoff valve to verify that valve closes.
- (4) Again depress left FIRE Switch. Pinhead light and ARMED lights will extinguish. Verify that valve has opened by observing the shutoff valve index pointer.
- (5) Perform steps (2), (3) and (4) twice more to verify proper valve operation.
- (6) Repeat steps (1) thru (5) using the right FIRE Switch, pinhead lights, ARMED lights and shutoff valve.
- (7) Set Battery Switch(es) to OFF and secure all accesses.



Fuel Supply Shutoff Valve Installation (Typical)
Figure 201

GATES LEARJET CORPORATION
maintenance manual

3. Inspection/Check

- A. Functional check of the fuel supply shutoff valves requires two mechanics, one in the cockpit and one in the tailcone access opening.

CAUTION: DO NOT DEPRESS EITHER ARMED LIGHT WHEN PERFORMING THIS CHECK AS THE FIRE EXTINGUISHER SYSTEM IS ARMED AFTER THE FIRE LIGHT IS DEPRESSED.

NOTE: On Aircraft Certified for Operation in the United Kingdom, the lower fuselage access panels must be removed to observe No. 2 shutoff valves as well as No. 1 shutoff valves.

- (1) Set Battery Switch(es) to ON.
- (2) Close left engine main fuel shutoff valve by depressing the left FIRE Switch. The pinhead light, located inboard of the FIRE Switch, will illuminate indicating a completed circuit to the shutoff valves. The left ARMED lights will also illuminate.
- (3) Observe the index pointer on the left main fuel shutoff valve to verify that valve closes. On Aircraft Certified for Operation in the United Kingdom, observe both left main fuel shutoff valves.
- (4) Again depress left FIRE Switch. Pinhead light and ARMED lights will extinguish. Verify that valve has opened by observing the shutoff valve index pointer.
- (5) Perform steps (2), (3) and (4) twice more to verify proper valve operation.
- (6) Repeat steps (1) thru (5) using the right FIRE Switch, pinhead lights, ARMED lights and shutoff valve.
- (7) Set Battery Switch(es) to OFF and secure all accesses.

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maintenance manual

MOTIVE FLOW VALVE - DESCRIPTION AND OPERATION

1. Description

- A. Each distribution system incorporates a motive flow valve. The valve is installed on the aft fuselage fuel tank wall above the fuel filters. The valves are two-position (open and closed) rotary actuator type and are electrically operated. On Aircraft 25-320 and Subsequent, and Prior Aircraft Modified per AMK 80-7, the motive flow valve is mounted in an enclosure box assembly which is attached to an overboard drain line.
- B. On Aircraft 25-320 and Subsequent, and Prior Aircraft Modified per AMK 80-7, the motive flow valve enclosure box assembly is designed to catch fuel and dump it overboard in the event of a motive flow valve leak.

2. Operation

- A. The left motive flow valve controls the fuel flow from the left engine-driven fuel pump to the left tip tank jet pump and left jet pump. The right motive flow valve controls the fuel flow from the right engine-driven fuel pump to the right tip tank jet pump and right jet pump. Operation is controlled by the Jet Pump Switches on the Fuel Control Panel and by the left and right START-GENERATOR Switches on the Power and Ignition Switch Panel.

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maintenance manual

MOTIVE FLOW VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

- A. Remove Motive Flow Valve (Aircraft 25-061, 25-070 thru 25-319 not Modified per AMK 80-7) (See figure 201)
- (1) Remove electrical power from aircraft.
 - (2) Lower tailcone access door.
 - (3) Disconnect electrical plug from motive flow valve.
 - (4) Remove fuel plumbing from motive flow valve.
 - (5) Remove safety wire, attaching parts, and motive flow valve from aircraft.
- B. Install Motive Flow Valve (Aircraft 25-061, 25-070 thru 25-319 not Modified per AMK 80-7) (See figure 201)
- (1) Install motive flow valve and secure with attaching parts. Safety wire bolts.
 - (2) Install fuel plumbing to motive flow valve.
 - (3) Connect electrical plug to motive flow valve.
 - (4) Check for leaks.
 - (5) Restore electrical power to aircraft.
 - (6) Close and secure tailcone access door.
- C. Remove Motive Flow Valve (Aircraft 25-320 and Subsequent and Prior Aircraft Modified per AMK 80-7) (See figure 201)
- (1) Set Battery Switches to OFF. Ensure that external power source is disconnected from aircraft.
 - (2) Gain access to the motive flow valve through the tailcone access door.
 - (3) Disconnect electrical plug from the motive flow valve.
 - (4) Scrape sealant off the mating surfaces between enclosure box and motive flow valve, the motive flow lines and valve fittings, and the attaching bolts at the top and bottom of the motive flow valve and enclosure box.
 - (5) Remove motive flow lines from valve fittings.
 - (6) Remove the bolts and washers which attach retainer, inspection plate, coverplate, and gasket.
 - (7) Remove retainer, inspection plate, coverplate and gasket.
 - (8) Remove fittings which are attached to the motive flow valve and protrude from the enclosure box.
 - (9) Remove lockwire from the attaching bolts (at the top and bottom) of the motive flow valve. Remove attaching bolts and washers and motive flow valve.
- D. Install Motive Flow Valve (Aircraft 25-320 and Subsequent and Prior Aircraft Modified per AMK 80-7) (See figure 201)
- (1) Place motive flow valve in the enclosure box and secure with bolts and washers and lockwire.
 - (2) Using Pro-Seal 890 sealant, form a faying seal between the bolt heads and the enclosure box.

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maintenance manual

- (3) Install new O-rings and install the fittings which protrude through the enclosure box. Using Pro-Seal 890 sealant, form a seal around the openings of the enclosure box where the motive flow line enters and exits.
- (4) Install a new gasket. Apply Pro-Seal 890 sealant sparingly around the perimeter of the gasket and install the inspection plate, coverplate, and retainer. Secure with attaching bolts and washers. Wipe away excess sealant.

CAUTION: WHEN FORMING A FAYING SEAL AROUND THE VALVE BODY, ENSURE THAT THE VALVE INDICATOR POINTER REMAINS FREE TO MOVE WITH THE INTERNAL VALVE AND THAT SEALANT PENETRATION INTO THE ENCLOSURE BOX IS MINIMAL.

- (5) Using Pro-Seal 890 sealant, form a seal where the valve body mates with the inspection plate and coverplate. Ensure that the motive flow valve indicator pointer remains free to move with the internal valve and that sealant penetration into the enclosure box is minimal.
- (6) Install motive flow lines on valve fittings which protrude from the enclosure box.
- (7) Connect the electrical plug to the motive flow valve.
- (8) Restore electrical power to aircraft.
- (9) Start engines in accordance with the FAA Approved Airplane Flight Manual.
- (10) Switch fuel flow to jet pump operation and check for fuel leaks at the motive flow valve outside drain line.
- (11) Shut down engines in accordance with the FAA Approved Airplane Flight Manual.
- (12) Close and secure tailcone access door.

E. Remove Enclosure Box Assembly (Aircraft 25-320 and Subsequent and Prior Aircraft Modified per AMK 80-7) (See figure 201)

- (1) Set Battery Switches to OFF. Ensure that external power source is disconnected.
- (2) Gain access to the motive flow valve enclosure box assembly through the tailcone access door.
- (3) Disconnect the electrical plug from the motive flow valve.
- (4) Remove the overboard drain line from the enclosure box assembly.
- (5) Remove the motive flow drain lines from the enclosure box assembly.
- (6) Scrape sealant off the attaching bolts. Remove the attaching bolts and washers and remove the enclosure box assembly from the aircraft.
- (7) Remove the motive flow valve from the enclosure box assembly (if necessary) in accordance with steps C.(4) thru C.(9).

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maintenance manual

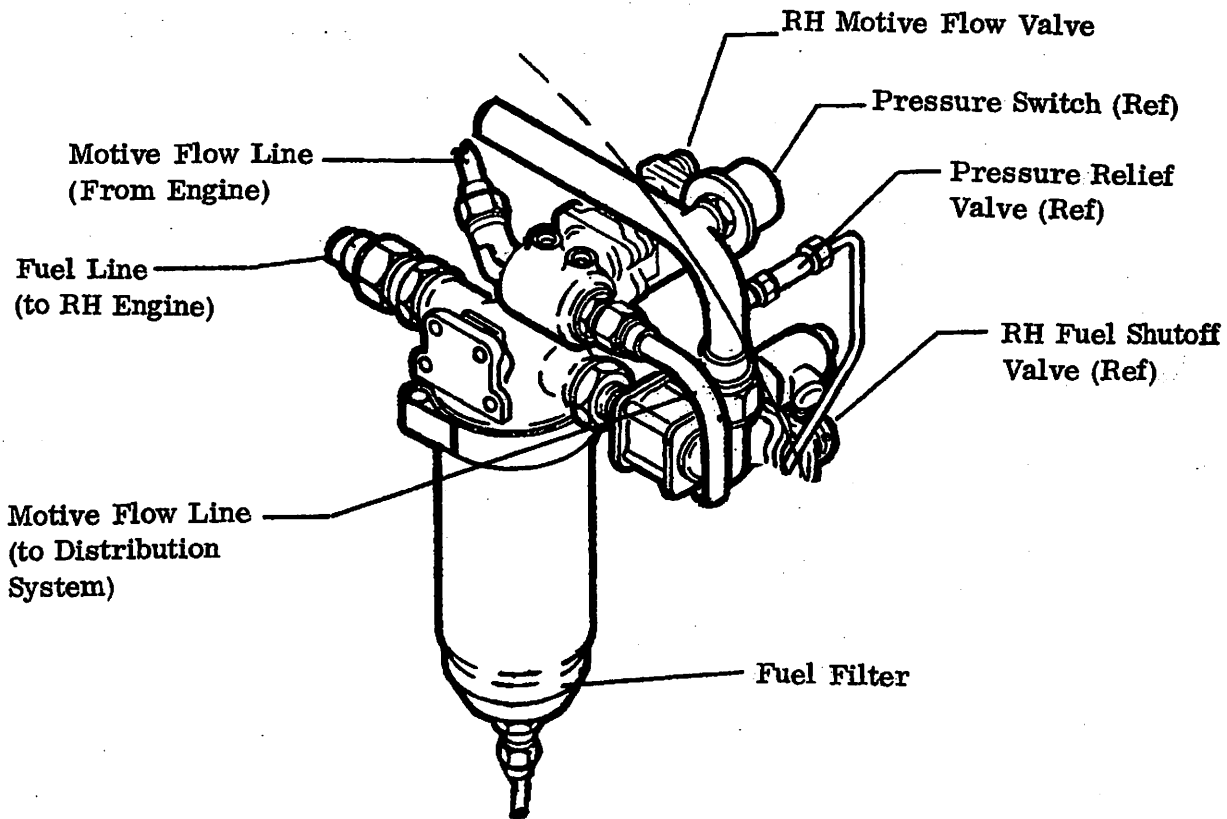
F. Install the Enclosure Box Assembly (Aircraft 25-320 and Subsequent and Prior Aircraft Modified per AMK 80-7) (See figure 201)

- (1) If the motive flow valve has been removed from the enclosure box assembly, install motive flow valve in enclosure box in accordance with steps D.(1) thru D.(7).
- (2) Position the enclosure box assembly between the mounting brackets, ensure that the flow arrow on the valve indicates proper position. Secure enclosure box assembly in place with attaching bolts and washers.

CAUTION: WHEN FORMING A SEAL AROUND THE HEAD OF EACH BOLT, ENSURE THAT SEALANT PENETRATION INTO THE ENCLOSURE BOX IS MINIMAL.

- (3) Using Pro-Seal 890 sealant, form a seal around the head of each attaching bolt.
- (4) Install the motive flow drain line on the enclosure box.
- (5) Install the motive flow lines on the motive flow valve fittings.
- (6) Connect the electrical plug to the motive flow valve.
- (7) Restore electrical power to the aircraft.
- (8) Start engines in accordance with the FAA Approved Airplane Flight Manual.
- (9) Switch fuel flow to jet pump operation and check for fuel leaks at the motive flow valve outside drain line.
- (10) Shut down engines in accordance with the FAA Approved Airplane Flight Manual.
- (11) Close and secure access door.

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maintenance manual



25-061, 25-070 thru 25-319 Not Modified per AMK 80-7,
"Installation of Motive Flow Valve Shrouds and Drain Lines"

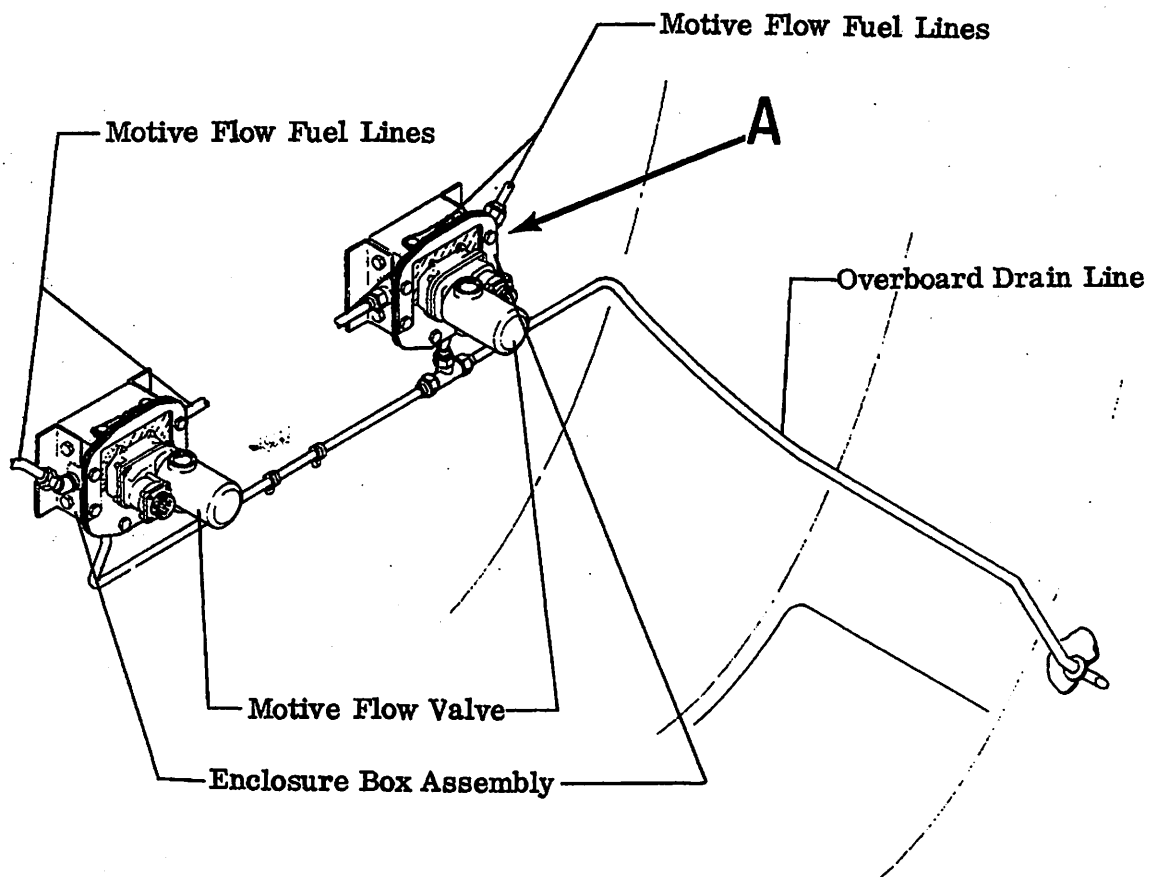
Motive Flow Valve Installation
Figure 201 (Sheet 1 of 3)

EFFECTIVITY: ALL

MM-98

28-20-05
Page 204
Dec 18/81

Gates Learjet Corporation
maintenance manual



25-320 and Subsequent and Prior Aircraft Modified per AMK 80-7,
"Installation of Motive Flow Valve Shrouds and Drain Lines"

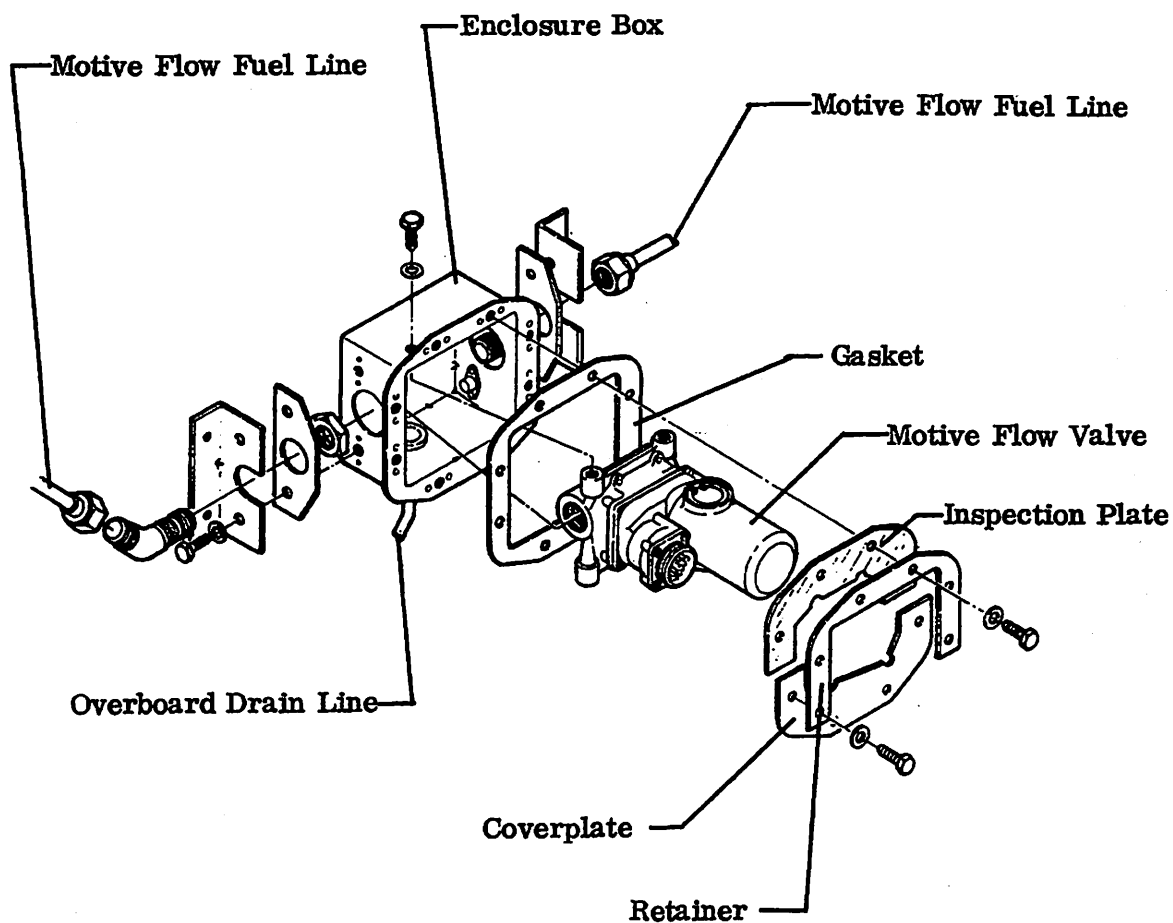
Motive Flow Valve Installation
Figure 201 (Sheet 2 of 3)

EFFECTIVITY: ALL

MM-98

28-20-05
Page 205
Dec 18/81

Gates Learjet Corporation
maintenance manual



25-320 and Subsequent and Prior Aircraft Modified per AMK 80-7,
"Installation of Motive Flow Valve Shrouds and Drain Lines"

Detail A

Motive Flow Valve Installation
Figure 201 (Sheet 3 of 3)

EFFECTIVITY: ALL

MM-98

28-20-05
Page 206
Dec 18/81

GATES LEARJET CORPORATION
maintenance manual

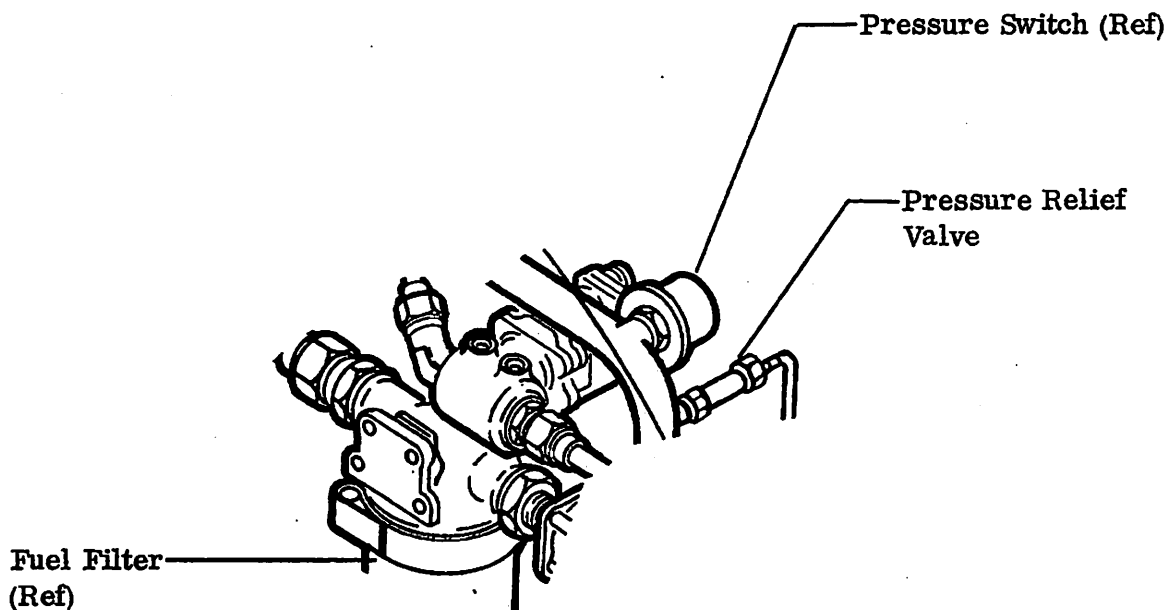
RELIEF VALVE - MAINTENANCE PRACTICES

1. General

- A. Maintenance practices consist of removal and installation of the relief valve.
- B. A high pressure relief valve is incorporated in each fuel distribution system. The relief valve is installed in the main fuel lines adjacent to the shutoff valve. The valves are installed to relieve any pressure buildup caused by thermal expansion of trapped fuel when the engine is shut down. The relief valve opens at 75 psi and vents fuel overboard.

2. Removal/Installation

- A. Remove Relief Valve (See figure 201)
 - (1) Lower tailcone access door.
 - (2) Disconnect drain line.
 - (3) Remove relief valve and O-ring from main fuel line.
- B. Install Relief Valve (See figure 201)
 - (1) Install O-ring and relief valve in main fuel line.
 - (2) Connect drain line.
 - (3) Secure tailcone access door.



**Relief Valve Installation
Figure 201**

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maintenance manual

FUEL CONTROL RELAY PANEL - MAINTENANCE PRACTICES

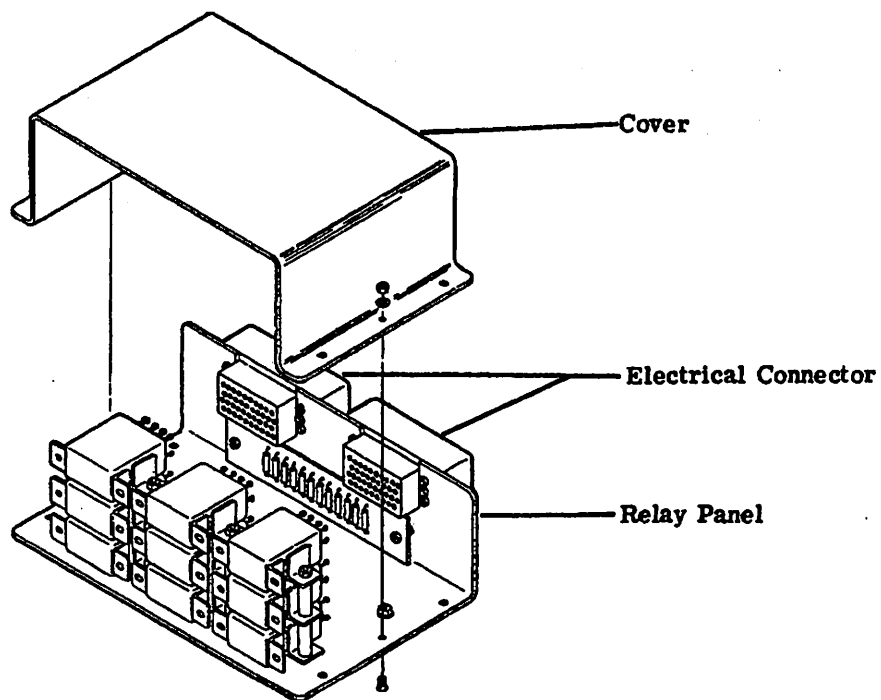
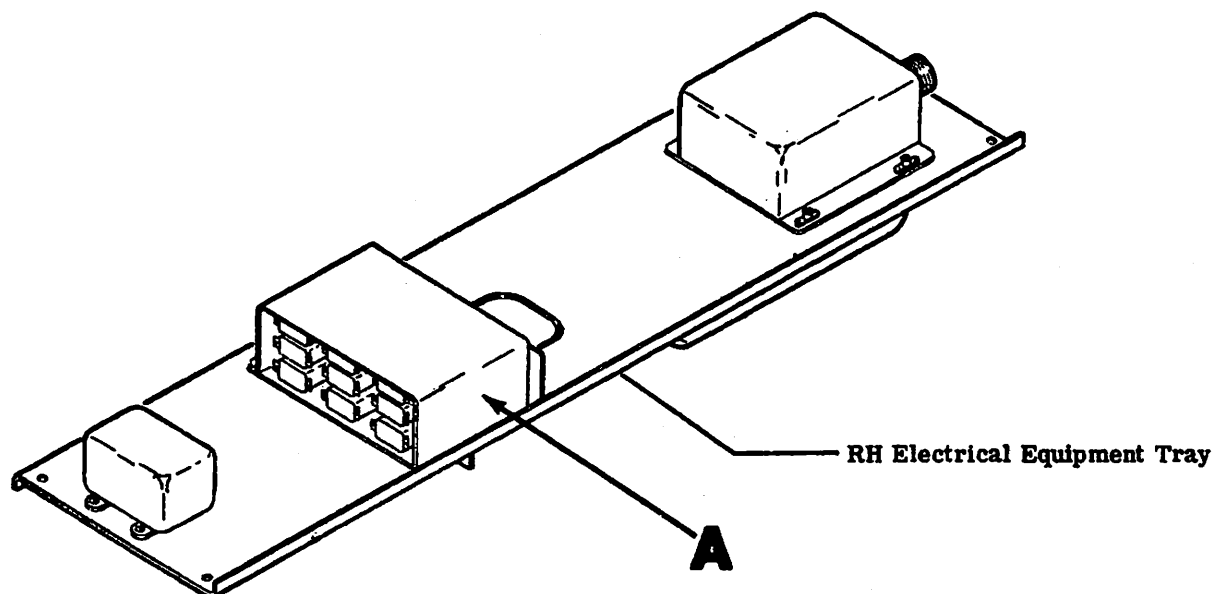
1. General

- A. The fuel control relay panel is located on the RH electrical equipment tray. The electrical equipment tray is located on the RH side of the tailcone.
- B. Maintenance procedures consist of electrical continuity checks and replacing components.
- C. Refer to 28-13-00 of Wiring Manual for panel wiring.

2. Removal/Installation

- A. Remove Fuel Control Relay Panel (See figure 201)
 - (1) Lower tailcone access door.
 - (2) Disconnect aircraft batteries.
 - (3) Disconnect electrical plugs from fuel control relay panel.
 - (4) Loosen and remove attaching parts and fuel control relay panel from electrical equipment tray.
- B. Install Fuel Control Relay Panel (See figure 201)
 - (1) Install fuel control relay panel and secure with attaching parts.
 - (2) Connect electrical plugs to relay panel.
 - (3) Connect aircraft batteries.
 - (4) Raise and secure tailcone access door.

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maintenance manual



Detail A

Fuel Control Relay Panel
Figure 201

EFFECTIVITY: ALL

28-20-07
Page 202
Dec 2/77

Gates Learjet Corporation

maintenance manual

FUEL SUPPLY LINE CHECK VALVES - MAINTENANCE PRACTICES

1. Description

- A. A fuel supply line check valve is installed adjacent to, and downstream from, each jet pump and fuel standby pump in the wing. These check valves prevent fuel from being pumped backwards through each pump by its adjacent pump. The flapper-type check valves each have a small orifice in the flapper which allows fuel to drain back to the engines after engine shutdown. Inspect the fuel supply line check valves in accordance with the intervals described in Chapter 5.

2. Inspection/Check

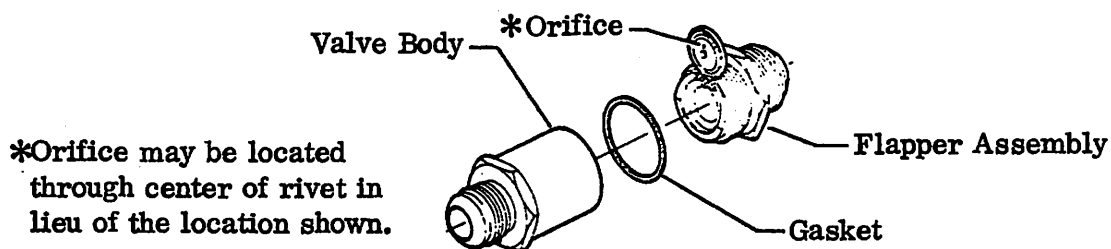
- A. Inspect Fuel Supply Line Check Valve

NOTE: It is desirable to remove and install the check valves with the aircraft defueled. Refer to Chapter 12 for defueling.

WARNING: OBSERVE ALL SAFETY PRACTICES DESCRIBED IN 28-00-00. SPECIAL CARE SHOULD BE TAKEN TO PROPERLY VENTILATE THE FUEL CELLS PRIOR TO PERFORMING MAINTENANCE PROCEDURES.

- (1) Remove lower fuselage access panels to gain access to the fuel supply line check valves.
- (2) Disconnect plumbing from check valves and remove check valves from tee.

NOTE: Disassembly of the check valve is not normally required unless the check valve is extremely contaminated. If valve must be disassembled, it must be reassembled using a new gasket.



Fuel Supply Line Check Valve
Figure 201

Gates Learjet Corporation

maintenance manual

- (3) Hold valve assembly next to a light source and look through the center of the valve to assure that orifice is not obstructed. If necessary, clean orifice and thoroughly flush valve using kerosene.

NOTE: The orifice in the flapper is either next to the rivet or through the rivet.

- (4) Install check valves on tee using new O-ring package.
(5) Connect plumbing to check valves.
(6) Inspect plumbing and check valves for leaks.
(7) Install lower access panels.

 GATES LEARJET CORPORATION
maintenance manual

FUEL JETTISON - DESCRIPTION AND OPERATION

1. General

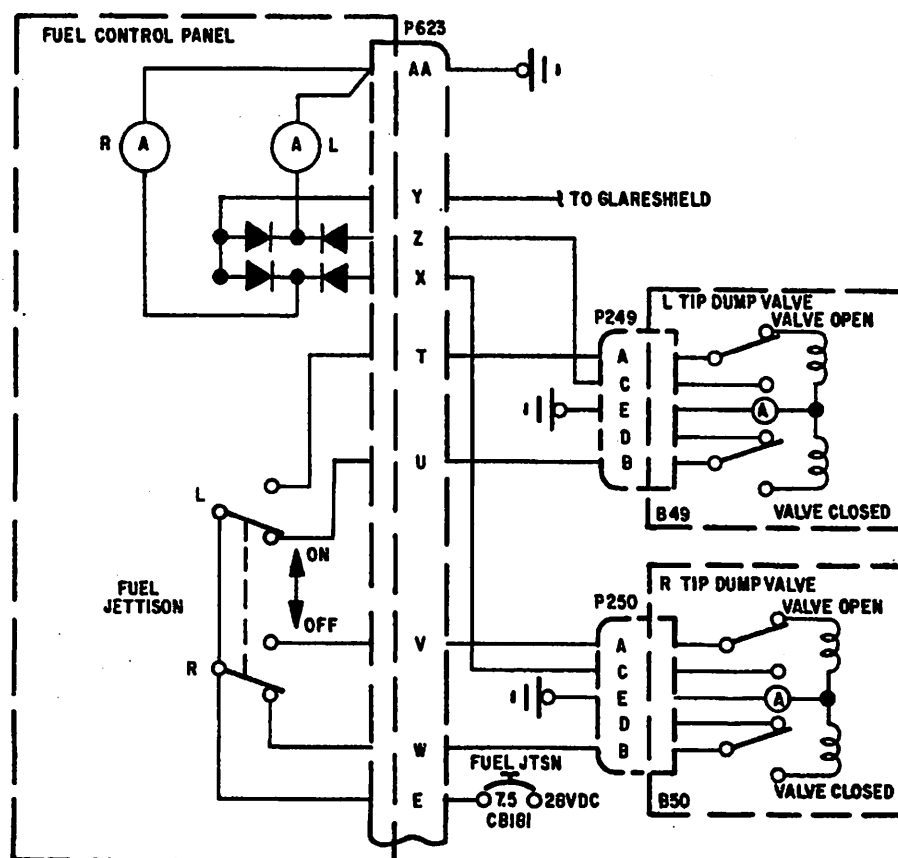
A. The fuel jettison system is installed in each tip tank and allows tip tank fuel to be jettisoned during an emergency situation.

2. Description

A. The system consists of two (amber) indicator lights, a system switch, a shutoff valve in each tip tank and a 7.5 ampere circuit breaker.

3. Operation (See figure 1)

A. Setting the Fuel Jettison Switch, on the fuel control panel, to ON applies 28 vdc power to open the shutoff valves in each tip tank. This allows the fuel to gravity flow from each tip tank. When the shutoff valves are in the open position, the two indicator lights, adjacent to the switch, will light giving a positive indication of system operation.



Fuel Jettison Electrical Control Schematic
Figure 1

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maintenance manual

FUEL JETTISON - MAINTENANCE PRACTICES

1. General

- A. Maintenance practices consist of removal and installation of the shut-off valve.**

2. Removal/Installation

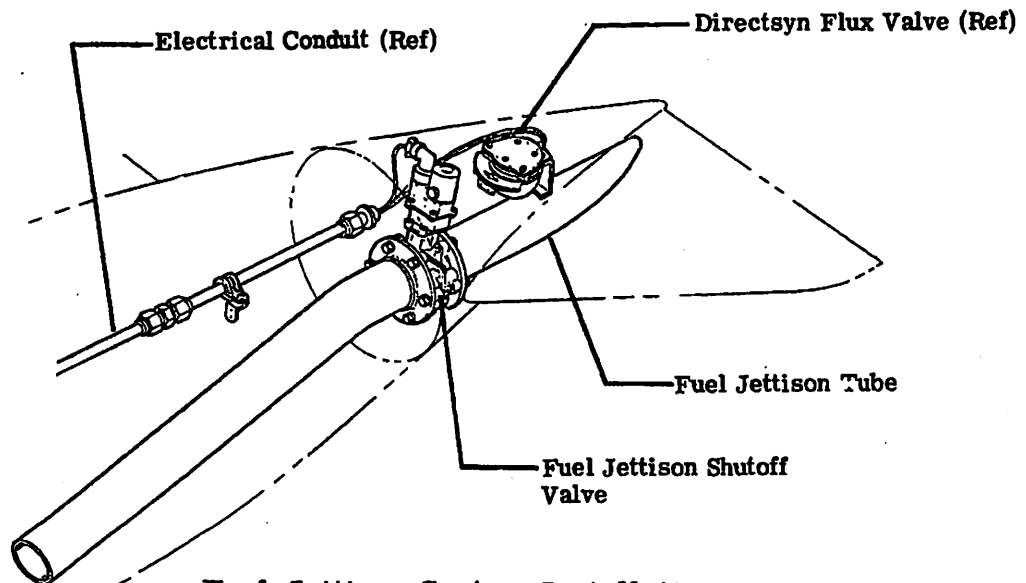
NOTE: The following procedure is applicable to either fuel jettison system.

A. Remove Shutoff Valve (See figure 201)

- (1) Drain tip tank. (Refer to 12-00-00)
- (2) Remove attaching parts and remove directsyn valve from bracket.
- (3) Remove tip tank tailcone and access cover.
- (4) Disconnect electrical plug from shutoff valve.
- (5) Remove attaching parts and shutoff valve from tip tank.

B. Install Shutoff Valve (See figure 201)

- (1) Install O-rings, shutoff valve and tube assembly and secure with attaching parts.
- (2) Connect electrical plug to shutoff valve.
- (3) Install directsyn valve and secure with attaching parts.
- (4) Apply a thin continuous coat of anti-corrosion grease to access cover (refer to 28-11-00). Install access cover.
- (5) Fuel tip tank and check for leaks.
- (6) Install tip tank tailcone.



**Fuel Jettison System Installation
Figure 201**

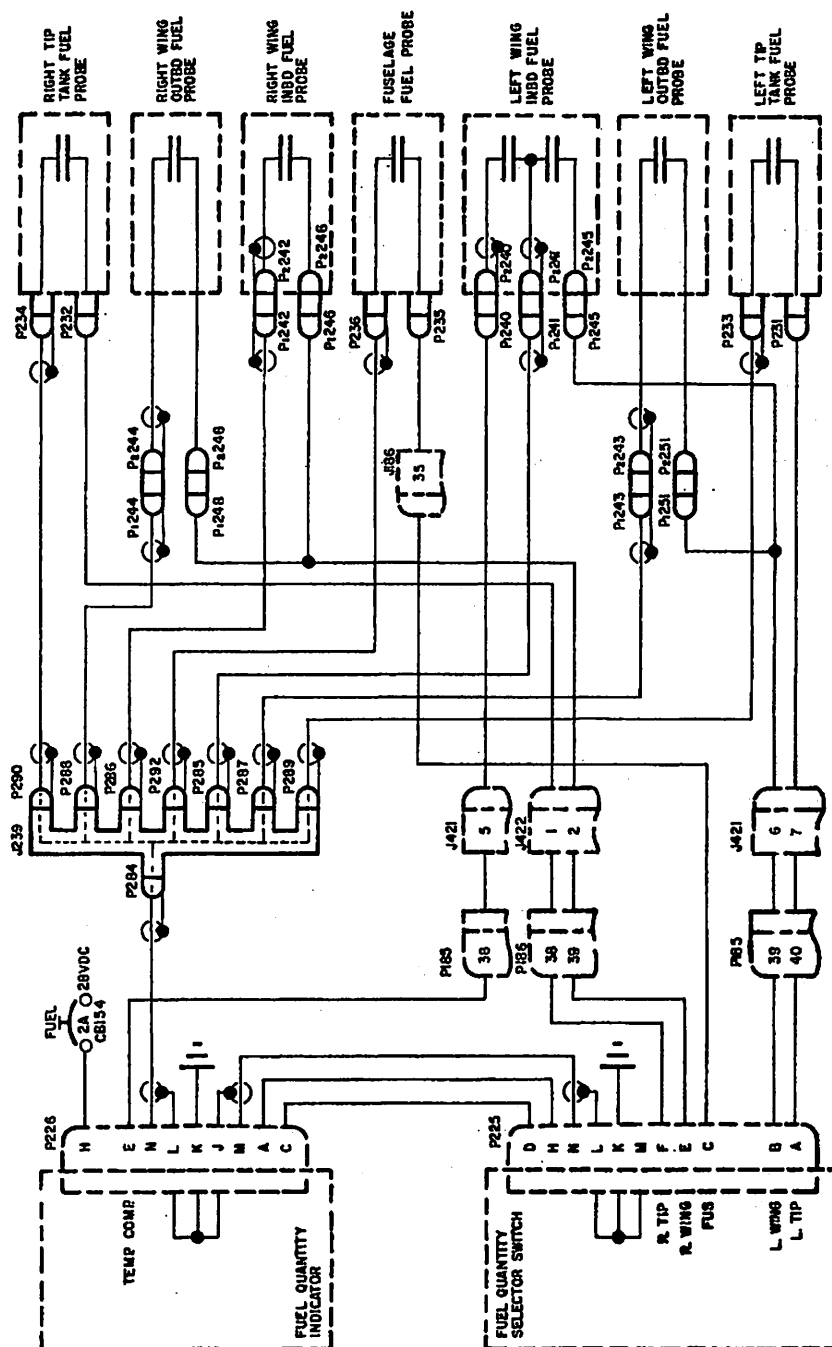
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INDICATING - DESCRIPTION AND OPERATION

1. Description

- A. A capacitance-type fuel quantity indicating system, manufactured by Minneapolis-Honeywell, is installed in the aircraft. The system components consist of an indicator and a selector switch on the fuel control panel and seven tank unit probes. Quantity and location of probes is: one in each tip tank, two in each wing tank, and one in the fuselage tank.
- B. Fuel system pressure is monitored by a pressure switch installed in the main distribution lines.
- C. The fuel probes are variable capacitors located in each fuel tank to sense fuel quantity. Each fuel probe consists of two concentric electrodes which are rigidly spaced at intervals throughout their length by insulating centering spacers. The inner electrode consists of a main body of insulating material with two separate insulating conducting surfaces around its outside face. One conducting surface is grounded and the other is connected to the amplifier input of the system bridge circuit located in the indicator. The outer electrode is an aluminum tube coated on the outside with insulating materials. Openings in the probes permit fuel to flow between the electrodes to the same level as that in the tank. Electrical connections to the fuel probes are made through quick-disconnect electrical connectors. The fuel probes in each tank are connected in parallel. The fuel is the variable dielectric factor of the capacitor. Capacitance of the fuel probes provide a continuous signal to the fuel quantity indicator. The fuel probes are part of the system bridge circuit which receives power from the secondary winding of a power transformer located in the fuel quantity indicator.
- D. The indicator basically consists of a hermetically sealed combination transistor power unit and indicator. Its function is to amplify a signal caused by unbalance in a bridge circuit and to convert this signal into a visual indication of fuel quantity in pounds. The unit, of subminiature construction, includes a self-balancing bridge circuit, amplifier stages, motor, gear train, and an indication dial. Electrical connection is made through a single multi-pin connection on the back of the unit. FULL and EMPTY adjustment screws are located on the back of the indicator.
- E. The switch is a cylindrical-shaped six-position rotary unit, located adjacent to the fuel quantity indicator. It permits reading of each tank quantity separately as well as the total quantity of the system.

 GATES LEARJET CORPORATION
maintenance manual



Fuel Quantity Indicating Electrical Schematic
Figure 1

EFFECTIVITY: ALL

MM-98

28-40-00
Page 2
Dec 2/77

Gates Learjet Corporation
maintenance manual

INDICATING - TROUBLE SHOOTING

1. Description

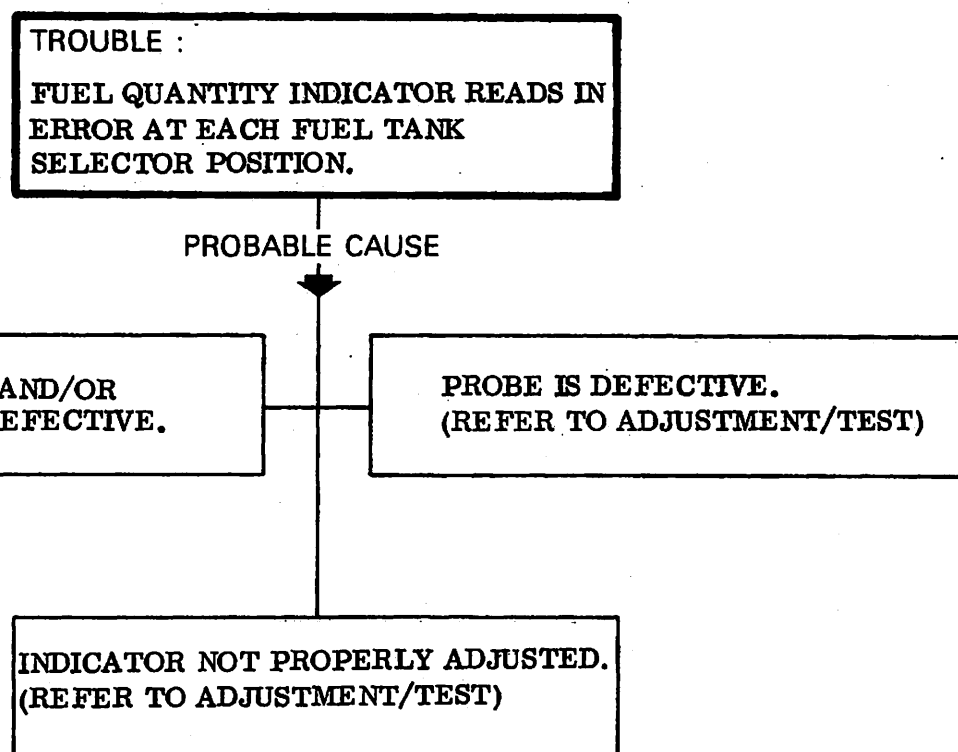
- A. Fuel quantity indicating errors can be attributed to a number of causes. The fuel quantity indicating system is a capacitance system that uses the fuel as the dielectric to vary the capacitance of the probe. This is a comparison between a full tank of fuel and an empty tank. The capacitance signal is directed to a self-balancing bridge circuit contained within the indicator. Also, the differences in fuels affect the accuracy in the system. Fuel quantity indicating errors are overcome by using a compensating probe in the LH inboard wing tank position. The compensating probe is similar to the uncompensated probes except it has an additional plate at the bottom end of the probe. The compensating plate measures the fuel density and directs a correcting signal in the balancing bridge circuit within the indicator. For example: a fuselage tank is filled with fuel but the compensating probe in the left wing tank is not covered with fuel. The fuel quantity reading will be in error because the compensator is unable to correct for the difference in fuel. This can be remedied by fueling the left wing tank. The most common causes of fuel quantity indicating errors are defective connectors where the wires are loose, broken, or shorted. Shorting or grounding the shielding will also result in quantity indicator errors. The shielding is grounded in the indicator only.

2. Trouble Shooting

- A. Visually inspect the system for loose, broken, or shorted wires; proper installation of connectors, probes, indicator, and selector switch, and shorts or grounding of wire shields. After correcting all of the preceding probable cause of trouble, if the system is not operating properly, perform the functional test of the system.
- B. See figure 101 for trouble shooting procedures.

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maintenance manual

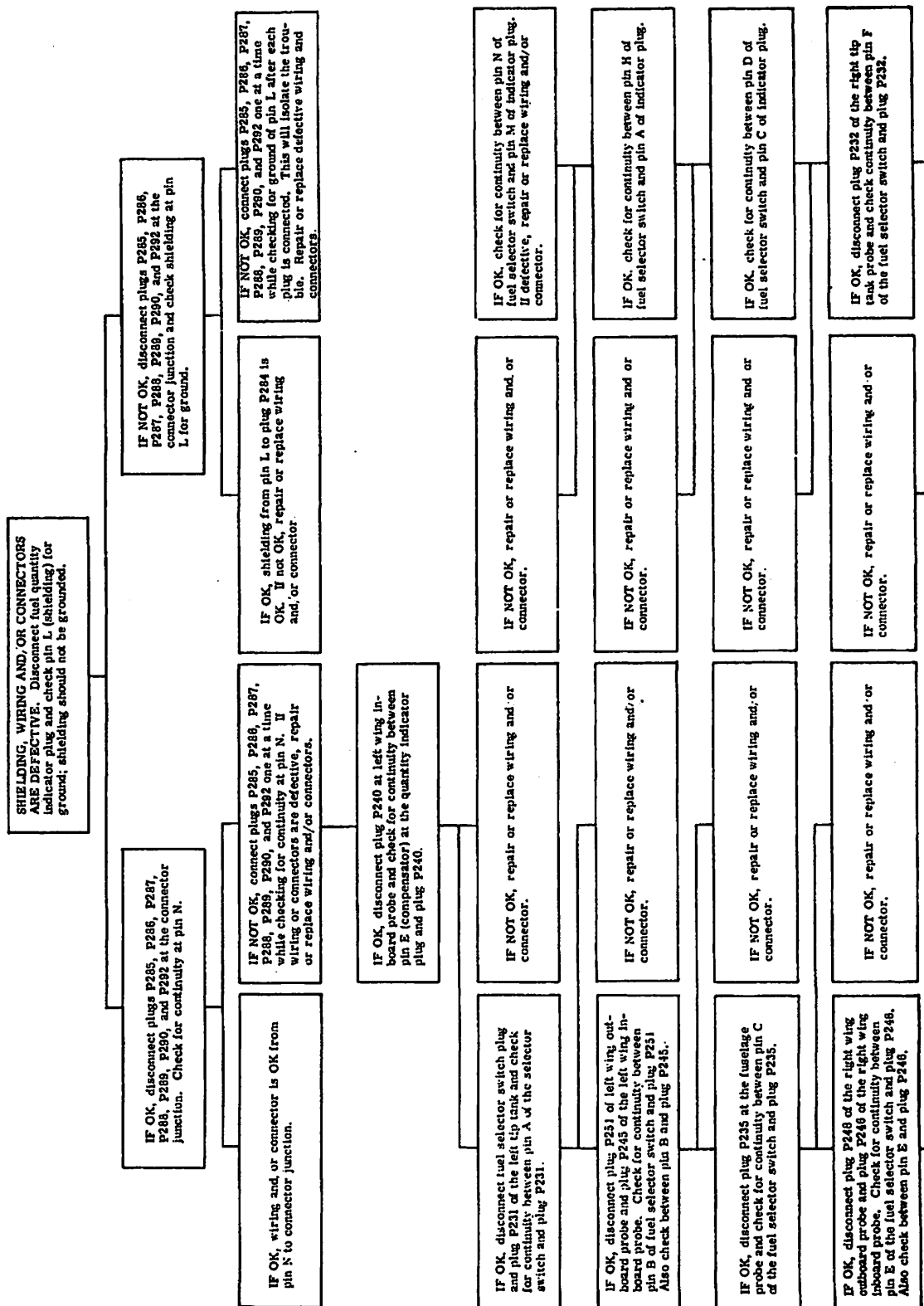


Fuel Quantity Indicating Trouble Shooting
Figure 101 (Sheet 1 of 2)

EFFECTIVITY: ALL

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Fuel Quantity Indicating Trouble Shooting
Figure 101 (Sheet 2 of 2)

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INDICATING - MAINTENANCE PRACTICES

1. Adjustment/Test

NOTE: One of the following functional tests (utilizing the TF-20 or 2548-G test sets) is to be used if extensive work has been accomplished on the fuel gaging system and/or as a trouble shooting guide.

A. Tools and Equipment

NAME	NUMBER	MANUFACTURER	USE
Liquid Quantity Test Set	Model TF-20	Consolidated Airborne Systems, Inc. 900 Third Ave. New Hyde Park, L. I., N. Y.	Functional Test of fuel gaging system.
or			
Test Set	Model 2548G	Barfield Instrument Corp. 1478 Central Ave. Atlanta, Ga. 30344	Functional Test of fuel gaging system.
Adapter Harness	101-00438	Barfield Instrument Corp.	
Coaxial Cables (1 ea. Red, White & Blue)	101-01008	Barfield Instrument Corp.	

B. Functional Test of Fuel Gaging System using Model TF-20 Test Set

- (1) Defuel the aircraft. (Refer to Chapter 12.)
- (2) Connect the TF-20 test set to the aircraft 115 volts AC power source and connect TF-20 test set ground binding post to the aircraft ground.

NOTE: All calibrations and checks of fuel quantity system must be made with 28.0 to 28.5 volts DC available at the generator control panel battery charging bus. One hundred fifteen (115) volts AC is provided by disconnecting the Primary Vertical Gyro electrical plug and inserting TF-20 power lead into the receptacle providing 115 volts AC. (Refer to applicable avionics wiring diagrams.)

- (3) Set Battery and Inverter Switches to on.
- (4) Set TF-20 ON-OFF Switch to ON and allow warmup time of at least 5 minutes.

EFFECTIVITY: ALL

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maintenance manual

- (5) Set ZERO ADJ controls as follows:
 - (a) Rotate FUNCTIONAL SELECTOR to CAP INC-ZERO CAL position and CAP-RES CHECK SELECTOR to CAP.
 - (b) Rotate capacitance RANGE SELECTOR to x 1 position.
 - (c) Check that the two pointers on the CAPACITANCE UUF dial are on their respective index marks.
 - (d) If not on zero, remove protective cover from the front panel ZERO ADJ control and, using a screwdriver, vary the adjustments as required to zero the indicators.
 - (e) Rotate RANGE SELECTOR to x 3, x 10, and x 50 positions. Check that the pointers do not shift from zero.
- (6) Set HIGH ADJ controls as follows:
 - (a) Rotate FUNCTIONAL SELECTOR TO CAP IND-HIGH CAL position.
 - (b) Check the value of capacitance shown on the front panel metal strip for the STANDARD CAPACITANCE MMF designation.
 - (c) Set RANGE SELECTOR to the lowest multiplier that will permit reading the STANDARD CAPACITANCE UUF value on the CAPACITANCE UUF dial.

Example: If the metal strip shows a value of 968.9, rotate RANGE SELECTOR to the x 10 position to permit reading this value on the CAPACITANCE UUF dial.

- (d) The CAPACITANCE UUF pointer should not deflect to the calibrating value shown on the metal strip. If the dial pointers fail to deflect to this value, remove protective cover from the front panel HIGH ADJ control. Using a screwdriver, adjust the controls to obtain the required capacitance indication on the CAPACITANCE UUF dial.
- (7) Repeat steps (5) and (6) to assure that calibration of one control has not affected the other.
- (8) Calibrate megohmmeter as follows:
 - (a) Rotate megohmmeter RANGE SELECTOR to ZERO CAL position.
 - (b) Check that pointer deflects to ZERO CALIBRATION. If not on zero, remove cover from front panel ZERO ADJ control. Using a screwdriver, adjust controls until pointer indicates ZERO CALIBRATION.
 - (c) With the CAP-RES CHECK switch in any position except CAP, rotate the megohmmeter RANGE SELECTOR to MIDSCALE CAL.
 - (d) The megohmmeter pointers will deflect to MIDSCALE CALIBRATE index line. If pointer is not on index line, remove cover from MIDSCALE ADJ. Using a screwdriver, adjust controls until pointer is on index line.

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maintenance manual

- (9) Repeat step (8) to assure that calibration of one control has not affected the other.
- (10) Remove attaching parts, disconnect electrical plug from fuel quantity indicator and remove fuel control panel sufficiently to rest fuel control panel on pedestal.
- (11) Connect one multi-pin connector from the TF-20 adapter harness (indicator side) to the receptacle at the rear of the fuel quantity indicator. Connect the second multi-pin from TF-20 adapter harness (tank unit side) to aircraft wiring. (See figure 201.)
- (12) Connect two branch groups of leads to proper receptacles on the TF-20 test unit as follows:
 - (a) Connect three leads in branch marked (TANK UNIT) to the appropriate TANK UNIT COMP-COAX-UNSH receptacles at the top center of the test unit.

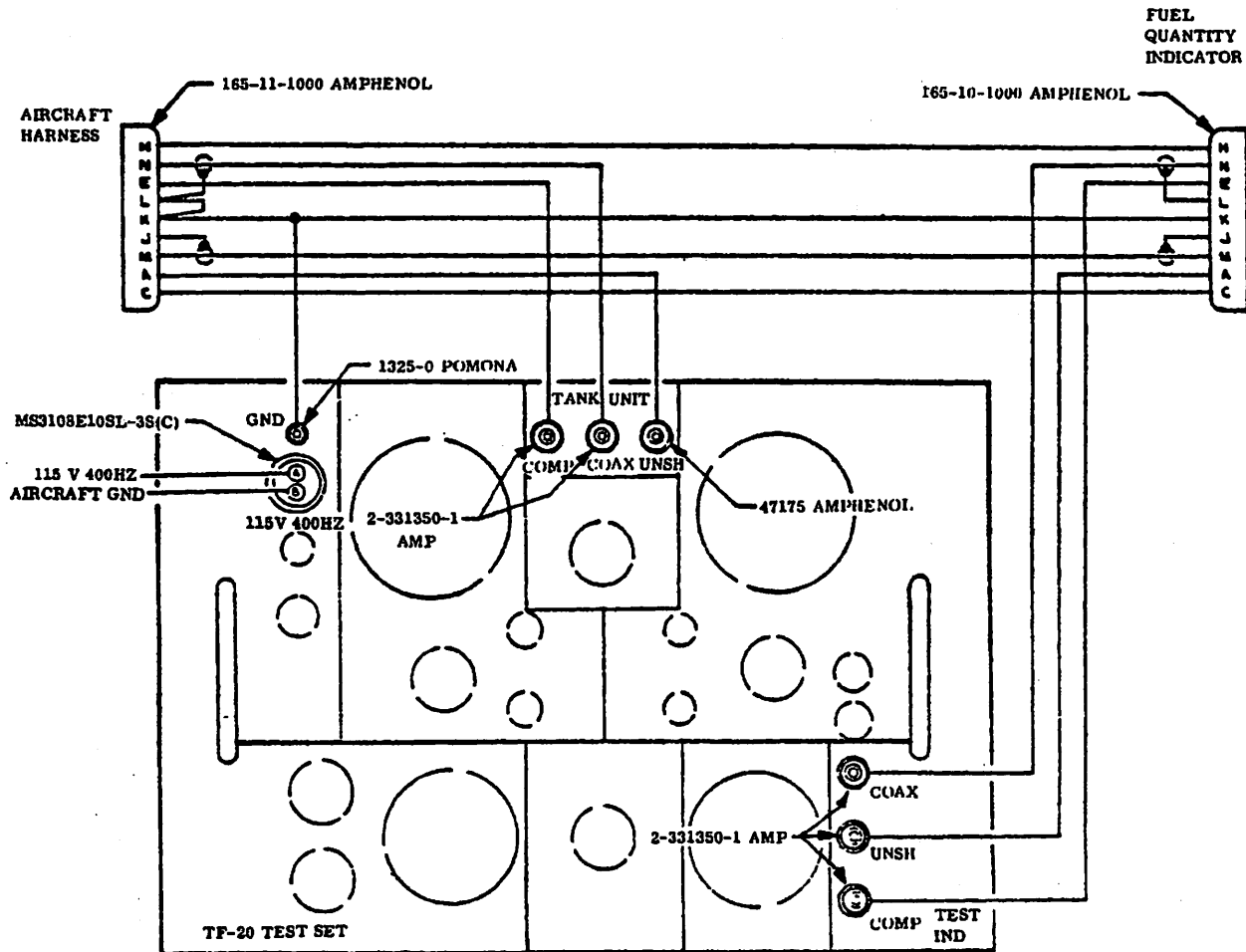
NOTE: A fourth lead is supplied for grounding purposes; connect this lead to the "GND" terminal (upper left side of test unit). Remove ground previously attached in step A. (2).

- (b) Connect three leads in the second branch marked (INDICATOR) to the appropriate TEST IND "COAX" - "UNSH" - "COMP" receptacles (lower right side of test unit).
- (13) Check fuel probe resistances as follows (fuel tanks must be empty):
 - (a) Rotate FUNCTION SELECTOR to TANK UNIT TEST-UNSH position and set CAP-RES CHECK Switch to A-B (lower set of letter symbols).

NOTE: In measuring resistances of the probes, always set megohmmeter RANGE SELECTOR to the multiplier factor where the pointer of the MEGOHMS meter deflects closest to mid-scale.

- (b) Rotate Fuel Quantity Selector Switch to each tank position and check for allowable leakage resistance tolerance of 50 megohms minimum. This check measures the resistance between the probe elements.
 - (c) Set CAP-RES CHECK Switch to A-GND. Rotate Fuel Quantity Selector Switch to each tank position and check for minimum resistance of 30 megohms. This check measures the resistance between the center conductor of the coaxial probe.
 - (d) Set CAP-RES CHECK Switch to B-GND. Rotate Fuel Quantity Selector Switch to each tank position and check for minimum resistance of 30 megohms. This check measures the resistance between the unshielded lead to ground.

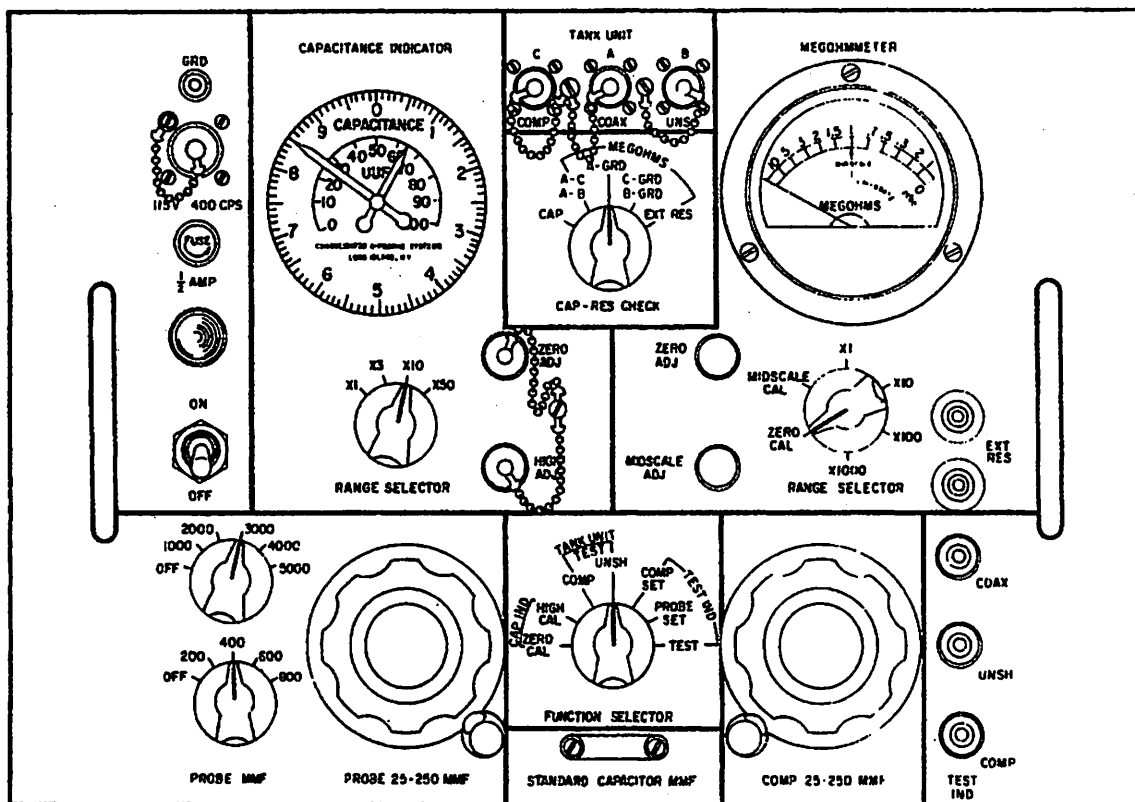
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Fuel Quantity Probe Test Setup
Figure 201

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Fuel Quantity Test Box
Figure 202

EFFECTIVITY: ALL

MM-98

28-40-00
Page 205
Dec 18/81

Gates Learjet Corporation

maintenance manual

- (14) Check fuel probe capacitance as follows:
- (a) With FUNCTION SELECTOR in the TANK UNIT TEST-UNSH position set CAP-RES CHECK Switch to its extreme counter-clockwise CAP position.
 - (b) Rotate Fuel Quantity Selector Switch to each tank position and check capacitance reading. (Refer to Table 201.)

EMPTY	25B/D	25C/F
L. Wing Tip	27.7 (± 0.67)	27.7 (± 0.67)
Left Wing	20.5 (± 0.67)	20.5 (± 0.67)
Fuselage	26.4 (± 0.67)	48.2 (± 0.67)
Right Wing	20.5 (± 0.67)	20.5 (± 0.67)
R. Wing Tip	27.7 (± 0.67)	27.7 (± 0.67)
Total	122.8 (± 1.5)	144.6 (± 1.5)

Table 201
Fuel Probe Capacitance Reading - MMF

- (15) Check the compensator (left wing inboard probe), by rotating the FUNCTION SELECTOR to TANK UNIT TEST-COMP position and check for capacitance of 30.3 (± 0.7) UUF (fuel sump dry). If wet, capacitance is 63.5 (± 0.7) UUF.
- (16) Disconnect test unit electrical plugs from fuel quantity indicator and aircraft wiring. Connect electrical plug (P226) to fuel quantity indicator.
- (17) Rotate Fuel Quantity Selector Switch to TOTAL and adjust "E" (empty) adjustment on the back of fuel quantity indicator for zero pounds indication.
- (18) Connect TF-20 test unit to aircraft wiring per figure 202.

CAUTION: CHECK SELECTOR SWITCH AND FUEL QUANTITY INDICATOR PART NUMBERS. COMBINATIONS SHOWN IN TABLE 202 ARE THE ONLY APPLICATIONS APPLICABLE TO THIS TEST. ANY OTHER COMBINATIONS SHOULD NOT BE USED.

NOTE: Ensure that the TF-20 test unit is warmed up and zeroed per steps A. (2) thru A. (9).

- (19) Set CAP-RES CHECK switch to CAP position and rotate FUNCTION SELECTOR to TEST-IND-COMP SET position.
- (20) Loosen COMP capacitor knob clamp and rotate knob to obtain the desired reading on the CAPACITANCE UUF dial. (Refer to Table 202 for desired reading.)

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- (21) Tighten clamp to lock COMP capacitor knob in place.

NOTE: Rotate the RANGE SELECTOR to the lowest multiplier factor that will permit an accurate reading of the CAPACITANCE UUF dial.

- (22) Rotate the FUNCTIONAL SELECTOR to the TEST IND PROBE SET position.
- (23) Set "MMF in hundreds" switch to 200 MMF and set the "MMF in thousands" switch to OFF.
- (24) Loosen PROBE capacitor knob clamp and rotate knob to obtain the desired reading of the main probe capacitance on the CAPACITANCE UUF dial. (Refer to Table 202, for desired reading.)
- (25) Rotate FUNCTION SELECTOR to the TEST IND TEST position. This applies capacitance to the indicator, the indicator should indicate a full scale reading indicating the fuel tanks are full.
- (26) If the indicator does not have a full scale deflection, adjust the "F" full adjustment on the back of fuel quantity indicator for a full scale deflection.
- (27) Rotate FUNCTION SELECTOR to PROBE SET and check quantity indicator "E" (Empty) setting. (Refer to steps A. (16) and A. (17).)
- (28) Remove power from the test setup and disconnect the test leads from the aircraft.
- (29) Restore the aircraft to operational configuration.
- (30) Fuel aircraft. (Refer to Chapter 12.)

Step	Model	Reqd Range Selector Switch Position	*See Part Numbers Below		*See Part Numbers Below		Full Scale Indication
			UUF Indicator Reading for x 3 Position	Full Scale Capacitance	UUF Indicator Reading for x 3 Position	Full Scale Capacitance	
(20)	25B/C/D/F	N/A	63.5 MMF	N/A	62.9 MMF	N/A	N/A
(25)	25B/D	x 3	83.57 UUF	250.7 UUF	85.9 UUF	257.6 UUF	6600 Lbs
(25)	25C/F	x 3	99.87 UUF	299.6 UUF	101.9 UUF	305.8 UUF	7000 Lbs

*25B/D - RG55B118 Selector Switch with JG911A104 Indicator.

*25C/F - RG55B23 Selector Switch with JG911A105 Indicator.

*25B/D - RG55B18 Selector Switch with JG911A4 Indicator.

- RG55B18 Selector Switch with JG911A104 Indicator.

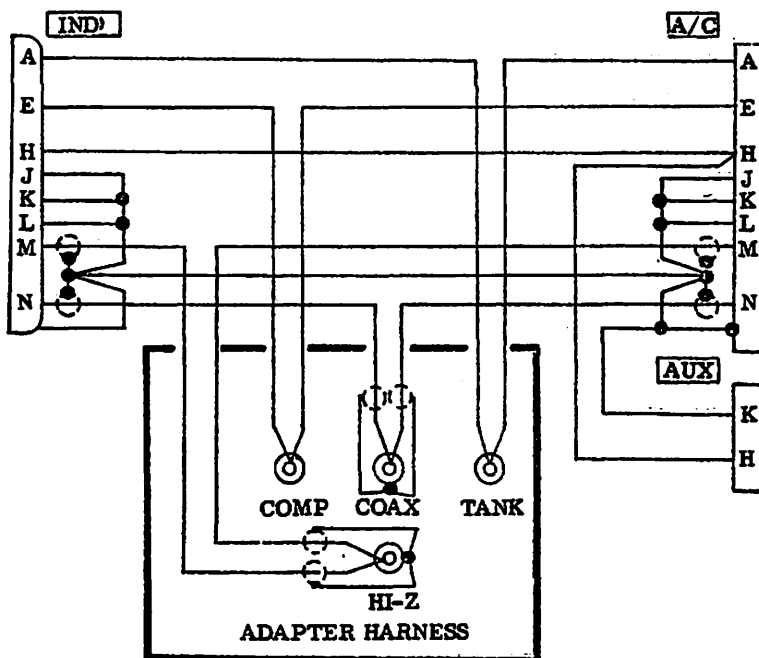
*25C/F - RG55B23 Selector Switch with JG911A5 Indicator.

Full Scale Capacitance Reading
Table 202

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- C. Functional Test of Fuel Gaging System using Model 2548-G Test Set
(See figure 203)
- (1) Defuel the aircraft. (Refer to Chapter 12.)
 - (2) Zero the test set meter as follows:
 - (a) With test set OFF, set TEST FUNCTION to CAP BRIDGE.
 - (b) Set TANK CAP thumbwheels to 62.9 pf.
 - (c) Turn test set ON and depress BATT TEST button. Meter pointer should read above BAT OK radial line.
 - (d) Press COMP CAL and adjust COMP SIM for a zero reading on uA scale.
 - (e) Turn test set OFF.



Test Set and Adapter Harness
Figure 203

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- (3) Pull fuel quantity circuit breaker and connect test set and adapter per figure 204.
- (4) Test fuel quantity indicator as follows.
 - (a) Set TANK CAP to 122.8 (25B/D) or 144.6 (25C/F).
 - (b) Set TEST FUNCTION to CAP SIM-1.
 - (c) Actuate the fuel quantity circuit breaker and turn test set ON.
 - (d) Adjust empty potentiometer "E" or Fuel Quantity Indicator to bring pointer to zero graduation.
 - (e) Reset TANK CAP pf to FULL value 250.7 pf (25B/D) or 299.6 pf (25C/F) and adjust fuel potentiometer "F" on Fuel Quantity Indicator for POUNDS FULL indication - 6,600 pounds (25B/D) or 8,000 pounds (25C/F).
 - (f) Reset TANK CAP pf for 122.8 (25B/D) or 144.6 (25C/F). Indicator valve shall return to zero.
 - (g) Pull fuel quantity circuit breaker and turn test set OFF
- (5) Test system insulation as follows:
 - (a) Disconnect red, white, and blue from test set.
 - (b) Set TEST FUNCTION to INSULATION and turn test set ON.
 - (c) Turn METER ADJUST to set meter pointer to full scale infinity mark on MEGOHMS scale.
 - (d) Turn test set OFF and connect test set and adapter per figure 205.
 - (e) Turn test set ON.
 - (f) Set INS TEST POINTS to each of the positions in Table 203, switching the Fuel Quantity Selector Switch through each of its positions.

NOTE: Meter may drift after initial setting. Wait for steady reading or until reading exceeds minimum resistance allowable for each of the insulation tests.

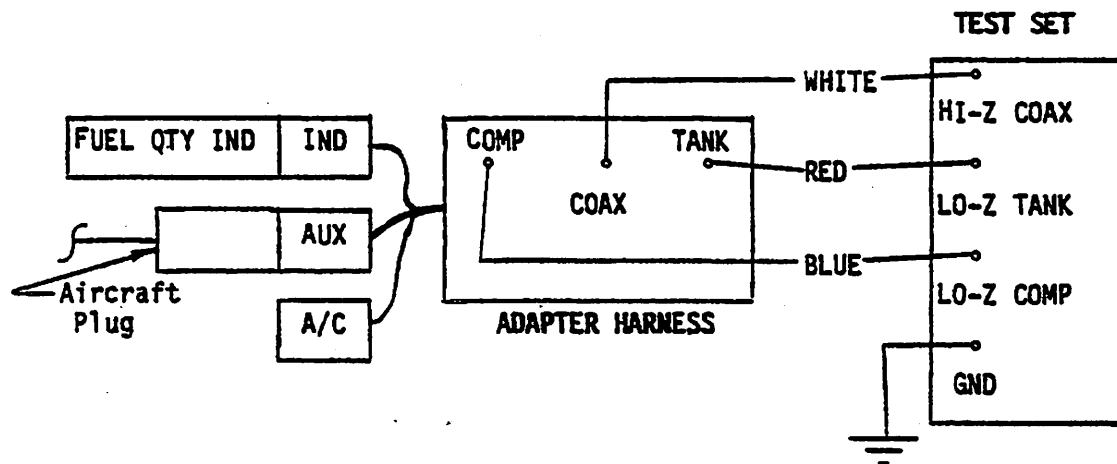
TEST POINTS	MEGOHMS MINIMUM
GND/LO-Z	10
GND/HI-Z	10
SH/LO-Z	10
SH/HI-Z	10
LO-Z/HI-Z	100

Table 203

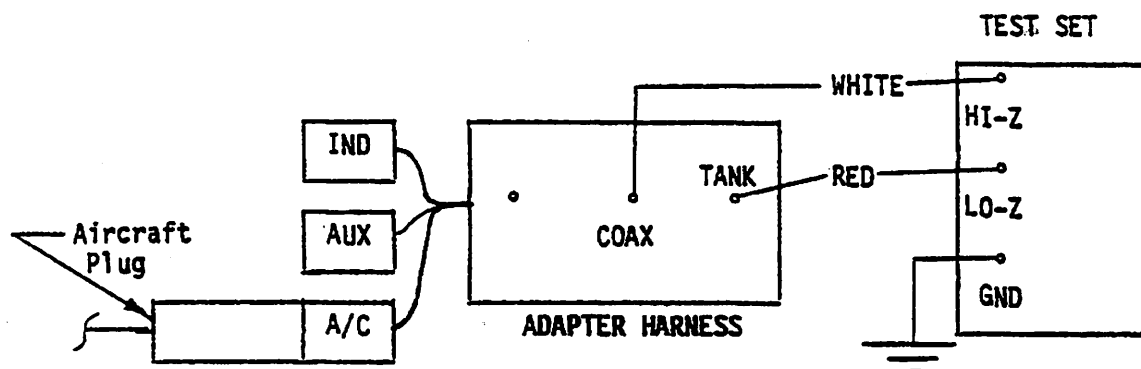
- (g) Turn test set OFF and connect red coded coax to COMP of adapter harness and repeat steps (5)(e) and (5)(f).

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Test Set and Adapter Connections
Figure 204



Test Set and Adapter Connections
Figure 205

EFFECTIVITY: ALL

MM-98

28-40-00
Page 210
Dec 18/81

Gates Learjet Corporation

maintenance manual

- (6) Test Fuel Quantity Selector Switch insulation as follows:
- (a) Turn test set OFF and connect test set and adapter harness per figure 206.
 - (b) Perform steps (5)(e) and (5)(f).
- (7) Test tank unit probe as follows:

NOTE: AC inverters should be OFF during test set capacitance measurements.

- (a) Turn test set OFF and connect test set and adapter harness per figure 207.
- (b) Set Fuel Quantity Switch to desired position per Table 204.

AIRCRAFT MODEL	FUEL SELECTOR	NOM. CAP. pf
25B, 25D	L & R Tip	27.7
	L & R Wing	20.5
	Fus	26.4
TOTAL		122.8
25C, 25F	L & R Tip	27.7
	L & R Wing	20.5
	Fus	48.2
TOTAL		144.6

Table 204

- (c) Set TANK CAP pf thumbwheels to corresponding pf value.
- (d) Set test set TEST FUNCTION to CAP BRIDGE.
- (e) Turn test set ON, depress PRESS TO MEASURE, and adjust TANK CAP pf for panel meter null of 100 μ A or less. (A very low reading indicates an open, unable to null indicates a short, and 200 μ A or more indicates a high resistance connection.)

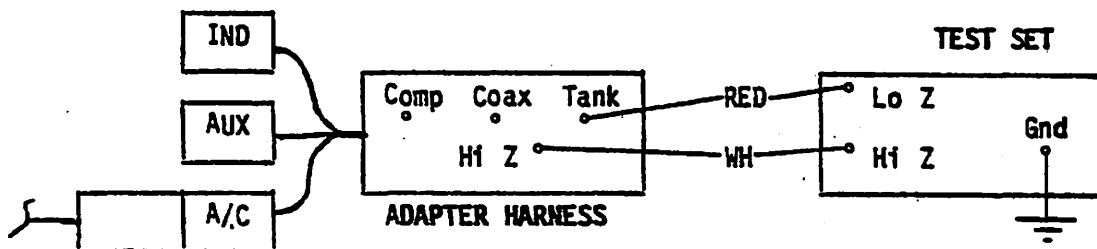
NOTE: Thumbwheel readings should be -1.0 to +1.5 for individual probes, and -4.5 to +5.5 for total.

- (f) Repeat steps c, d, and e at all Fuel Quantity Switch positions.
 - (g) Turn test set OFF.
- (8) Measure compensator capacitance as follows:
- (a) Interchange red coded coax and shorting plug on COMP and TANK jacks of adapter harness.
 - (b) Turn test set ON.

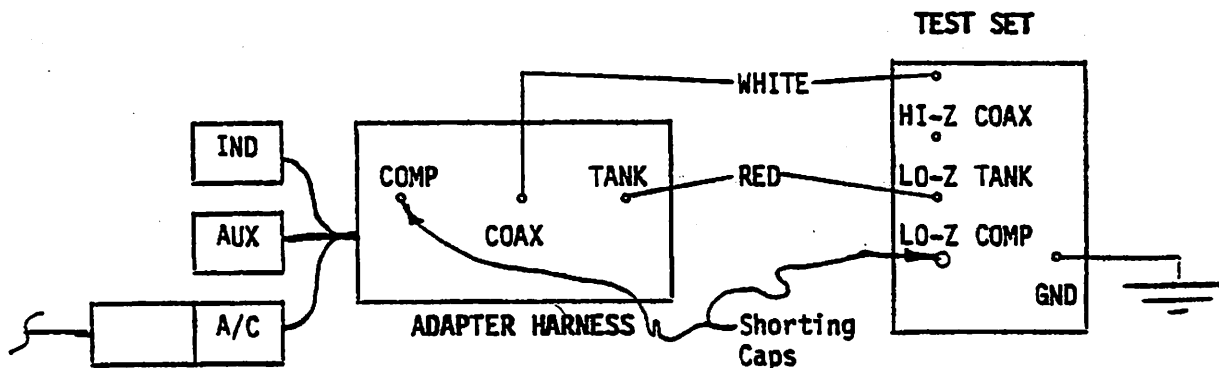
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- (c) Set thumbwheels to 30.3 pf.
- (d) Depress PRESS TO MEASURE, and adjust thumbwheels for null of 100 μ A or less.
- (e) Compensator should read 30.3 -1.0 +1.5 pf.
- (f) Turn test set OFF.



Test Set and Adapter Connections
Figure 206



Test Set and Adapter Connections
Figure 207

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- (9) Measure Selector Switch capacitance as follows:
- (a) Turn test set OFF and connect test set and adapter harness per figure 208.
 - (b) Set Fuel Quantity Selector Switch to desired position per Table 205.
 - (c) Set TANK CAP pf thumbwheel to corresponding pf value.
 - (d) Set test set TEST FUNCTION to CAP BRIDGE.

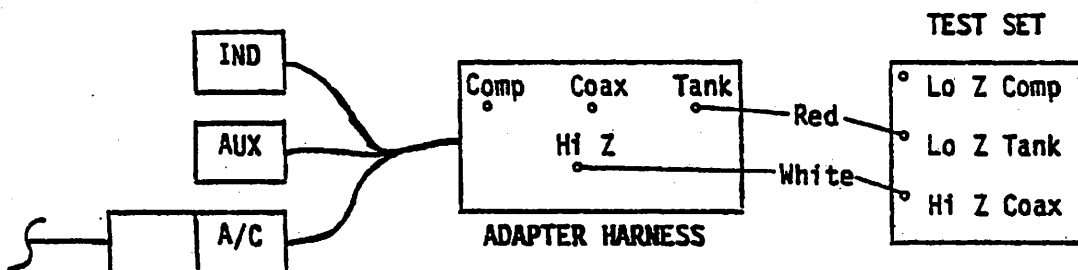
MODEL	FUEL SELECTOR	NOM. CAP pf	
25B, 25D	L & R Tip	95.1	96.1
	L & R Wing	102.3	103.3
	Fus	96.4	97.4
25C, 25F	L & R Tip	116.9	
	L & R Wing	124.1	
	Fus	96.4	

Table 205

- (e) Turn test set ON.
- (f) Depress PRESS TO MEASURE and adjust TANK CAP pf for panel meter null of 100 μ A or less. (A very low reading indicates an open, unable to null indicates a shot, and 200 μ A or more indicates a high resistance connection.)

NOTE: Thumbwheel readings should be -1.0 to +1.5 for individual probes and -4.5 to +5.5 for total.

- (g) Repeat steps c, d, e, and f at all Fuel Quantity Switch positions.
- (h) Turn test set OFF.



Test Set and Adapter Connections
Figure 208

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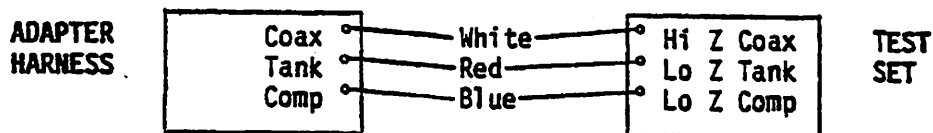
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- (10) Connect adapter harness IND plug to fuel quantity indicator, and A/C receptacle to indicator plug.
- (11) Set TEST FUNCTION TO CAP BRIDGE.
- (12) Set TANK CAP pf to 32.6 and turn test set ON.
- (13) Press COMP CAL and adjust COMP SIM for minimum meter reading.
- (14) Turn test set OFF.
- (15) Connect color coded coax per figure 209 and disconnect GND lead between test set and ground.
- (16) Set TANK CAP pf to 000.0 and TEST FUNCTION to CAP SIM 1.
- (17) Set Fuel Quantity Selector Switch to TOTAL.
- (18) Depress Fuel Quantity circuit breaker and set empty potentiometer to zero. (Test set OFF.)

MODEL	ADD FOR FUEL	FULL POUNDS
25B/D	127.9 pf	6,600
25C/F	155.0 pf	8,000

Table 206

- (19) Set TANK CAP pf to ADD FOR FULL value listed in Table 206.
- (20) Turn test set ON, and adjust FULL potentiometer on indicator to bring pointer to FULL POUNDS value listed in Table 206.
- (21) Turn test set OFF. Fuel Quantity Indicator should return to zero. If not, repeat steps (16) through (20) until no further improvement can be made.
- (22) Pull Fuel Quantity circuit breaker.
- (23) Disconnect test set and adapter harness from airplane and connect aircraft connector to indicator.
- (24) Depress Fuel Quantity circuit breaker.



Test Set and Adapter Connections
Figure 209

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2. Inspection/Check

A. Operational Check of Fuel Gaging System

NOTE: The following operational check is to be used to calibrate the fuel quantity indicator or check the system if any system component has been replaced.

- (1) Defuel the aircraft. (Refer to Chapter 12.) Remaining fuel must also be drained from the sumps and siphoned from tip tanks.
- (2) Remove attaching parts and fuel control panel sufficiently to gain access to fuel quantity adjustment potentiometers.
- (3) Set Battery Switches on and set Fuel Quantity Selector switch to TOTAL. Allow system to warm up. Adjust empty potentiometer marked "E" on back of indicator for zero indication.
- (4) Rotate Fuel Quantity Selector Switch to each individual tank position. The fuel quantity indicator must read -50 to +100 lbs. (25B/D) or -100 to +100 lbs. (25C/F) at each tank position.
- (5) Fuel the aircraft. (Refer to Chapter 12.)
- (6) Set Fuel Quantity Selector Switch to TOTAL. Adjust full potentiometer marked "F" on back of indicator for TOTAL pounds indication.
- (7) Rotate Fuel Quantity Selector Switch to each individual tank position. The fuel quantity indicator must read within ± 100 pounds at each tank position.
- (8) If any individual tank reading is out of tolerance, adjust full potentiometer on back of indicator until reading is within the ± 100 pound tolerance. If any adjustments were made on individual tank readings, set Fuel Quantity Selector Switch to TOTAL and adjust fuel potentiometer to within ± 100 pounds of TOTAL fuel quantity.
- (9) If individual tank readings are within tolerance, install and secure fuel control panel.
- (10) Set Battery Switches to OFF.

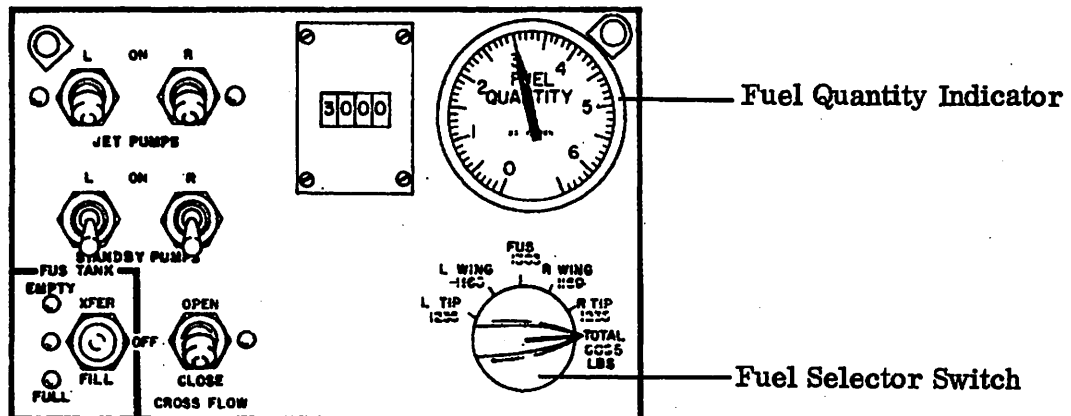
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FUEL QUANTITY INDICATOR - MAINTENANCE PRACTICES

1. Removal/Installation

- A. Remove Fuel Quantity Indicator and Fuel Selector Switch (See figure 201)
- (1) Remove attaching parts and fuel control panel from pedestal.
 - (2) Disconnect electrical connector from fuel control panel.
 - (3) Disconnect electrical connectors from selector switch and indicator.
 - (4) Remove attaching parts and selector switch and fuel quantity indicator from fuel control panel.
- B. Install Fuel Quantity Indicator and Fuel Selector Switch (See figure 201)
- (1) Install selector switch and fuel quantity indicator and secure with attaching parts.
 - (2) Connect electrical connectors.
 - (3) Install fuel control panel in pedestal.
 - (4) Fuel the aircraft. (Refer to Chapter 12.)
 - (5) Rotate the Fuel Quantity Selector Switch to each tank position. The fuel quantity indicator must read within ± 100 pounds of each tank value as shown on fuel control panel. TOTAL value must be within ± 100 pounds.



Fuel Control Panel (Typical)
Figure 201

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FUEL PROBE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Fuselage Fuel Probe (See figure 201)

- (1) Defuel fuselage tank. (Refer to Chapter 12.)
- (2) Remove access cover from top of aircraft.
- (3) Disconnect electrical connectors.
- (4) Remove attaching parts and fuel probe from tank.

NOTE: Lower fuel probe fitting need not be removed unless replacement is required.

B. Install Fuselage Fuel Probe (See figure 201)

- (1) Install fuel probe and secure with attaching parts. Torque attaching bolts to 50 (+5) inch-pounds. Assure that bonding jumper is secure.
- (2) Connect electrical connectors.
- (3) Fuel the fuselage tank and check for leaks. (Refer to Chapters 12 and 20.)
- (4) Install access cover and secure with attaching parts.

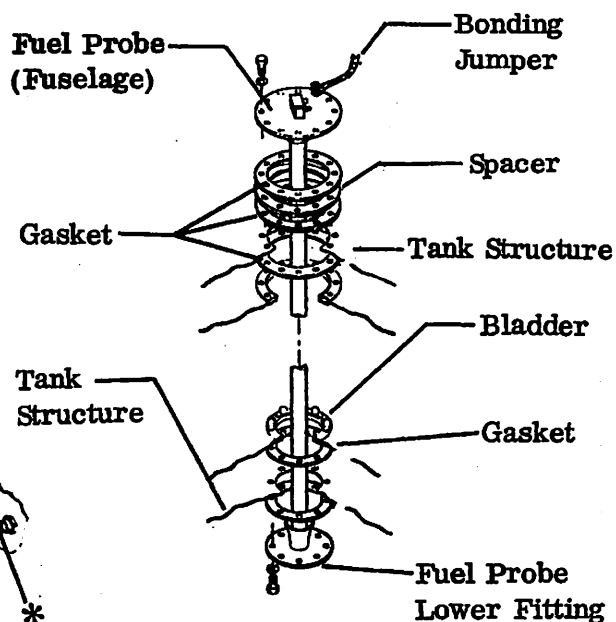
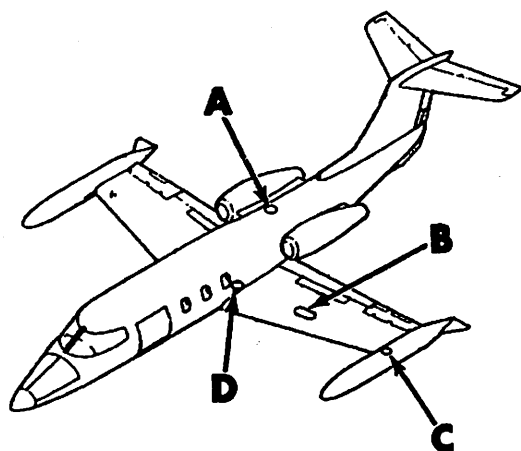
C. Remove Wing Fuel Probes (Typical Outboard Wing) (See figure 201)

- (1) Defuel the aircraft. (Refer to Chapter 12.)
- (2) Disconnect electrical connector from probe receptacle at spar 7.
- (3) Remove attaching parts and two access covers aft of fuel probe access panel.
- (4) Remove attaching parts securing probe electrical receptacle to spar 7.
- (5) Remove attaching parts and clamps securing probe wiring at spars 4, 5, and 6.
- (6) Remove attaching parts and probe access panel from aircraft.
- (7) Remove attaching parts and probe from access cover.
- (8) Install replacement probe on access panel using existing attaching parts.

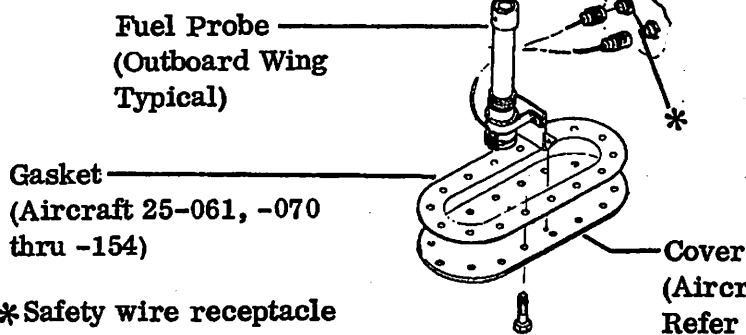
D. Install Wing Fuel Probe (Typical Outboard Wing) (See figure 201)

- (1) Apply lightning strike protective seal to access cover. (Refer to 28-12-00.)
- (2) Route probe wiring through spars, 4, 5, and 6 and install fuel probe access cover.
- (3) Secure probe wiring with clamps at spars 4, 5, and 6.
- (4) Install O-ring and receptacle on spar 7 and secure with attaching parts.
- (5) Connect electrical connector to receptacle.
- (6) Apply lightning strike protective seal to access covers. (Refer to 28-12-00.) Install access covers.
- (7) Perform Operational Check of Fuel Gaging System, this section.

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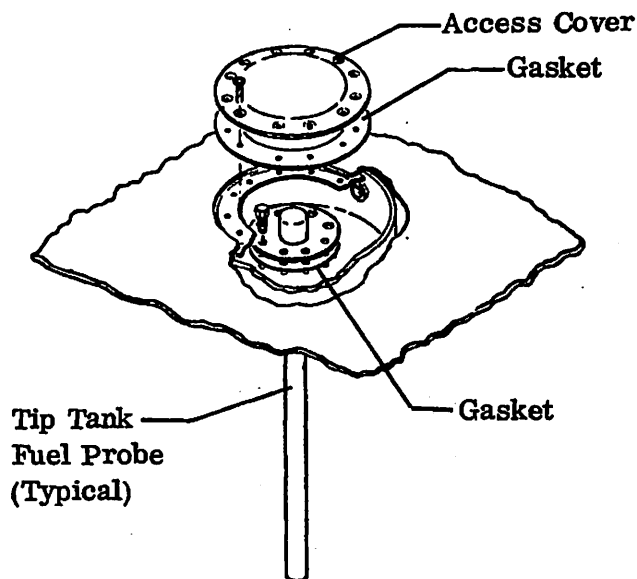


Detail A

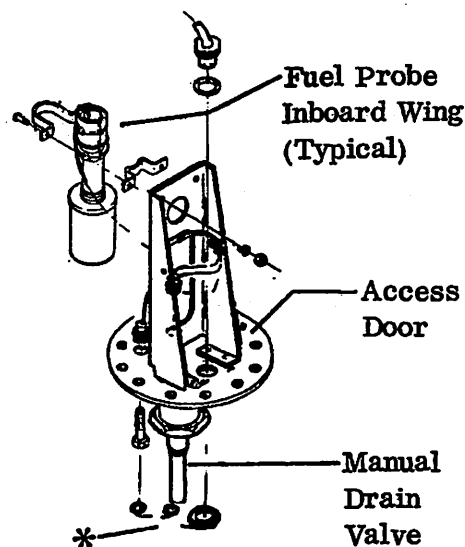


*Safety wire receptacle mounting nuts to retain torque.

Detail B



Detail C



Detail D

Fuel Probe Installation
Figure 201

EFFECTIVITY: ALL

MM-98

28-40-02
 Page 202
 Dec 18/81

Gates Learjet Corporation
maintenance manual

E. Remove Fuel Probe (Inboard Wing) (See figure 201)

NOTE: Removal and installation procedures for the LH and RH fuel probes are identical, except the LH fuel probe has three electrical plugs connected to the access cover whereas the RH fuel probe has only two.

- (1) Defuel the aircraft. (Refer to Chapter 12.)
- (2) Remove lower fuselage access covers to gain access to fuel probe installation.
- (3) Disconnect electrical connectors from access covers.
- (4) Remove attaching parts and access cover from wing.
- (5) Remove attaching parts and probe receptacles from access cover.
- (6) Remove attaching parts and probe from access cover.
- (7) Install replacement probe on access cover and secure with existing attaching parts.

F. Install Fuel Probe (Inboard Wing) (See figure 201)

- (1) Install gasket and access cover and secure with attaching parts.
- (2) Connect electrical plugs to probe receptacles.
- (3) Perform fuel quantity indicating calibration check. (Refer to 28-40-00.)
- (4) Install lower fuselage access covers.

G. Remove Tip Tank Fuel Probe (See figure 201)

- (1) Drain tip tank.
- (2) Remove attaching parts and junction box access cover.
- (3) Disconnect electrical plugs from fuel probe.
- (4) Remove attaching parts and fuel probe from tip tank.
- (5) Inspect and clean fuel probe.

H. Install Tip Tank Probe (See figure 201)

- (1) Install gasket, fuel probe and secure with attaching parts.
- (2) Connect electrical plugs to fuel probe.
- (3) Install gasket, junction box access cover and secure with attaching parts.
- (4) Check fuel quantity indicating system calibration.

2. Inspection/Check

A. Inspect Fuel Probe

- (1) Visually inspect outer electrode for small nicks or scratches in protective insulating coating.
- (2) Check all parts for evidence of physical damage, distortion and deterioration.

Gates Learjet Corporation

maintenance manual

3. Cleaning/Painting

A. Clean Fuel Probe

- (1) Fill a suitable container with Stoddard solvent. Using a soft bristle brush and back-and-forth motion, remove loose deposits from probe. Avoid using brush between probe electrodes.
- (2) Thoroughly dry fuel probe.

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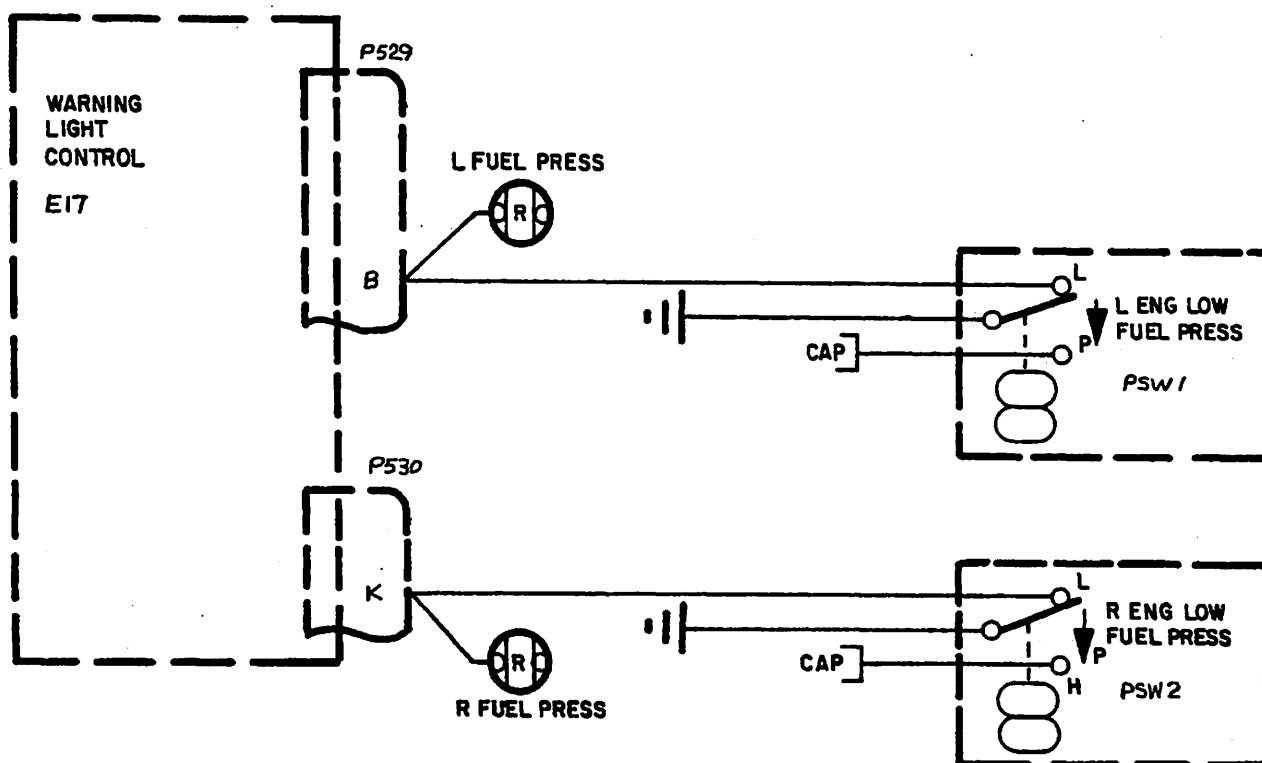
PRESSURE SWITCH - DESCRIPTION AND OPERATION

1. General

A. The switches are located between the relief valve and the engine in the engine main fuel lines (RH and LH). Access to the switches is gained through the tailcone access door.

2. Operation (See figure 1)

A. The engine low fuel pressure switches are installed to alert the pilot of jet pump failure. The function of an engine low fuel pressure switch is to sense pressure from the main fuel line and energize either the R or L FUEL PRESS indicator light on the glareshield. The switch closes to energize the light when fuel pressure drops below .25 psi and reopens at 1.0 psi.



Engine Fuel Pressure Warning Light Electrical Control Schematic
Figure 1

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PRESSURE SWITCH - MAINTENANCE PRACTICES

1. **General**

A. Maintenance practices consist of removal and installation of the pressure switch.

2. **Removal/Installation**

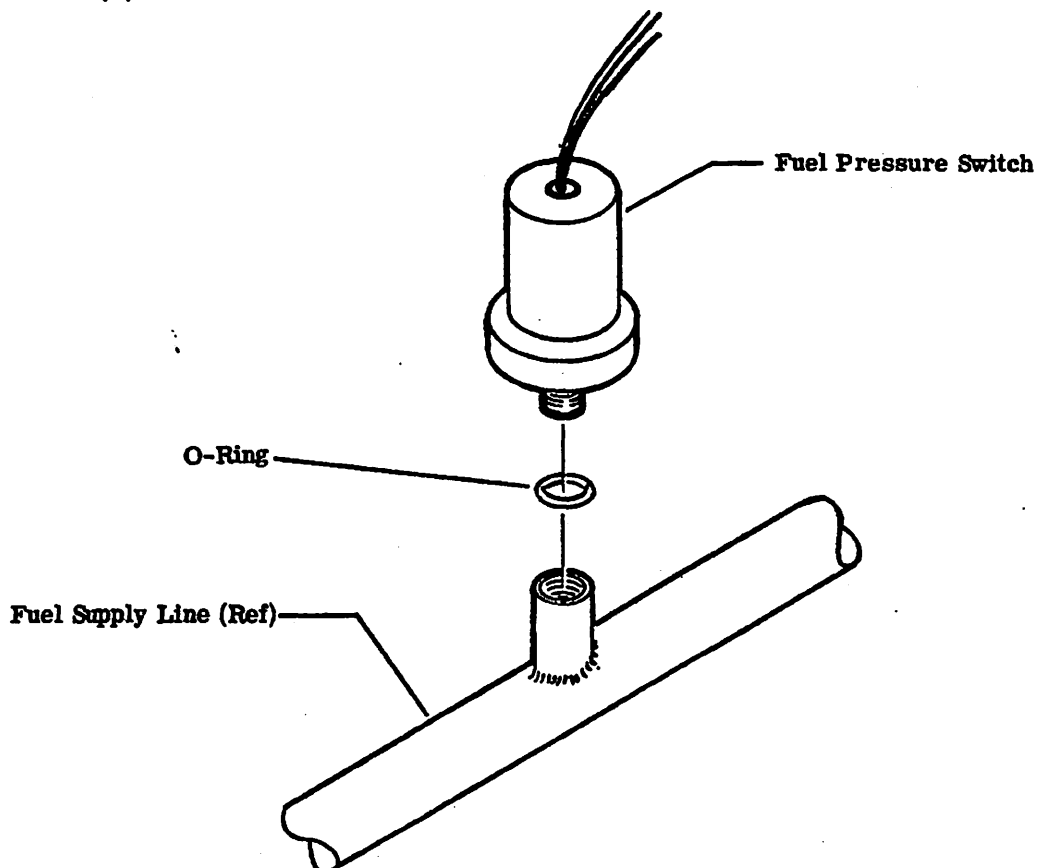
NOTE: The following procedures are applicable to either pressure switch.

A. Remove Pressure Switch (See figure 201)

- (1) Lower tailcone access door.
- (2) Disconnect pressure switch wiring at splice.
- (3) Remove pressure switch and O-ring from main fuel line.

B. Install Pressure Switch (See figure 201)

- (1) Install O-ring and pressure switch in main fuel line.
- (2) Connect electrical wiring at splice.
- (3) Check pressure switch using applicable standby boost pump.
- (4) Secure tailcone access door.



**Fuel Pressure Switch Installation
Figure 201**



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FUEL COUNTER SYSTEM - MAINTENANCE PRACTICES

1. General

- A. The fuel counter system consists of an indicator on the fuel control panel and utilizes an output from the fuel monitor computer.
- B. The fuel counter is an electromechanical device that indicates total pounds of fuel used. The counter can be reset by depressing the small button adjacent to the indicator.

2. Operation

- A. The engine fuel flowmeters supply an alternating voltage to the fuel monitor computer which is converted into DC electrical signals. These DC electrical signals are applied to the fuel counter system and the fuel flow indicators. (Refer to 73-31-00)